



Design and Deployment of UC Services in SD-WAN

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BRKCOL-2078

Connect Users
to the World

Simplify Network
Operations



Webex App

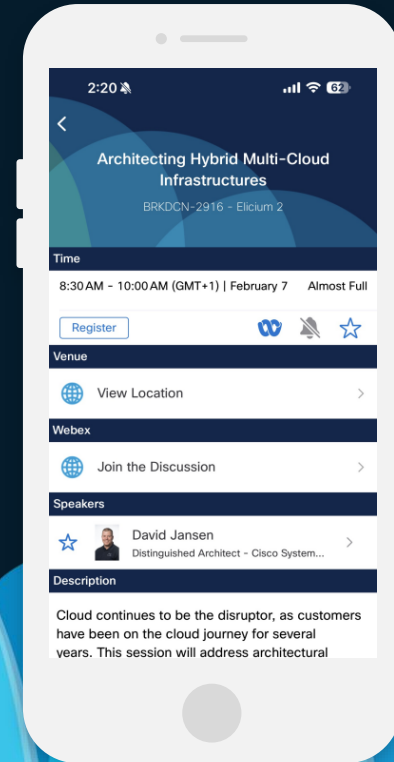
Questions?

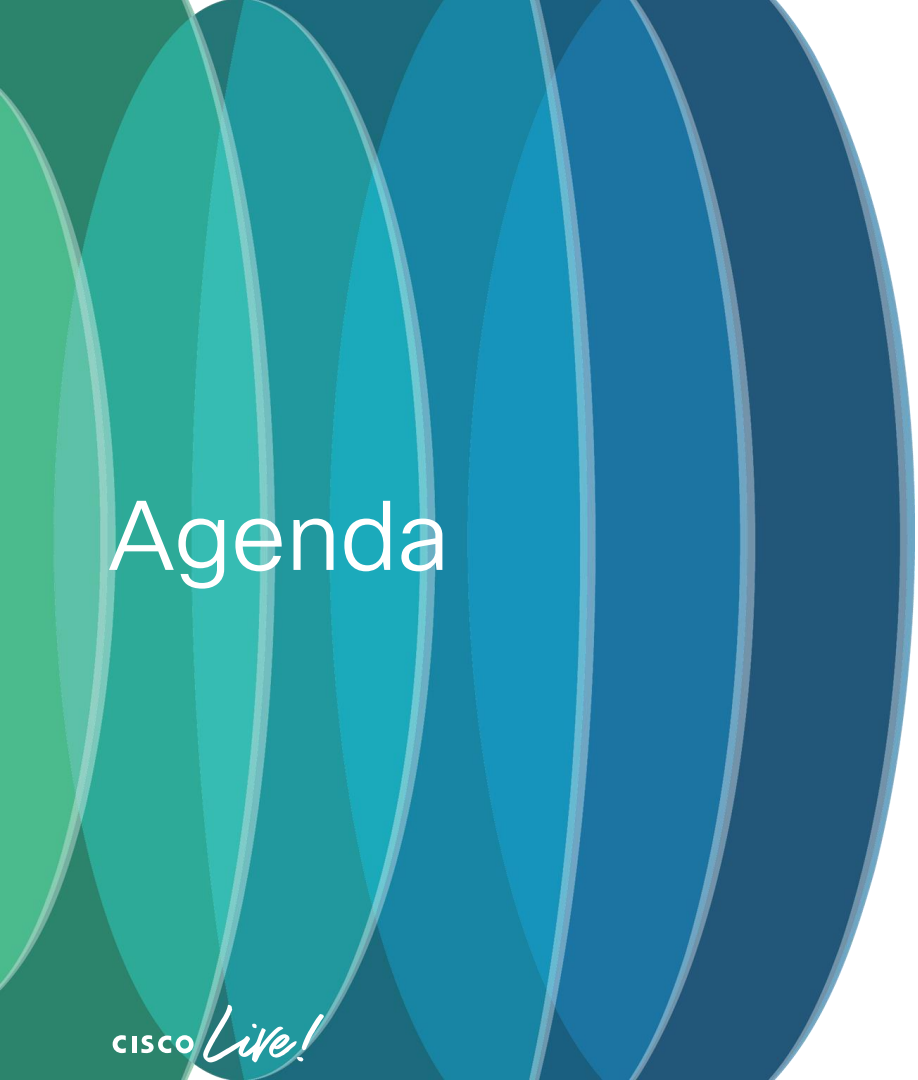
Use the Webex app to chat with the speaker after the session

How

- 1 Find this session in the Cisco Events mobile app
- 2 Click “Join the Discussion”
- 3 Install the Webex app or go directly to the Webex space
- 4 Enter messages/questions in the Webex space

Webex spaces will be moderated by the speaker until February 28, 2025.

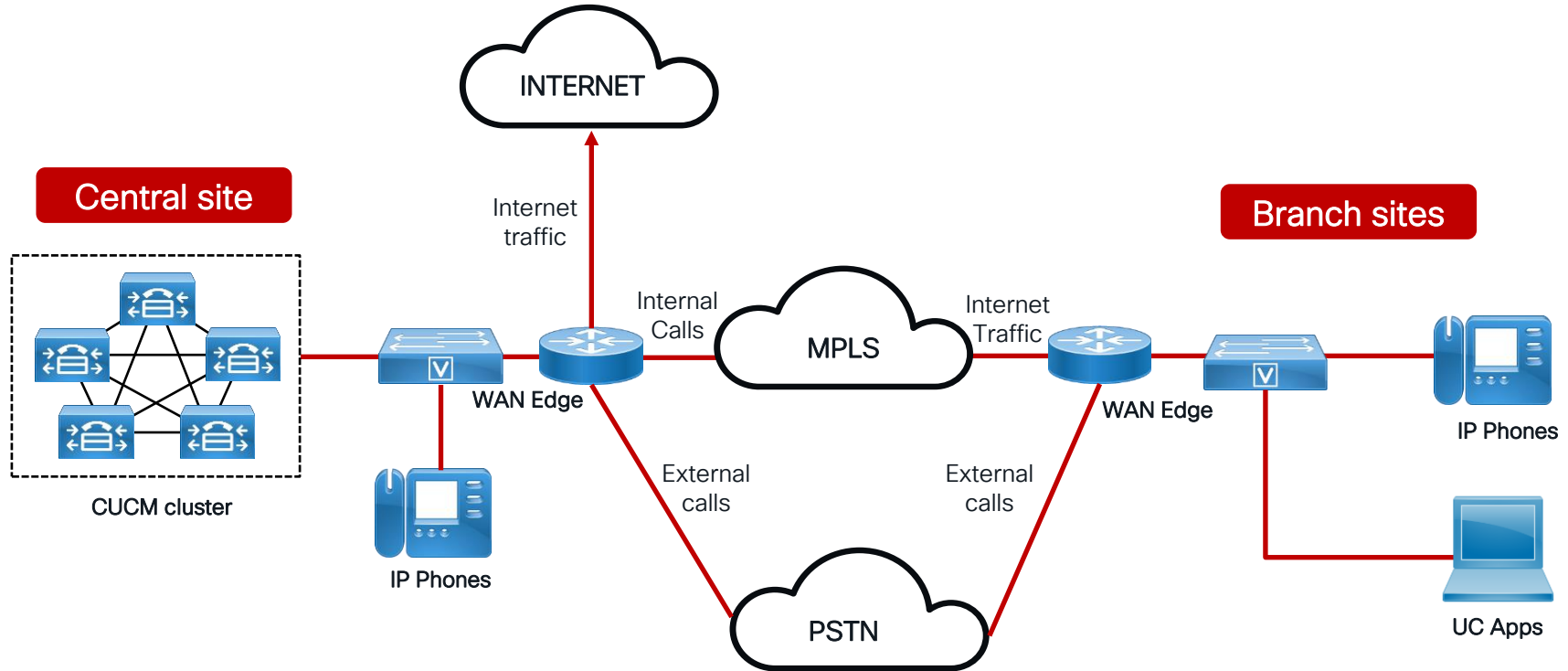




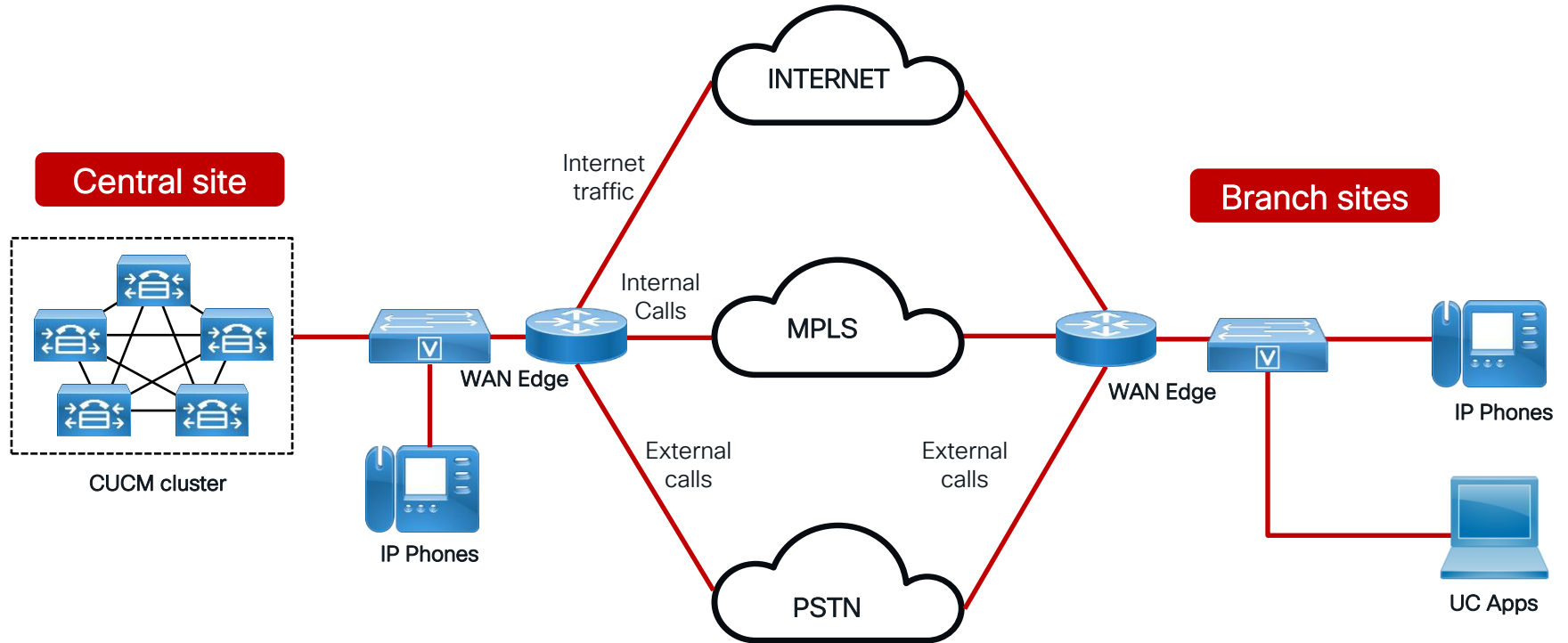
Agenda

- Architecture & Use Cases
- Design & Deployment
 - ❖ Direct Internet Access
 - ❖ SIP TDM Gateway
 - ❖ Webex Calling Local Gateway
- Path Insights & Optimization
 - ❖ ThousandEyes
 - ❖ Cloud OnRamp for Webex

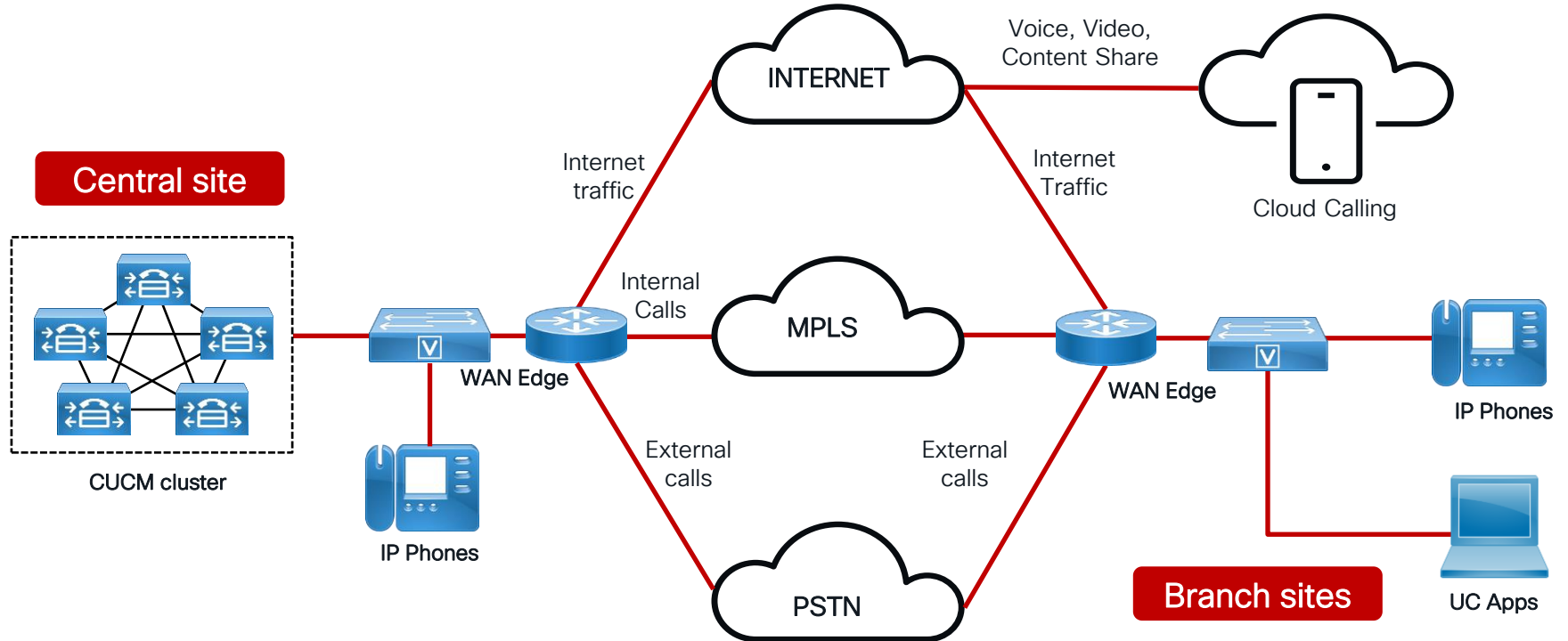
Traditional WAN Architecture



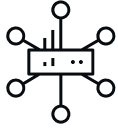
Network Connectivity Changes



UC Service Delivery Evolves



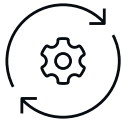
New Business Needs



Utilize high-speed Internet



Optimize path to UC Cloud Services

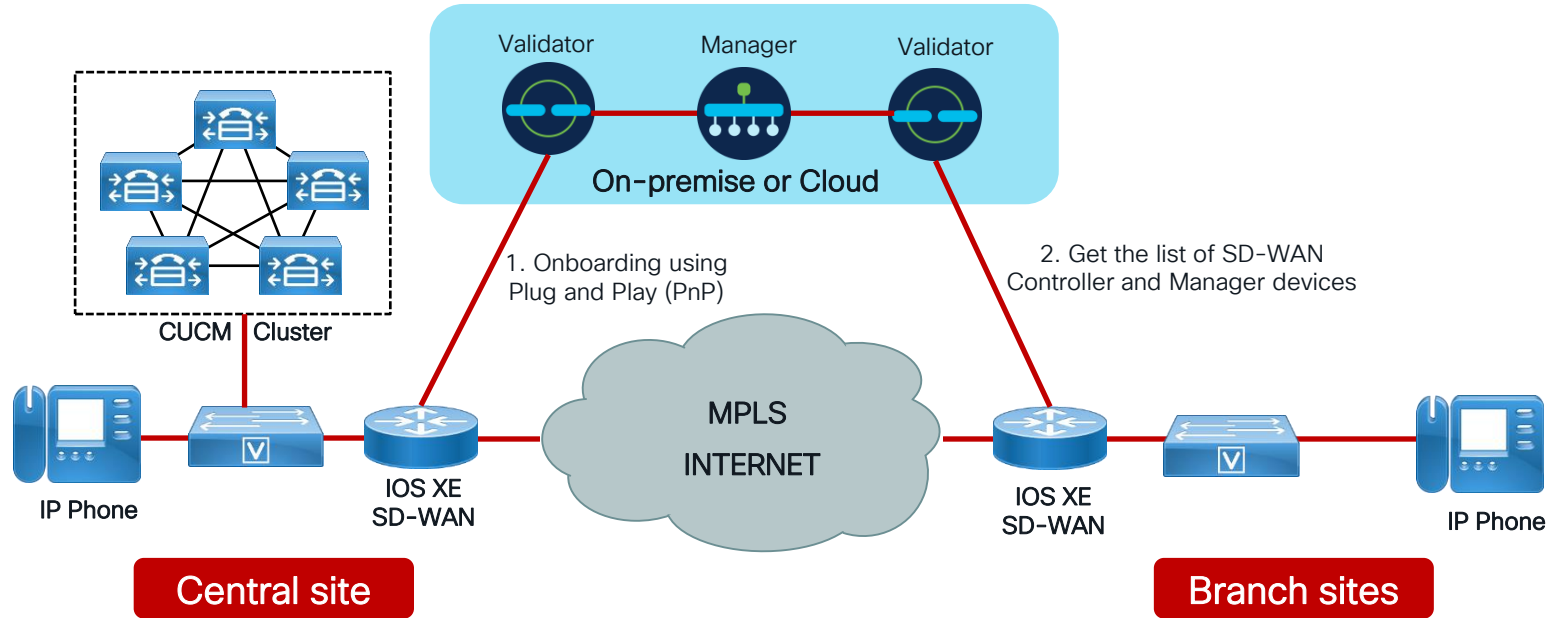


Automate provisioning processes

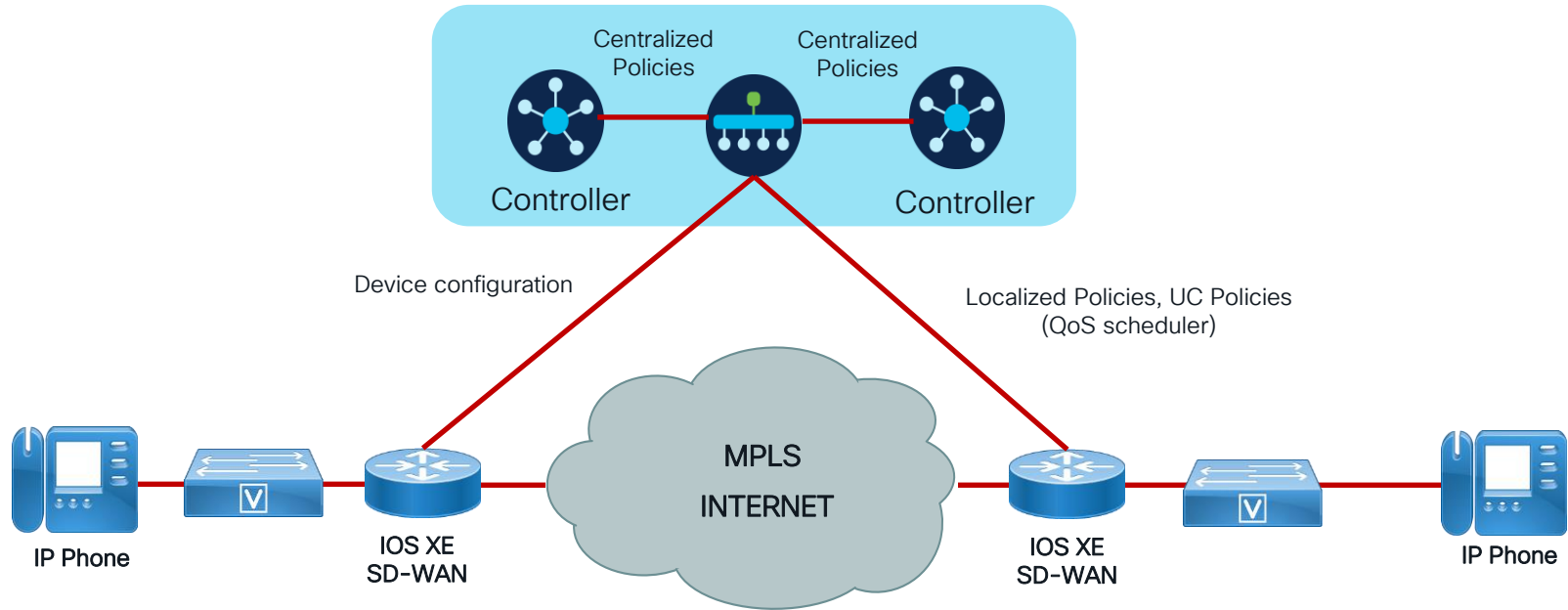


Manage consistent configurations

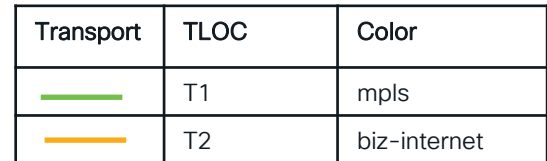
Catalyst SD-WAN Architecture



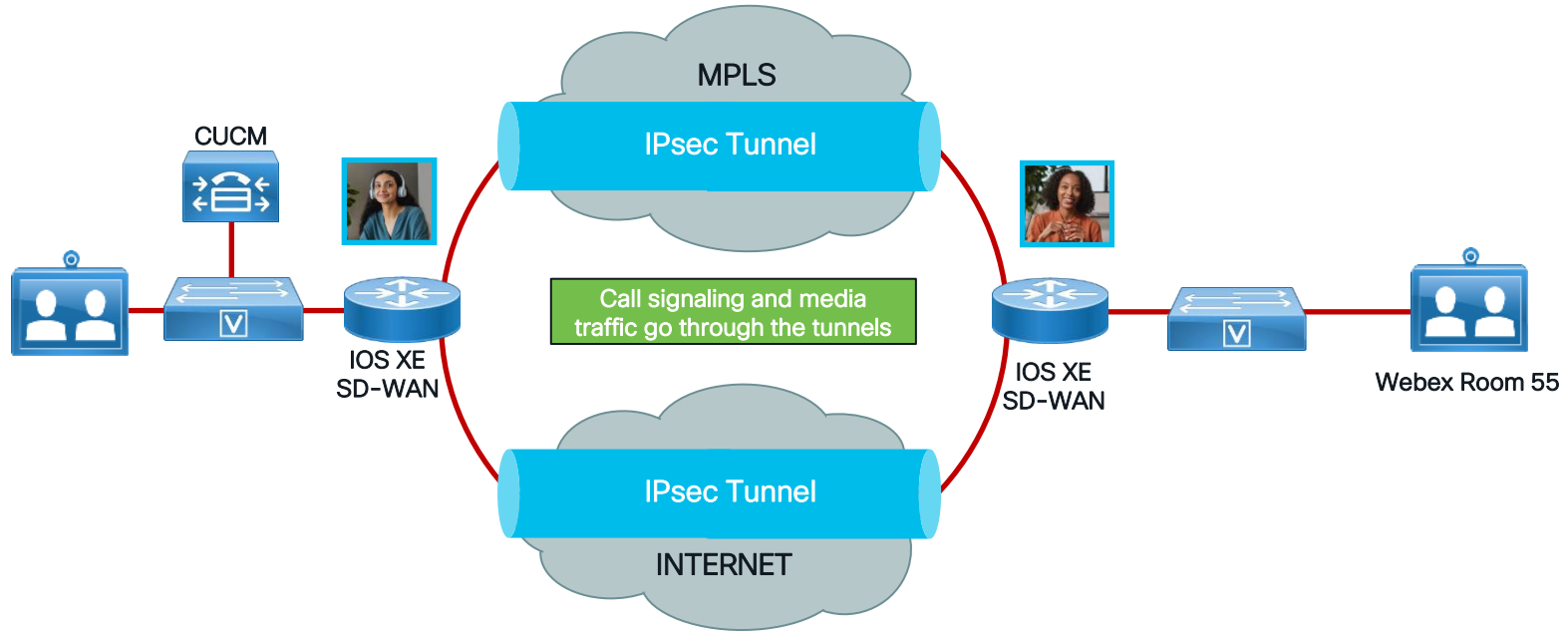
SD-WAN Orchestration and Management Plane



CISCO *Live!*



SD-WAN Data Plane

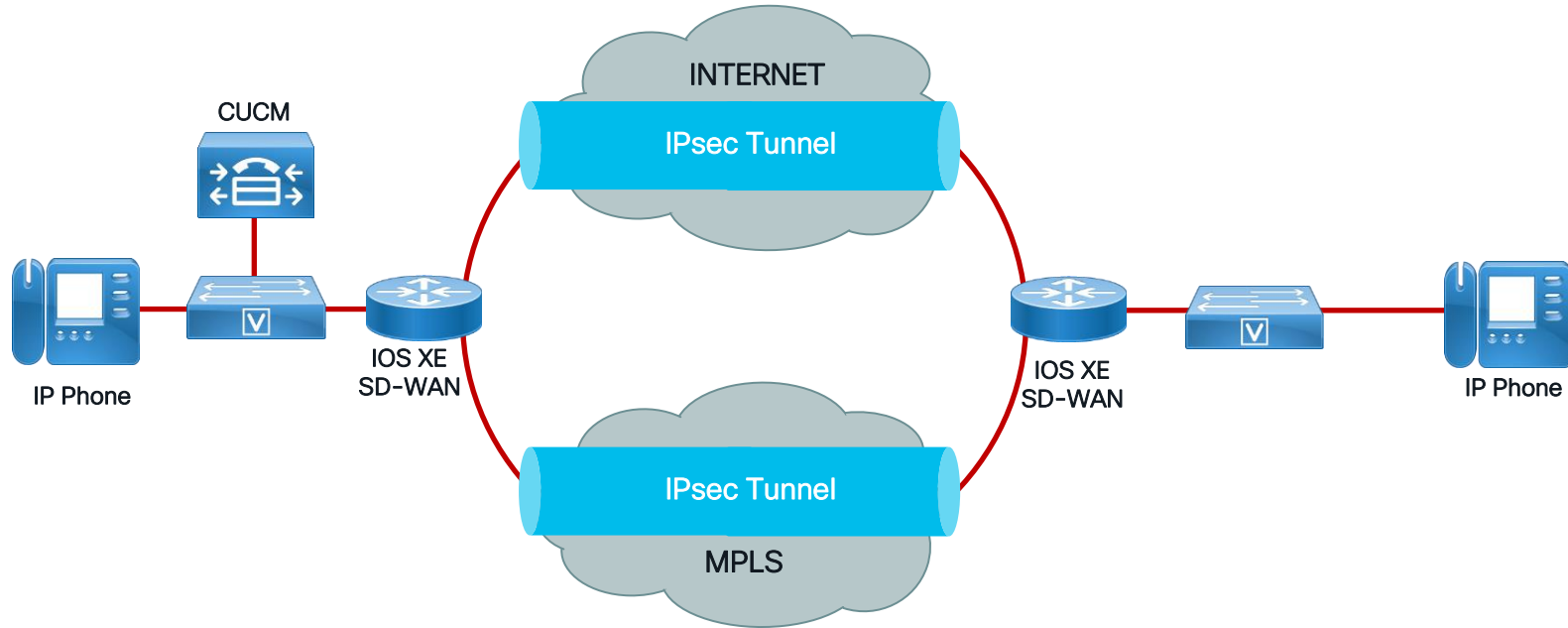


Catalyst 8000 Edge Platform Family

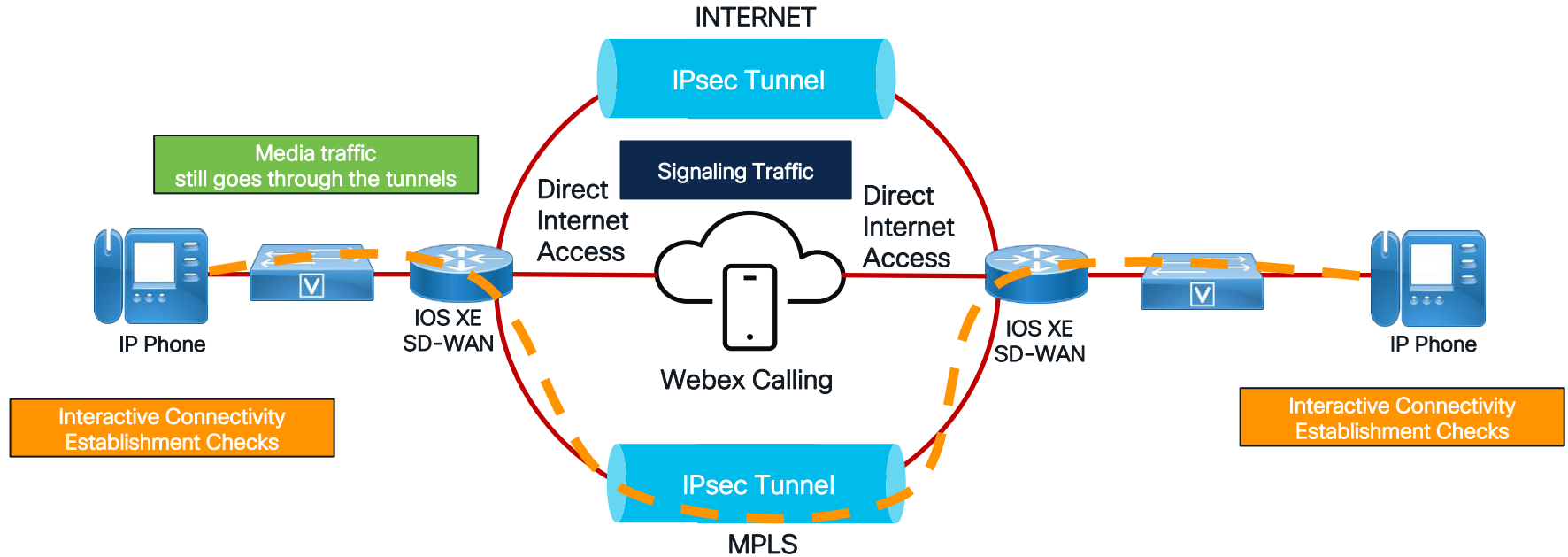


Use Cases

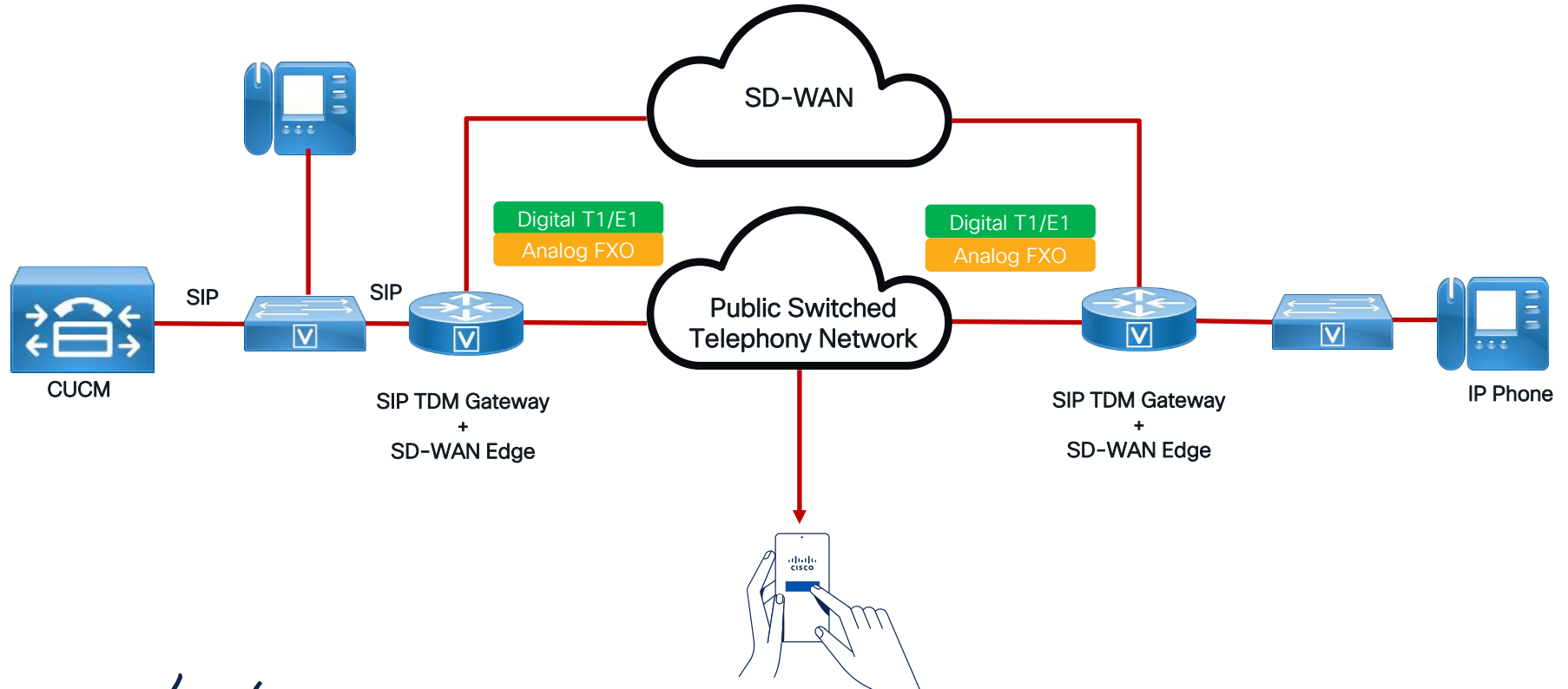
On-Premise UC: Site-to-Site Calls



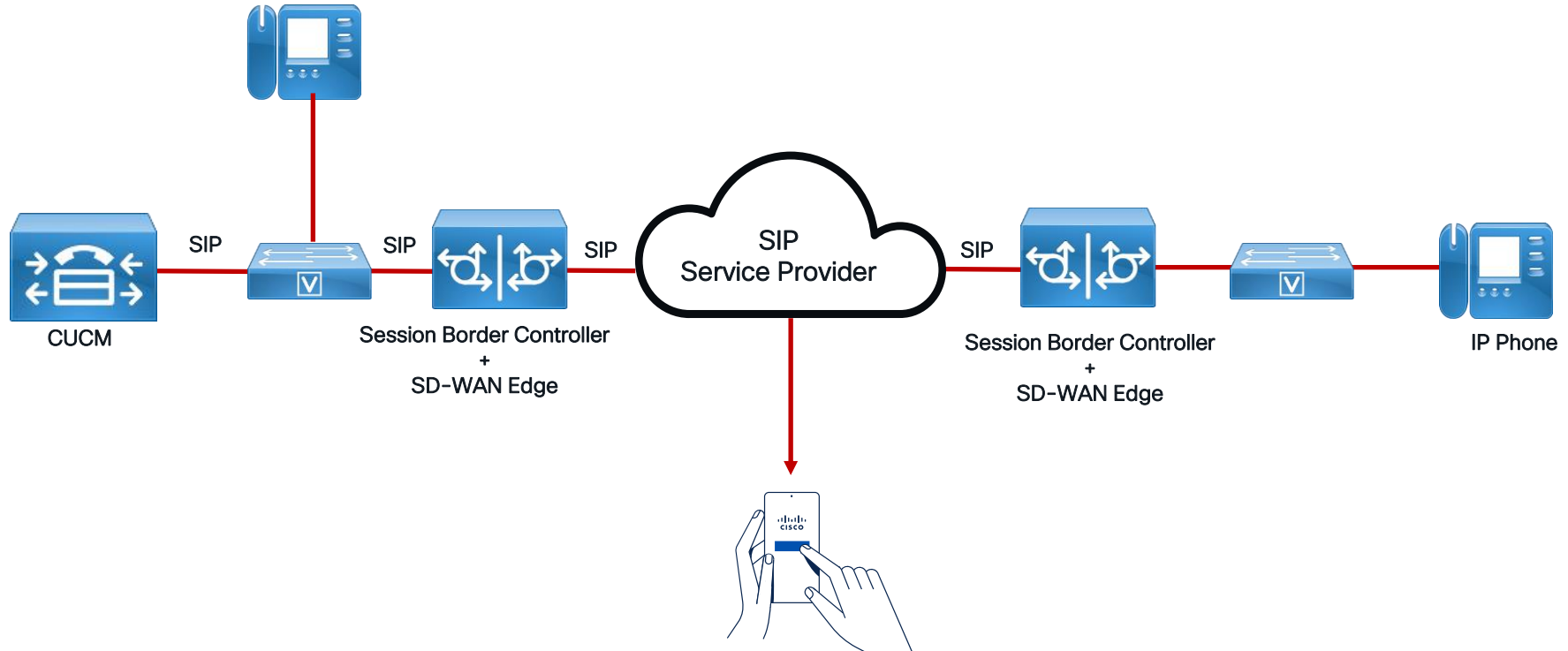
Webex Calling: Site-to-Site Calls



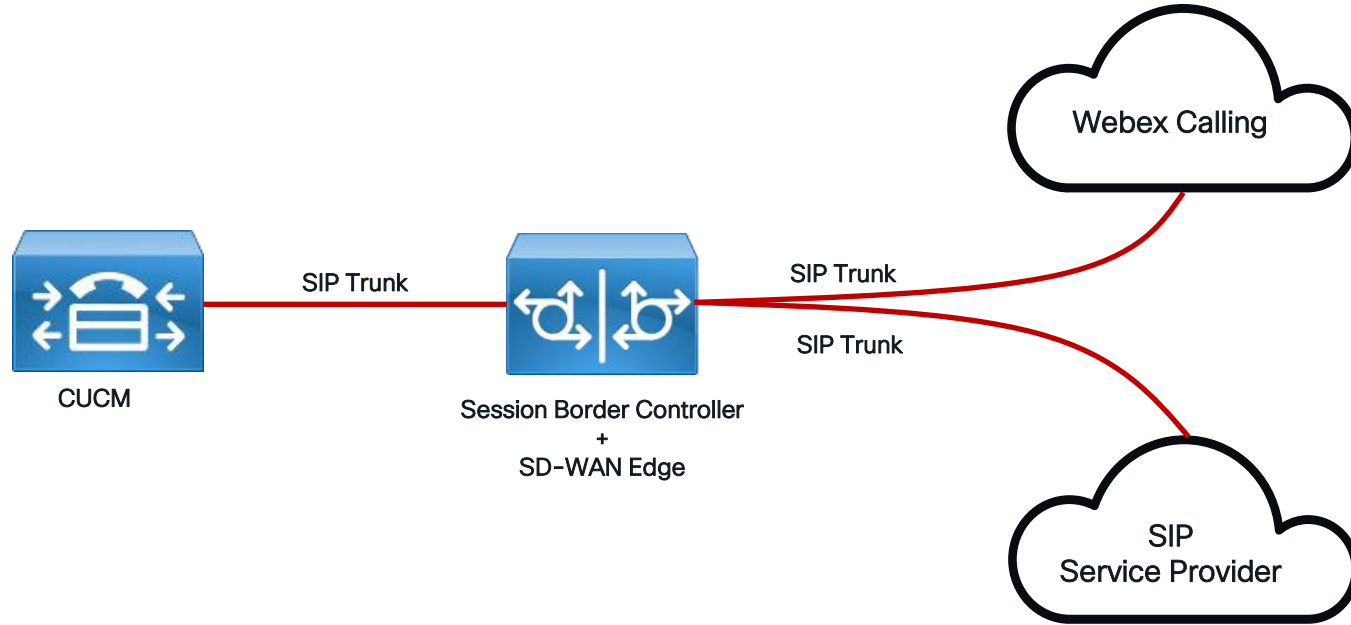
On-Premise UC: External Calls Via PSTN



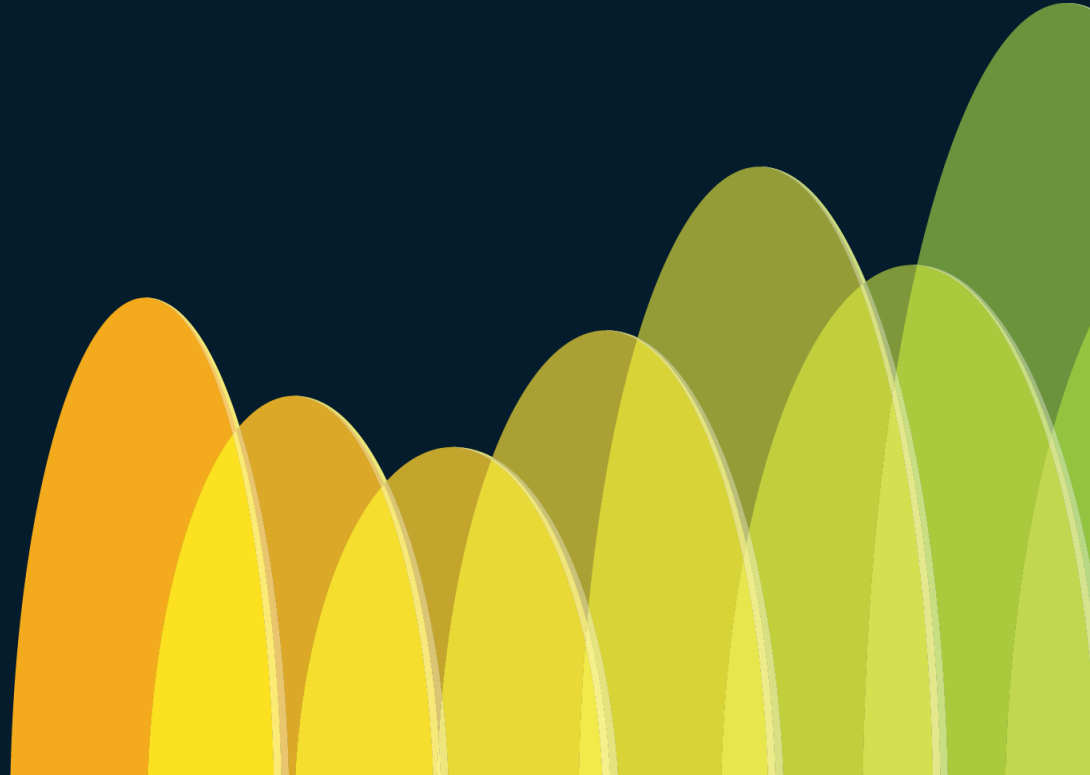
On-Premise UC: External Calls Via SIP Provider



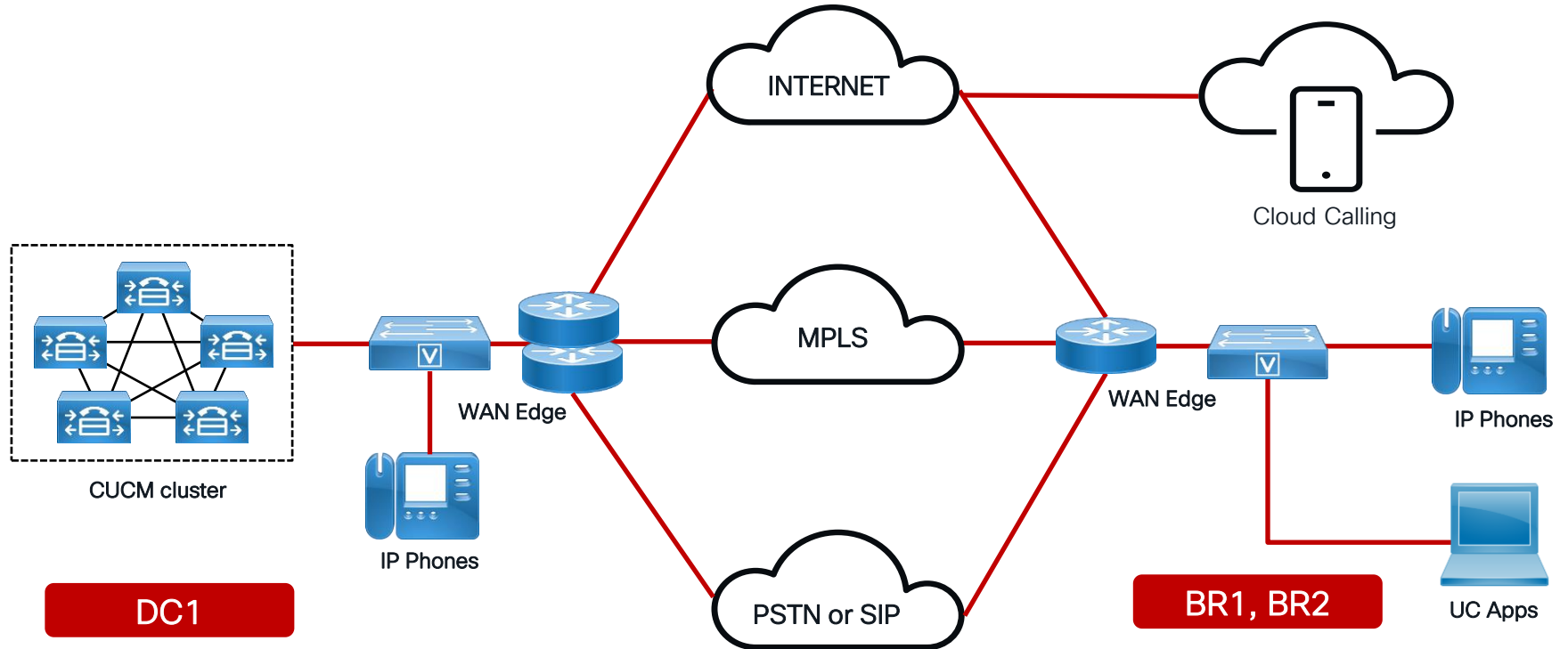
Webex Calling: Local Gateway



Network Design & Deployment



Sample Network Topology

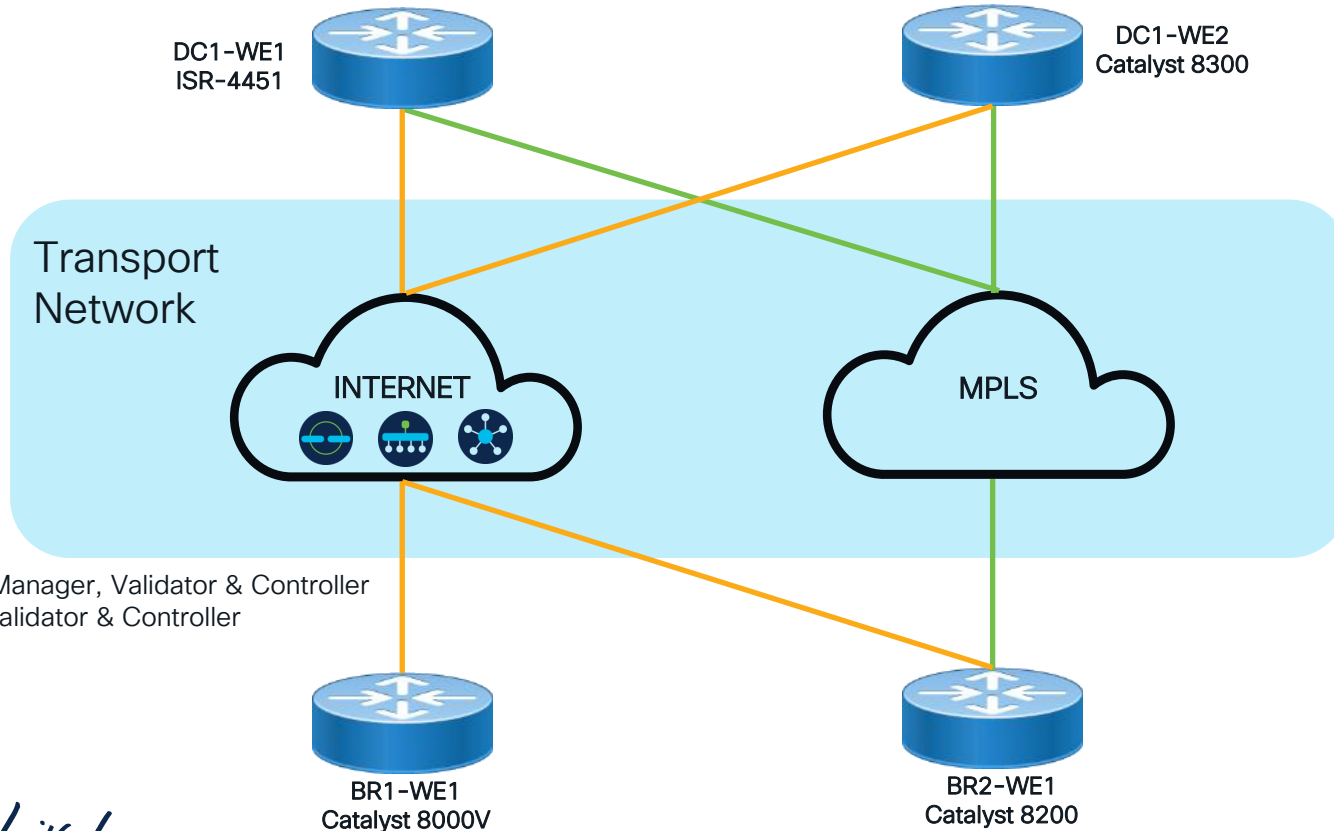


Collaboration Endpoint Directory Numbers



Site	Registered to	Extension
DC1	CUCM	2xxx
BR1	CUCM	3xxx
BR2	CUCM	4xxx
DC1	Webex Calling	5xxx
BR1	Webex Calling	6xxx
BR2	Webex Calling	7xxx

Sample Network Topology

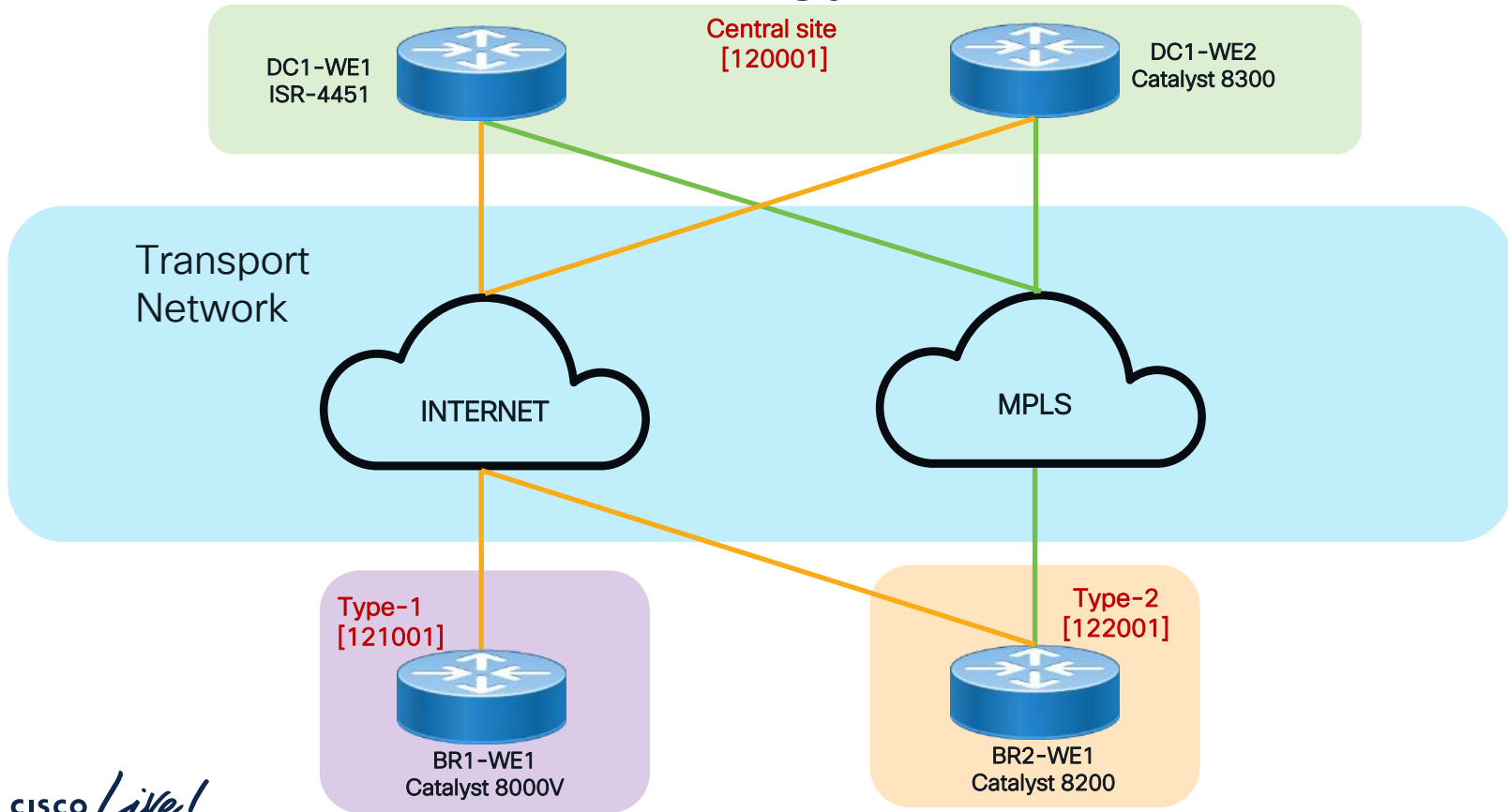


Site ID Design

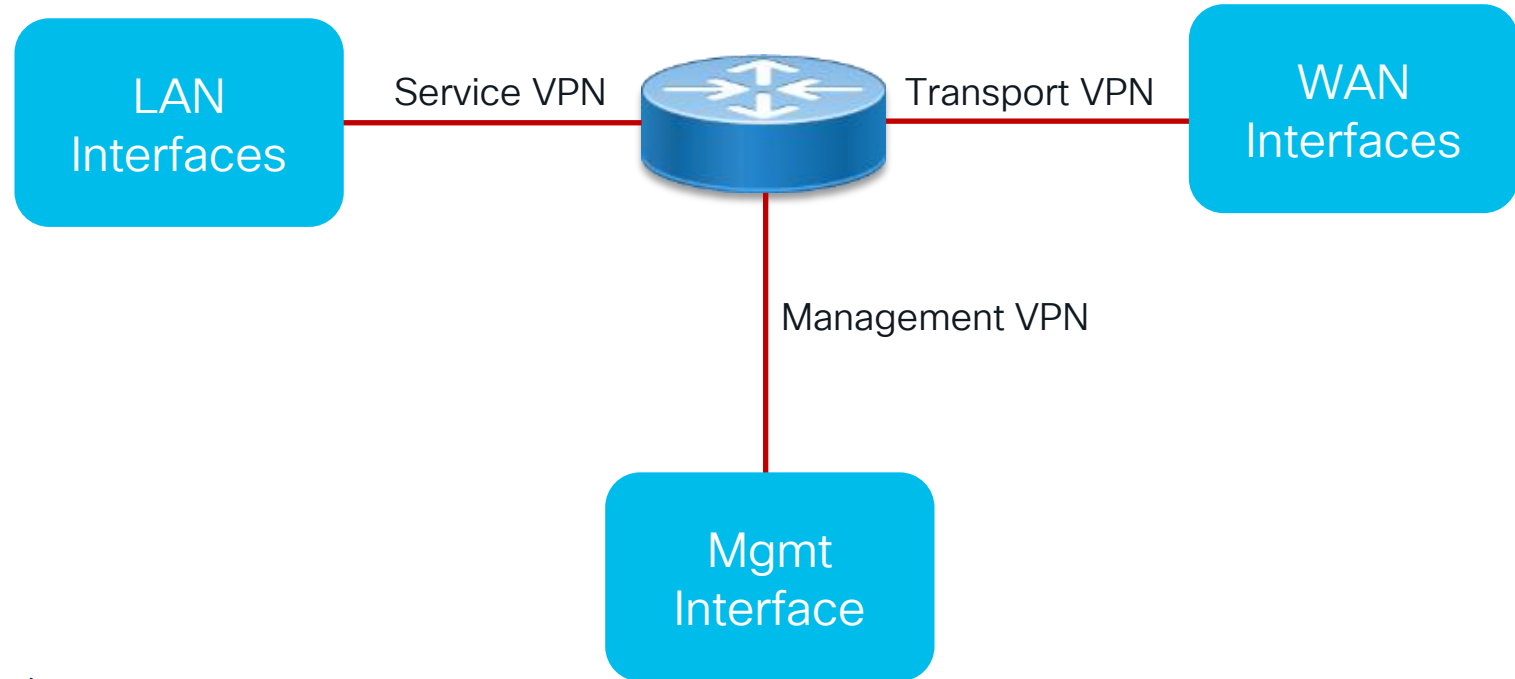
Digit 1	Country
Digit 2	Region within the country
Digit 3-6	Site types
	0000 - 0099 = Central sites
	1000 - 1999 = Type 1 sites
	2000 - 2999 = Type 2 sites
	3000 - 3999 = Type 3 sites
	4000 - 4999 = Type 4 sites
	5000 - 5999 = Type 5 sites
	*Types are defined based on your needs
	This session uses Central, Type-1 and Type-2



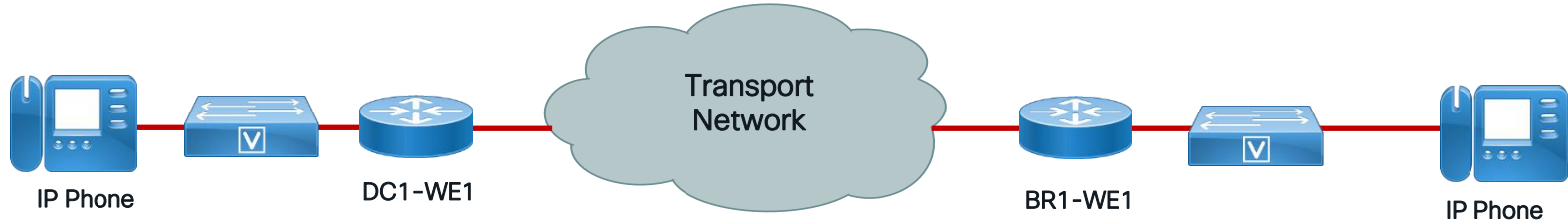
Sample Network Topology



Network Segmentation @ SD-WAN Edge

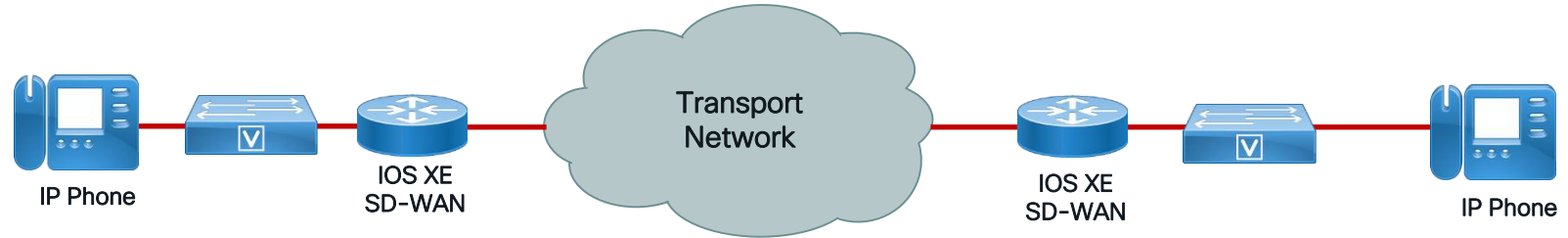


Transport VPN: Network Addressing



cEdge	Interface	IP Address	Description	Tunnel	Color
DC1-WE1	GigabitEthernet0/0/1 GigabitEthernet0/0/2	192.0.2.153 203.0.113.153	MPLS Interface Internet Interface	Tunnel1 Tunnel2	mpls biz-internet
DC1-WE2	GigabitEthernet0/0/1 GigabitEthernet0/0/2	192.0.2.154 203.0.113.154	MPLS Interface Internet Interface	Tunnel1 Tunnel2	mpls biz-internet
BR1-WE1	GigabitEthernet1	203.0.113.187	Internet Interface	Tunnel1	biz-internet
BR2-WE2	GigabitEthernet0/0/1 GigabitEthernet0/0/2	192.0.2.193 203.0.113.193	MPLS Interface Internet Interface	Tunnel1 Tunnel2	mpls biz-internet

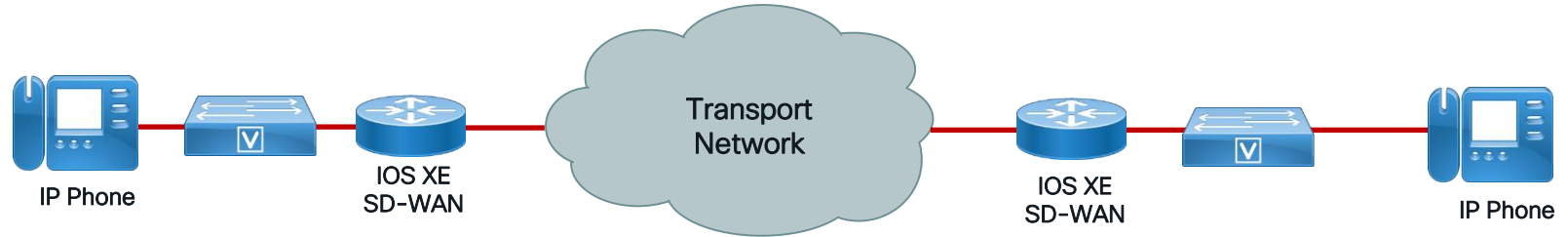
Service VPN: Network Addressing



VLAN	Subnet	Description
110	172.16.110.0/24	DC1_Servers
106	172.16.106.0/24	DC1_UC_WxC_Data
107	172.16.107.0/24	DC1_UC_WxC_Voice
108	172.16.108.0/24	DC1_UC_OnPrem_Data
109	172.16.109.0/24	DC1_UC_OnPrem_Voice

VLAN	Subnet	Description
115	172.16.115.0/24	BR1_UC_Edge
116	172.16.116.0/24	BR1_UC_WxC_Data
117	172.16.117.0/24	BR1_UC_WxC_Voice
118	172.16.118.0/24	BR1_UC_OnPrem_Data
119	172.16.119.0/24	BR1_UC_OnPrem_Voice

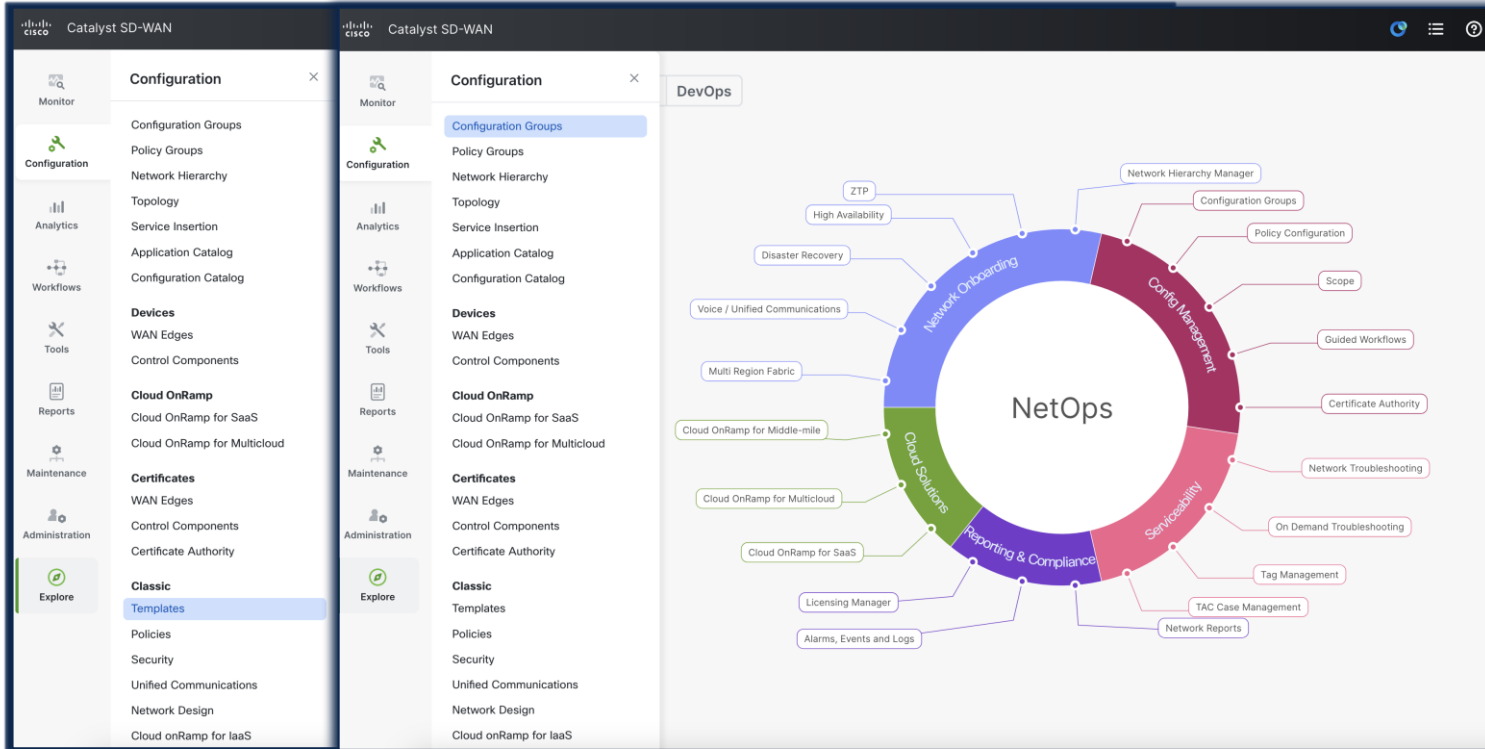
Management VPN: Network Addressing



cEdge	IP Address
DC1-WE1	192.168.1.153
DC1-WE2	192.168.1.154
BR1-WE1	192.168.2.187
BR2-WE2	192.168.3.193

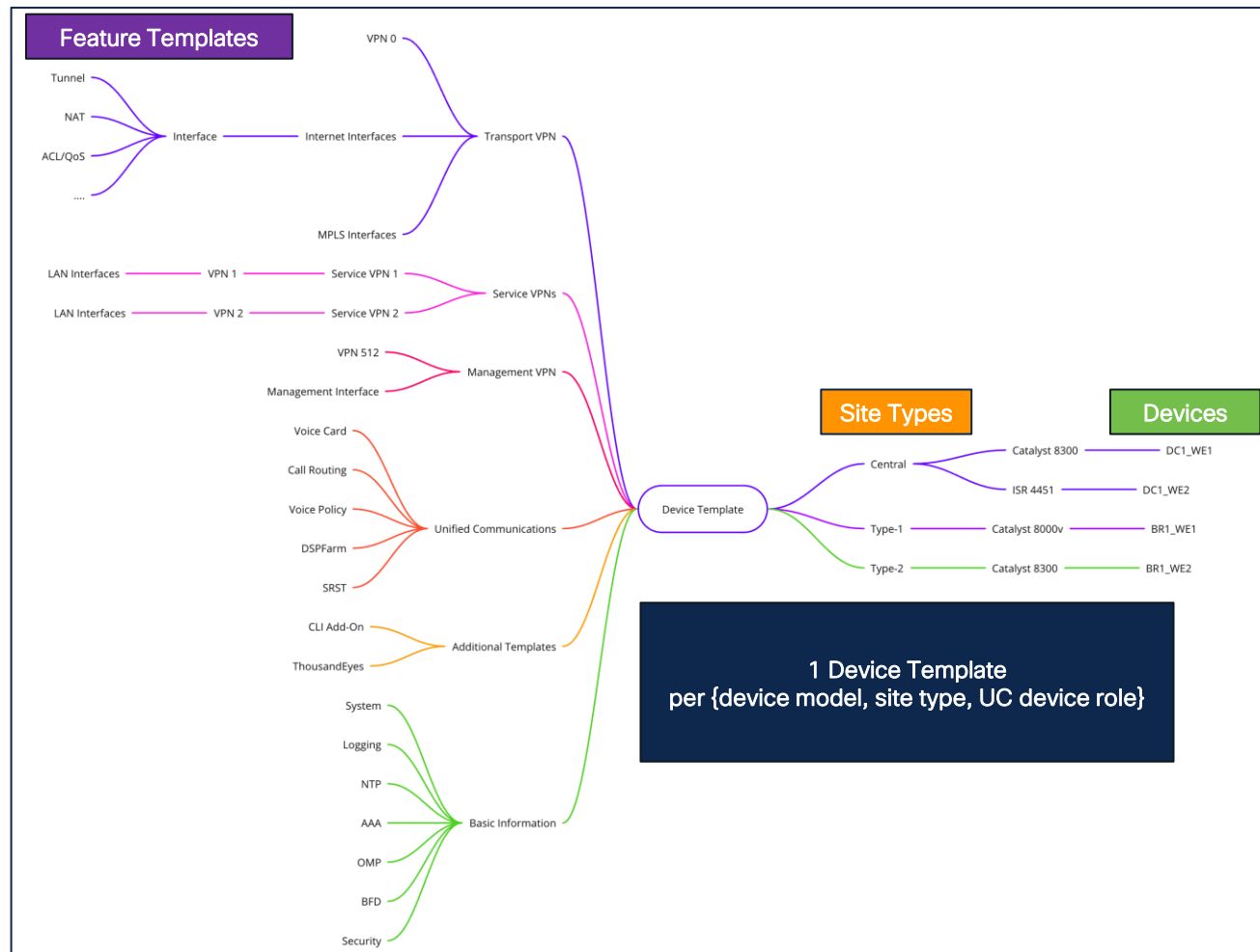
SD-WAN Manager

Central place For Device and Policy Configurations



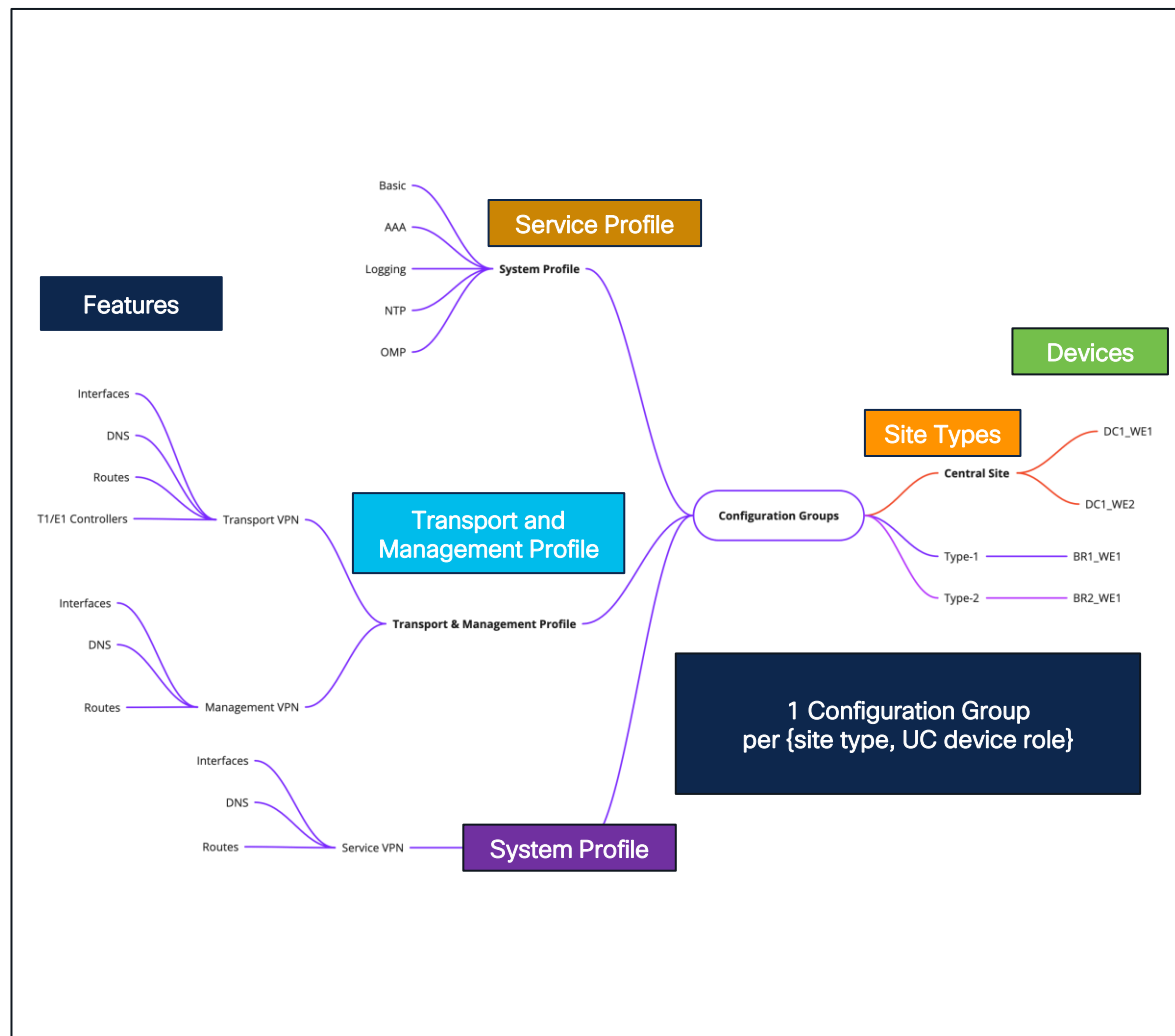
Device Templates

Define once,
Reuse everywhere!

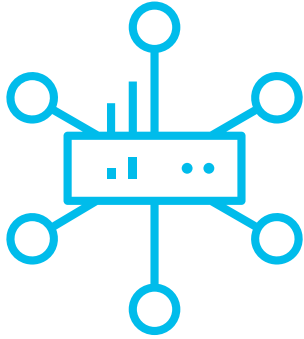


Configuration Groups

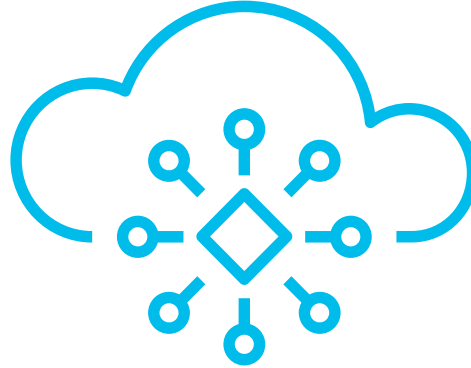
Define once *easily*,
Reuse everywhere!



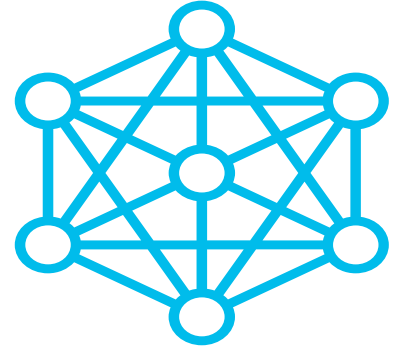
cEdge Deployment Steps



Sync Devices to
SD-WAN Manager

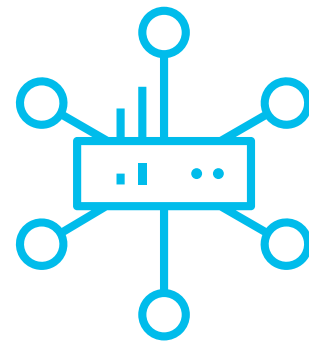


Control Plane
Establishment



Data Plane
Establishment

Sync Devices



Manager <-> PnP Portal

cEdge: Add Device to Plug-N-Play Portal

```
Router# controller-mode enable
```

! After reboot

```
Router # show pnp version
```

```
----- show pnp version -----
```

PnP Schema Version: 1.0, Baseline Tracking: rel35.1.20201014

Device UDI: PID:<PID>,VID:V01,SN:<SN>

UDI Checking: Yes

SID: AuthOnly:No, MustValidate:No, MustRenew:No

Security Enforced: No, PostReloadPriv'd Profile: N/A

Startup Config: LoadingDone: Yes, FoundAtBootup: Yes, NowPresent: Yes

SUDI Certificate: CISCO_IDEVID_SUDI

Device SUDI: PID:<PID>,SN:<SUDI SN>

```
Router#
```

```
Router# show sdwan certificate serial
```

Chassis number: C8300-2N2S-6T-<SN> Board ID serial number: <certificate serial number> Subject S/N: <SN>

Add device in PnP portal using

- a. SN
- b. PID
- c. SUDI SN, certificate serial number (this is for ISR 4K only)

cEdge: Add Device to SD-WAN Manager

Quick Connect Workflow

The screenshot shows the Cisco Catalyst SD-WAN Manager interface. The top navigation bar includes the Cisco logo, the text "Catalyst SD-WAN", and user information "admin". A left sidebar contains navigation icons for Monitor, Configuration, Analytics, Workflows, Tools, Reports, Maintenance, Administration, and Explore. The main content area is titled "Quick Connect" and shows a five-step workflow. Step 1, "Sync your device inventory", is highlighted. It includes instructions to "Import device serial numbers from PnP or upload file with serial numbers" and three radio button options: "Log into your Smart Account", "Upload file with serial numbers" (which is selected and highlighted with a blue box), and "Skip for now". Below these options, it says "Upload a signed file (.vptela file) from Cisco Plug and Play Or an un-signed file (.csv file), prepared from the Sample CSV". A dashed box contains an upload icon and the text "Click or drag to this area to upload" and "Accepted files: .vptela, text/csv". At the bottom right, there are "Review" and "Next" buttons.

Cisco Catalyst SD-WAN Manager

admin

Monitor

Configuration

Analytics

Workflows

Tools

Reports

Maintenance

Administration

Explore

← Workflows Home

Quick Connect

- 1 Sync your device inventory
- 2 Select devices to bring up
- 3 Add and Review Device Configur...
- 4 (Optional) Tag Devices
- 5 Summary

Sync your device inventory

Import device serial numbers from PnP or upload file with serial numbers

Import device serial numbers

- ☐ Log into your Smart Account ⓘ
- ☒ Upload file with serial numbers ⓘ
- ☐ Skip for now ⓘ

Upload a signed file (.vptela file) from Cisco Plug and Play
Or an un-signed file (.csv file), prepared from the
[Sample CSV](#)

Click or drag to this area to upload

Accepted files: .vptela, text/csv

Cancel Review Next

cEdge: Add Device to SD-WAN Manager

The screenshot displays the Cisco Catalyst SD-WAN Manager interface. The top navigation bar includes the Cisco logo, 'Catalyst SD-WAN', and user information 'admin'. A left sidebar contains navigation icons for Monitor, Configuration, Analytics, Workflows, Tools, Reports, Maintenance, Administration, and Explore. The main content area is titled 'Quick Connect' and shows a five-step workflow: 1. Sync your device inventory, 2. Select devices to bring up (active), 3. Add and Review Device Configur..., 4. (Optional) Tag Devices, and 5. Summary. Under step 2, the section 'Select devices to bring up' explains that selected devices are added to the inventory and authorized to connect. A link 'View the Latest Sync Device Inventory Operation Task Details' is provided. Below this, a search bar shows 'SN' entered. A table titled 'Edge Devices (1/34)' displays one device with columns for Chassis Number, Device Model, and Serial No./Token. The device's Chassis Number is 'PID-SN', Device Model is 'C8200-1N-4T', and Serial No./Token is 'Token'. The table indicates 0 selected items and a timestamp of 'As of: Feb 02, 2025 02:57 PM'. At the bottom, there are 'Review', 'Back', and 'Next' buttons.

Cisco Catalyst SD-WAN

admin

Monitor

Configuration

Analytics

Workflows

Tools

Reports

Maintenance

Administration

Explore

← Workflows Home

Quick Connect

- ✓ Sync your device inventory
- 2 Select devices to bring up**
- 3 Add and Review Device Configur...
- 4 (Optional) Tag Devices
- 5 Summary

Select devices to bring up
Selected devices are added to the inventory and authorized to connect to the controllers.

[View the Latest Sync Device Inventory Operation Task Details](#)

Edge Devices (1/34)

0 selected As of: Feb 02, 2025 02:57 PM

☐	Chassis Number	Device Model	Serial No./Token
☐	PID-SN	C8200-1N-4T	Token

Items per page: 25 1 ~ 1 of 1

Review Back Next

cEdge: Add Device to SD-WAN Manager

The screenshot displays the Cisco Catalyst SD-WAN Manager web interface. The top navigation bar includes the Cisco logo, the text 'Catalyst SD-WAN', and user information 'admin'. A left-hand sidebar contains navigation links: Monitor, Configuration, Analytics, Workflows, Tools, Reports, Maintenance, Administration, and Explore. The main content area is titled 'Quick Connect' and shows a five-step workflow. Step 3, 'Add and Review Device Configuration', is the active step. Below the workflow steps, the 'Add and Review Device Configuration' section is shown, with a note that 'All settings are auto-generated. If you wish to edit, directly double-click the row.' A table titled 'Edge Devices (1)' contains one row with the following data: Chassis Number 'C8200-1N-4T-FJC2', Hostname 'BR2_WE2', System-IP '192.168.3.254', Site 'SITE_122001' (selected from a dropdown), and Dual Stack IPv6 Default 'False' (selected from a dropdown). The table has search and import/export buttons. At the bottom of the table are 'Cancel' and 'Apply' buttons. Pagination information shows 'Items per page: 25' and '1 - 1 of 1'. At the bottom right of the interface are 'Review', 'Back', and 'Next' buttons.

Cisco Catalyst SD-WAN Manager

← Workflows Home

Quick Connect

- ✓ Sync your device inventory
- ✓ Select devices to bring up
- 3 Add and Review Device Configur...**
- 4 (Optional) Tag Devices
- 5 Summary

Add and Review Device Configuration

All settings are auto-generated. If you wish to edit, directly double-click the row.

Edge Devices (1) [Import](#) [Export](#)

Search Table

Chassis Number	Hostname	System-IP	Site	Dual Stack IPv6 Default
C8200-1N-4T-FJC2	BR2_WE2	192.168.3.254	Select SITE_122001	False

[Cancel](#) [Apply](#)

Items per page: 25 1 - 1 of 1 |< < > >|

[Review](#) [Back](#) [Next](#)

cEdge: Add Device to SD-WAN Manager

The screenshot displays the Cisco Catalyst SD-WAN Manager web interface. The top navigation bar includes the Cisco logo, the text 'Catalyst SD-WAN', and user information 'admin'. A left sidebar contains navigation links: Monitor, Configuration, Analytics, Workflows, Tools, Reports, Maintenance, Administration, and Explore. The main content area is titled 'Quick Connect' and shows a progress list of steps: 'Sync your device inventory', 'Select devices to bring up', 'Add and Review Device Configur...', '(Optional) Tag Devices', 'Summary', and 'Congratulations!'. The 'Congratulations!' step is currently active. A large light blue callout box contains the text: 'Congratulations! For Dynamic IP devices, ZTP/PnP process happens automatically. For Static IP devices, Need to generate the bootstrap configuration for the devices in Manager. Then download this bootstrap configuration to the device manually and reboot.' To the right of the text is an illustration of two people celebrating a success.

Cisco Catalyst SD-WAN

admin

← Workflows Home

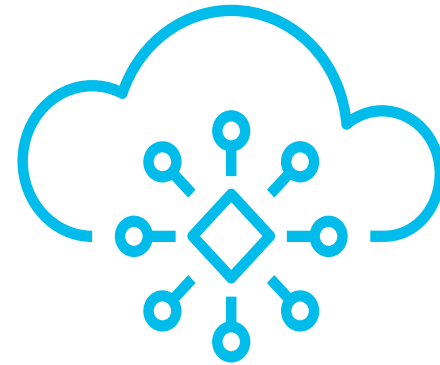
Quick Connect

- ✓ Sync your device inventory
- ✓ Select devices to bring up
- ✓ Add and Review Device Configur...
- ✓ (Optional) Tag Devices
- ✓ Summary
- ✓ Congratulations!

Congratulations!

For Dynamic IP devices, ZTP/PnP process happens automatically.
For Static IP devices, Need to generate the bootstrap configuration for the devices in Manager. Then download this bootstrap configuration to the device manually and reboot.

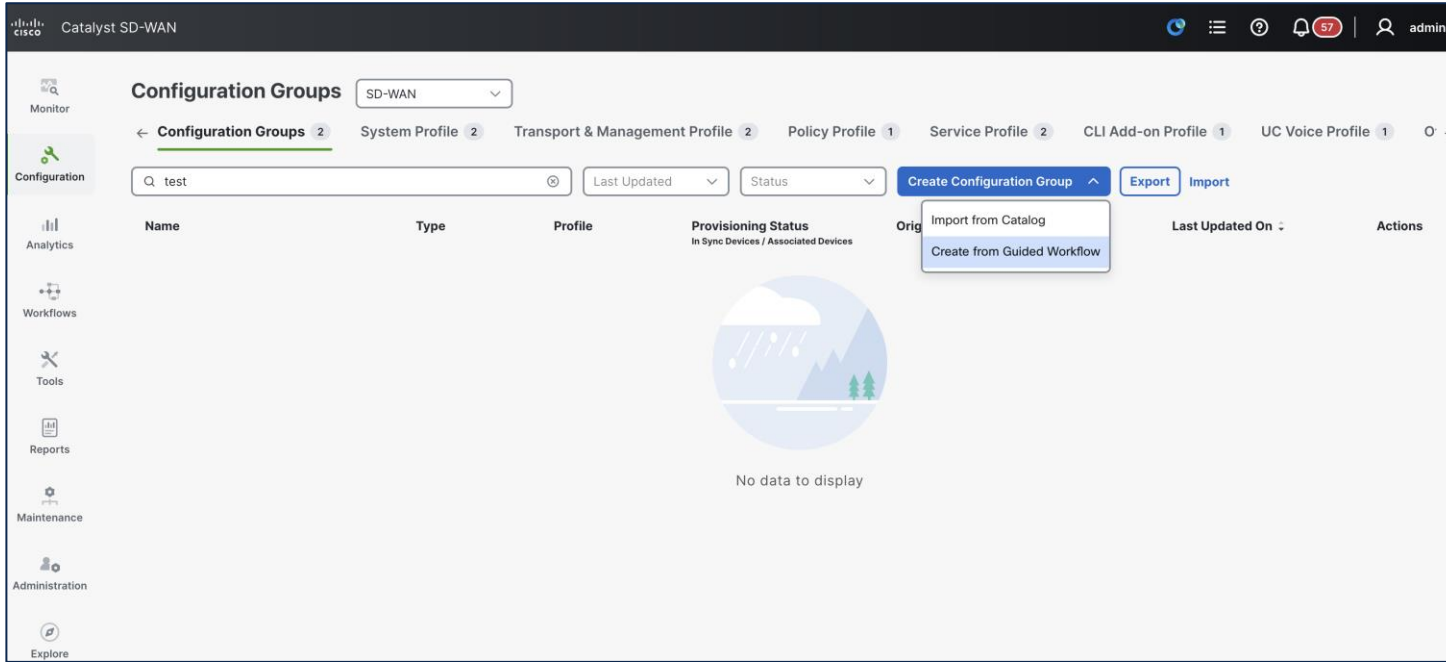
Control Plane Establishment



cEdge <-> Controller

Create Device Configurations in a Snap!

Use Guided Workflow in SD-WAN Manager



The screenshot displays the Cisco Catalyst SD-WAN Manager interface. The top navigation bar includes the Cisco logo, the text "Catalyst SD-WAN", and user information "admin". The left sidebar contains navigation links: Monitor, Configuration (active), Analytics, Workflows, Tools, Reports, Maintenance, Administration, and Explore. The main content area is titled "Configuration Groups" and features a dropdown menu set to "SD-WAN". Below this, a breadcrumb trail shows: Configuration Groups (2) > System Profile (2) > Transport & Management Profile (2) > Policy Profile (1) > Service Profile (2) > CLI Add-on Profile (1) > UC Voice Profile (1). A search bar with "test" and filters for "Last Updated" and "Status" are present. A "Create Configuration Group" button is highlighted, with a dropdown menu open showing two options: "Import from Catalog" and "Create from Guided Workflow". Other buttons for "Export" and "Import" are also visible. The table below has columns for Name, Type, Profile, Provisioning Status, Orig, Last Updated On, and Actions. The table is currently empty, displaying a "No data to display" message with a background illustration of a mountain and rain.

Name	Type	Profile	Provisioning Status	Orig	Last Updated On	Actions
No data to display						

Define a Configuration Group

Site Type

Create Configuration Group

Monitor
Configuration
Analytics
Workflows
Tools
Reports
Maintenance
Administration
Explore

Site Configurations [Expand All](#) [Collapse All](#)

Configuration Groups begin with smart default settings pre-populated for the user. You can modify or add to these settings as you wish.

Site Type

Configuration Group Type

☐ Single Router ☒ Dual Router

Site Settings

WAN Interfaces

☐ Transport Extension ☒ Full Mesh

DC1_WE1 DC1_WE2

☐ DHCP ☒ Static

Transport Name Interface Color

MPLS mpls

☐ Show Advanced

☒ DHCP ☐ Static

Transport Name Interface Color

biz-internet biz-internet

[Back](#) [Next](#)

Define a Configuration Group

Transport VPN = 0, WAN Interface & Routing Configurations

Create Configuration Group

Site Configurations

Configuration Groups begin with smart default settings pre-populated for the user. You can modify or add to these settings as you wish.

Site Type

Configuration Group Type

☐ Single Router ☒ Dual Router

Site Settings

WAN Interfaces

☐ Transport Extension ☒ Full Mesh

DC1_WE1 DC1_WE2

☐ DHCP ☒ Static

Transport Name Interface Color

MPLS mpls

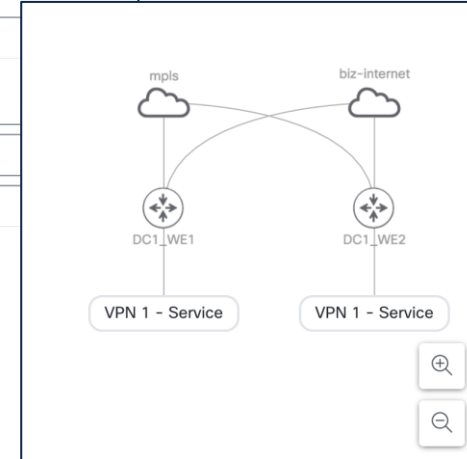
☐ Show Advanced

☒ DHCP ☐ Static

Transport Name Interface Color

biz-internet biz-internet

Cancel Back Next



Define a Configuration Group

Service VPN = 1, LAN Interface and Routing Configurations

Create Configuration Group

Monitor
Configuration
Analytics
Workflows
Tools
Reports
Maintenance
Administration
Explore

1 Create Configuration Group
2 Site Configurations
3 Additional Features
4 Summary

LAN & Service VPN Profile

Redundancy Protocol
☒ VRRP

Add Multiple VPNs at Once

Segment Name: LAN VPN: 1
Value between 1 - 65527

Number of Interfaces: 1

Add Routing
☒ Show Advanced

INTERFACE

Interface Name: {{ vpn_1_if_ }}
Description: <SYSTEM DEFAULT>
IP Address: {{ vpn_1_if__ipaddr }}
Subnet: {{ vpn_1_if__subnet }}

Cancel Back Next

Define a Configuration Group

Review & Create

Create Configuration Group

Summary [Expand All](#) [Collapse All](#) [Export to PDF](#)

Configuration Group Basic Details

Configuration Group [Edit](#)

Configuration Group Name: DC1_WE_Config_Group_1
Description: WAN Edge devices in Data Center

Site Configuration

Site Type [Edit](#)

Configuration Group Type: Dual Router

Site Settings [Edit](#)

BANNER

MOTD Message: System Default
Login Message: System Default

LOCAL DEVICE ACCESS

Default Password:

[Cancel](#) [Back](#) [Create Configuration Group](#)

Configuration Group is Created!

With System, Transport & Management, Service Profiles

The screenshot displays the Cisco Catalyst SD-WAN Configuration Groups interface. The top navigation bar includes the Cisco logo, the text "Catalyst SD-WAN", and user information "admin". The left sidebar contains navigation icons for Monitor, Configuration, Analytics, Workflows, Tools, Reports, and Maintenance. The main content area is titled "Configuration Groups" and shows a dropdown menu set to "SD-WAN". Below this, a breadcrumb trail indicates the current path: Configuration Groups (2) > System Profile (3) > Transport & Management Profile (3) > Policy Profile (1) > Service Profile (3) > CLI Add-on Profile (1) > UC Voice Profile (1). A search bar and filters for "Last Updated" and "Status" are present. A "Create Configuration Group" button is visible. The table below lists the configuration groups, with the first entry highlighted:

Name	Type	Profile	Provisioning Status	Origin	Updated By	Last Updated On	Actions
DC1_WE_Config_Group_1	WAN Edge devices in Data Center		In Sync Devices / Associated Devices		Updated by admin	Updated Jan 23, 2025	

The details for the selected configuration group "DC1_WE_Config_Group_1" are shown below the table. It is a "Dual Device" type. The profiles are:

- System Profile:** DC1_WE_Config_Group_1_Basic
- Transport & Management Profile:** DC1_WE_Config_Group_1_WAN
- Service Profile (optional):** DC1_WE_Config_Group_1_LAN

A "+ Add Profile" link is available at the bottom left. On the right, the "Deployment" section shows tags "DC1_WE1, DC1_WE2", an "Associated" link with a "+ Add" button, and a "Provisioning" status of "N/A". A "Deploy" button is located at the bottom right of the deployment section.

Transport & Management Profile is Created!

VPN = 0

Cisco Catalyst SD-WAN

← Configuration Groups (DC1_WE1_Config_Group_1a - Transport & Management Profile)

DC1_WE1_Config_Group_1a_WAN [Edit](#)

Description: DC1_WE1_Config_Group_1a WAN Transport Profile

Type	Device Name	DEVICE SOLUTION	MODIFIED BY	LAST UPDATED	SHARED
Dual Device	DC1_WE1, DC1_WE2	SD-WAN	Admin	Jan 25, 2025 03:31:55	1 Group

[All](#) [DC1_WE1](#) [DC1_WE2](#)

Transport VPN

- VPN0 [+](#)
- Ethernet Interface
 - Internet [+](#)
 - Ethernet Interface
 - MPLS [+](#)

[Back](#) No Changes Made

Add Feature

- Management VPN
- Transport VPN
- Cellular Controller
- Managed Cellular Activation - eSIM
- T1E1 Controller

Service Profile is Created!

VPN = 1

The screenshot displays the Cisco Catalyst SD-WAN configuration interface. The top navigation bar includes the Cisco logo, the text "Catalyst SD-WAN", and user information for "admin". A left-hand sidebar contains navigation links: Monitor, Configuration, Analytics, Workflows, Tools, Reports, Maintenance, Administration, and Explore. The main content area is titled "Configuration Groups (DC1_WE_Config_Group_1 - Service Profile)" and shows the configuration for "DC1_WE_Config_Group_1_LAN". Below the title, a description reads "Description: DC1_WE_Config_Group_1 LAN Service Profile". A table lists configuration details: Type (Dual Device), Device Name (DC1_WE1, DC1_WE2), DEVICE SOLUTION (SD-WAN), MODIFIED BY (Admin), LAST UPDATED (Jan 04, 2025 04:49:09), and SHARED (1 Group). Below the table, there are tabs for "All", "DC1_WE1", and "DC1_WE2", along with a search bar and a "+ Add New Feature" button. The configuration area is titled "Service VPN" and contains two sections: "Service" and "Ethernet Interface". The "Service" section has a dropdown menu showing "Service" and a "+" button. The "Ethernet Interface" section has a dropdown menu showing "VPN_1_Local_Network_Interface_0" and a "+" button. A "Back" button and the text "No Changes Made" are at the bottom of the configuration area.

Add Management VPN to Transport Profile

Click Add New Feature

The screenshot displays the Cisco Catalyst SD-WAN configuration interface. The main configuration area shows the 'DC1_WE1_Config_Group_1a_WAN' profile, which is a 'Dual Device' profile for 'DC1_WE1, DC1_WE2'. The profile is currently configured with 'Transport VPN' and 'Ethernet Interface' (Internet and MPLS). A 'Back' button indicates 'No Changes Made'.

The 'Add New Feature' dialog box is open on the right side of the screen. It lists the following features:

- Management VPN (highlighted with a red box)
- Transport VPN
- Cellular Controller
- Managed Cellular Activation - eSIM
- T1E1 Controller

The 'Add New Feature' button is located in the top right corner of the configuration area.

Add Management VPN to Transport Profile

VPN = 512

The screenshot displays the Cisco Catalyst SD-WAN configuration interface. The main heading is "DC1_WE_Config_Group_1_WAN" with an "Edit" link. Below it, the description is "DC1_WE_Config_Group_1 WAN Transport Profile". A table provides details: Type (Dual Device), Device Name (DC1_WE1, DC1_WE2), DEVICE SOLUTION (SD-WAN), MODIFIED BY (Admin), LAST UPDATED (Jan 23, 2025 01:25:40), and SHARED (1 Group). A search bar and "Add New Feature" button are present. The configuration is divided into two sections: "Transport VPN" and "Management VPN". The "Transport VPN" section includes "VPN0" with two "Ethernet Interface" options: "Internet" and "MPLS". The "Management VPN" section, highlighted with a red box, includes "VPN512" with a "Management Ethernet Interface" option: "Management". A "Back" button and "No Changes Made" status are at the bottom.

Cisco Catalyst SD-WAN

← Configuration Groups (DC1_WE_Config_Group_1 - Transport & Management Profile)

DC1_WE_Config_Group_1_WAN [Edit](#)

Description: DC1_WE_Config_Group_1 WAN Transport Profile

Type	Device Name	DEVICE SOLUTION	MODIFIED BY	LAST UPDATED	SHARED
Dual Device	DC1_WE1, DC1_WE2	SD-WAN	Admin	Jan 23, 2025 01:25:40	1 Group

[All](#) [DC1_WE1](#) [DC1_WE2](#) [+ Add New Feature](#)

Transport VPN

VPN0 [+](#)

Ethernet Interface [+](#)

Internet [+](#)

Ethernet Interface [+](#)

MPLS [+](#)

Management VPN

VPN512 [+](#)

Management Ethernet Interface [+](#)

Management [+](#)

[Back](#) No Changes Made

Associate cEdge Devices to Configuration Group

Add Devices, Assign Tags & Deploy

The screenshot displays the Cisco Catalyst SD-WAN Configuration Groups interface. The top navigation bar includes the Cisco logo, 'Catalyst SD-WAN', and user information. The left sidebar contains navigation links for Monitor, Configuration, Analytics, Workflows, Tools, Reports, and Maintenance. The main content area is titled 'Configuration Groups' and shows a dropdown menu for 'SD-WAN'. Below this, there are tabs for Configuration Groups (2), System Profile (3), Transport & Management Profile (3), Policy Profile (1), Service Profile (3), CLI Add-on Profile (1), and UC Voice Profile (1). A search bar and filters for 'Last Updated' and 'Status' are present. A table lists the configuration groups, with the first group being 'DC1_WE_Config_Group_1' (WAN Edge devices in Data Center). Below the table, the configuration details for this group are shown, including System Profile, Service Profile, and Transport & Management Profile. A 'Deployment' panel on the right shows tags 'DC1_WE1, DC1_WE2' and a 'Deploy' button.

Name	Type	Profile	Provisioning Status	Origin	Updated By	Last Updated On	Actions
DC1_WE_Config_Group_1	WAN Edge devices in Data Center		In Sync Devices / Associated Devices		Updated by admin	Updated Jan 23, 2025	

Type: Dual Device

System Profile

DC1_WE_Config_Group_1_Basic

Service Profile (optional)

DC1_WE_Config_Group_1_LAN

Transport & Management Profile

DC1_WE_Config_Group_1_WAN

Deployment

Tags: DC1_WE1, DC1_WE2

Associated: + Add

Provisioning: N/A

Deploy

Associate cEdge Devices to Configuration Group

Select Devices to Deploy

Deploy Configuration Group

Select Devices to Deploy

Select the devices below to deploy

Search

☒ Global (1 of 6 nodes) ▼

☒ SITE_120001

Edge Devices (1)

Search Table

1 selected

☒	Chassis Numbers	Site Name	Hostname	System IP	Tags	Last Configured
☒	C8300-2N2S-6T-FI	SITE_120001	DC1_WE2	192.168.1.154	DC1_WE2	-

Items per page: 25 1-1 of 1 |< < > >|

Cancel Next

Associate cEdge Devices to Configuration Group

Configure Device Specific Settings

Deploy Configuration Group

Add and Review Device Configuration

We've autogenerated minimal configurations to make it easier for you to bring up your devices. Modify them as needed and directly edit the table to add System IP and Site IDs.

[Import](#) [Export](#)

Search

Global

SITE_120001

Chassis Number:DC1_WE2 - C8300-2N2S-6T-FLM2815101G

DC1_WE_Config_Group_1_Basic (system)	1 Optional
UC_SIP_TDM_GW (uc-voice)	8 Optional
DC1_WE_Config_Group_1_WAN (transport)	3 Optional
DC1_WE_Config_Group_1_LAN (service)	

[Review](#) [Back](#) [Next](#)

Associate cEdge Devices to Configuration Group

Configure Device Specific Settings

Deploy Configuration Group

Add and Review Device Configuration

We've autogenerated minimal configurations to make it easier for you to bring up your devices. Modify them as needed and directly edit the table to add System IP and Site IDs.

[Import](#) [Export](#)

Chassis Number:DC1_WE2 - C8300-2N2S-6T-FLM2815101G

DC1_WE_Config_Group_1_Basic (system) 1 Optional

Basic (basic)

System IP: 192.168.1.154 Host Name: DC1_WE2 Site Id: 120001

Dual Stack IPv6 Default (...): False

AAA (aaa)

admin_password: [password field] [Show](#)

[Review](#) [Back](#) [Next](#)

Associate cEdge Devices to Configuration Group

Preview Configuration & Deploy

The screenshot shows the 'Deploy Configuration Group' interface in the Cisco Catalyst SD-WAN management console. The interface is divided into a sidebar on the left and a main content area. The sidebar contains navigation links for Monitor, Configuration, Analytics, Workflows, Tools, Reports, Maintenance, Administration, and Explore. The main content area has a 'Summary' section with a 'Please review the details.' message. Below this is a 'Selected Devices (1)' section with a 'Preview CLI' button highlighted. The 'Preview CLI' button is located next to the 'Edit Device List' button. Below the buttons is a table with columns: Device ID, System IP, Host Name, Site Id, Dual Stack IPv6 Default, Rollback Timer (sec), sourceInterfa_source_intf, and SIP_TDM_GW_Inbound_Pt. The table contains one record for device C8300-2N2S-6T-FLM. At the bottom of the interface are 'Back' and 'Deploy' buttons.

Deploy Configuration Group

Summary
Please review the details.

Selected Devices (1) **Preview CLI** Edit Device List

All DC1_WE2

Q Search Table

Device ID	System IP	Host Name	Site Id	Dual Stack IPv6 Default	Rollback Timer (sec)	sourceInterfa_source_intf	SIP_TDM_GW_Inbound_Pt
C8300-2N2S-6T-FLM	192.168.1.154	DC1_WE2	120001	false	300	GigabitEthernet0/0/3	91943808.

1 Record

Items per page: 25 1 - 1 of 1 |< < > >|

Back Deploy

Associate cEdge Devices to Configuration Group

Preview Configuration & Deploy

Review device configuration

Click a device to preview its CLI configuration

Global

SITE_120001

DC1_WE2

Inline Diff

Side by Side

Old Configuration

		Export	Copy
1	system		
2	ztp-status	success	
3	pseudo-confirm-commit	300	
4	personality	vedge	
5	device-model	vedge-C8300-2N2S-6T	
6	chassis-number	C8300-2N2S-6T-FLM2815101G	
7	system-ip	192.168.1.154	
8	overlay-id	1	
9	site-id	120001	
10	ipv6-strict-control	false	
11	role	edge-router	
12	port-offset	0	
13	control-session-pps	300	
14	admin-tech-on-failure		
15	sp-organization-name	calo-auto-lab	
16	organization-name	calo-auto-lab	
17	port-hop		
18	track-transport		

New Configuration

		Export	Copy
45	sdwan		
46	interface GigabitEthernet0/0/0		
47	exit		
48	interface GigabitEthernet0/0/1		
49	tunnel-interface		
50	encapsulation ipsec weight 1		
51	no border		
52	color mpls		
53	no last-resort-circuit		
54	no low-bandwidth-link		
55	no vbond-as-stun-server		
56	vmanage-connection-preference 5		
57	port-hop		
58	carrier	default	
59	nat-refresh-interval	5	
60	hello-interval	1000	
61	hello-tolerance	12	
62	no allow-service all		

Cancel

cEdge Device Deployed!

The screenshot displays the Cisco Catalyst SD-WAN Configuration Groups page. The top navigation bar includes the Cisco logo, the text "Catalyst SD-WAN", and user information "admin". The left sidebar contains navigation icons for Monitor, Configuration, Analytics, Workflows, Tools, Reports, and Maintenance. The main content area is titled "Configuration Groups" and features a dropdown menu set to "SD-WAN". Below this, a breadcrumb trail shows the navigation path: Configuration Groups (2) > System Profile (3) > Transport & Management Profile (3) > Policy Profile (1) > Service Profile (3) > CLI Add-on Profile (1) > UC Voice Profile (1). A search bar and filters for "Last Updated" and "Status" are present, along with buttons for "Create Configuration Group", "Export", and "Import".

Name	Type	Profile	Provisioning Status In Sync Devices / Associated Devices	Origin	Updated By	Last Updated On	Actions
DC1_WE_Config_Group_1	WAN Edge devices in Data Center				Updated by admin	Updated Jan 23, 2025	

Below the table, the configuration details for "DC1_WE_Config_Group_1" are shown. The "Type" is "Dual Device". The "System Profile" is "DC1_WE_Config_Group_1_Basic". The "Service Profile (optional)" is "DC1_WE_Config_Group_1_LAN". The "Transport & Management Profile" is "DC1_WE_Config_Group_1_WAN". The "UC Voice Profile (optional)" is "UC_SIP_TDM_GW". A "+ Add Profile" link is at the bottom left.

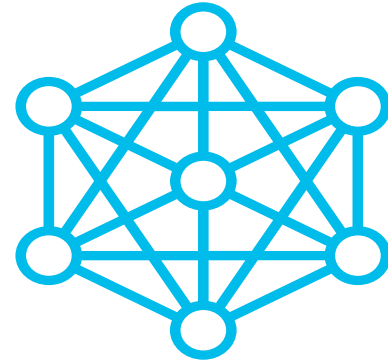
On the right, the "Deployment" section shows "Tags" as "DC1_WE1, DC1_WE2". It includes a summary: "Associated 1 device" and "Provisioning 0 out of sync". A "Deploy" button is at the bottom.

cEdge: Control Plane Established!

The screenshot displays the Cisco Catalyst SD-WAN management interface. The left sidebar contains navigation menus for Monitor, Configuration, Analytics, Workflows, Tools, Reports, Maintenance, Administration, and Explore. The 'Control Connections' option under the 'Explore' menu is highlighted with a blue box. The main content area is titled 'Devices > Control Connections'. It shows a dropdown for 'Select Device' with 'DC1_WE2 | 192.168.1.154' selected. Below this, it indicates 'Site Name 120001' and 'Device Model: C8300-2N2S-6T'. The 'Controller Control Connections' section shows 'Expected: 4 | Actual: 2'. A diagram illustrates the network topology with a central 'biz-internet' node connected to two green star nodes and a blue server node. Below the diagram, a table lists the connections:

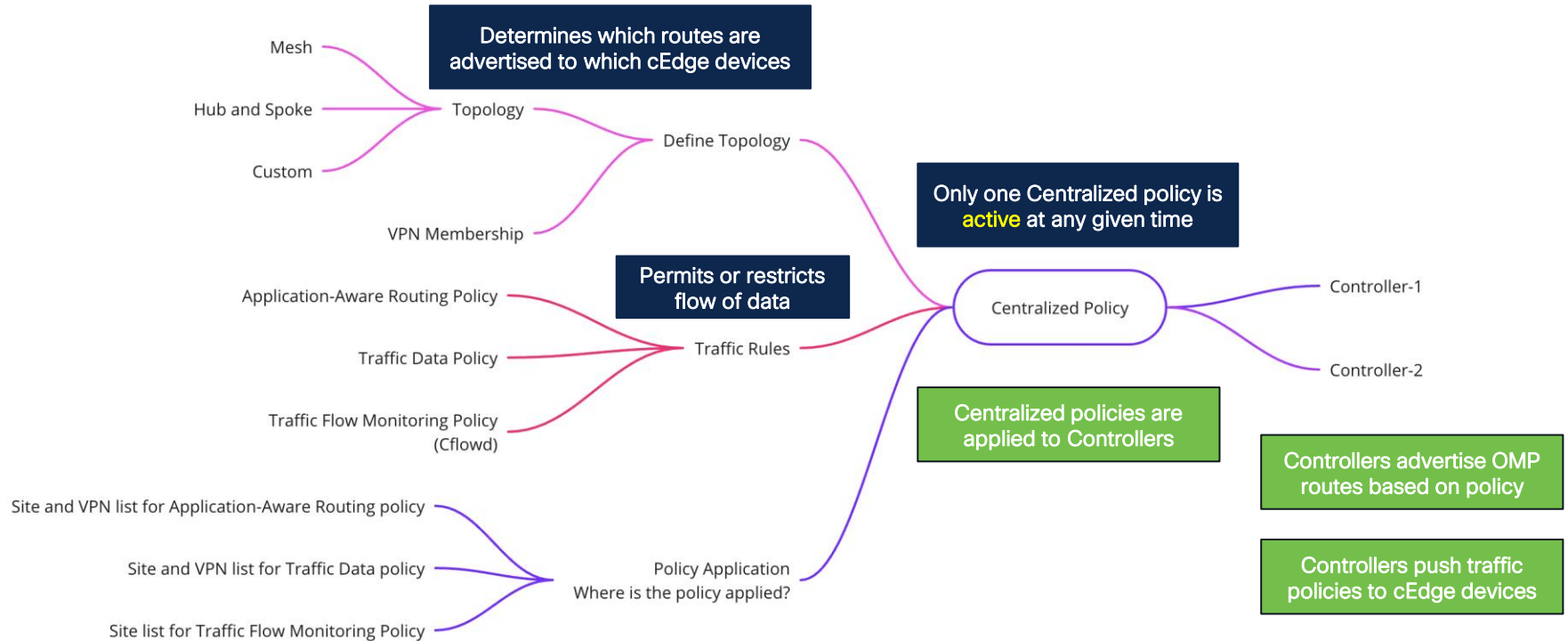
Peer Type	Peer System IP	Peer Protocol	Private Port	Public Port	Controller Group ID	Last Updated
Controller	10.2.0.11	dtls	12546	12546	0	07 Feb 2025 6:55:27 AM E
Controller	10.1.0.11	dtls	12546	12546	0	05 Feb 2025 6:43:32 PM E
Manager	1.1.1.7	dtls	12546	12546	0	05 Feb 2025 6:43:32 PM E

Data Plane Establishment

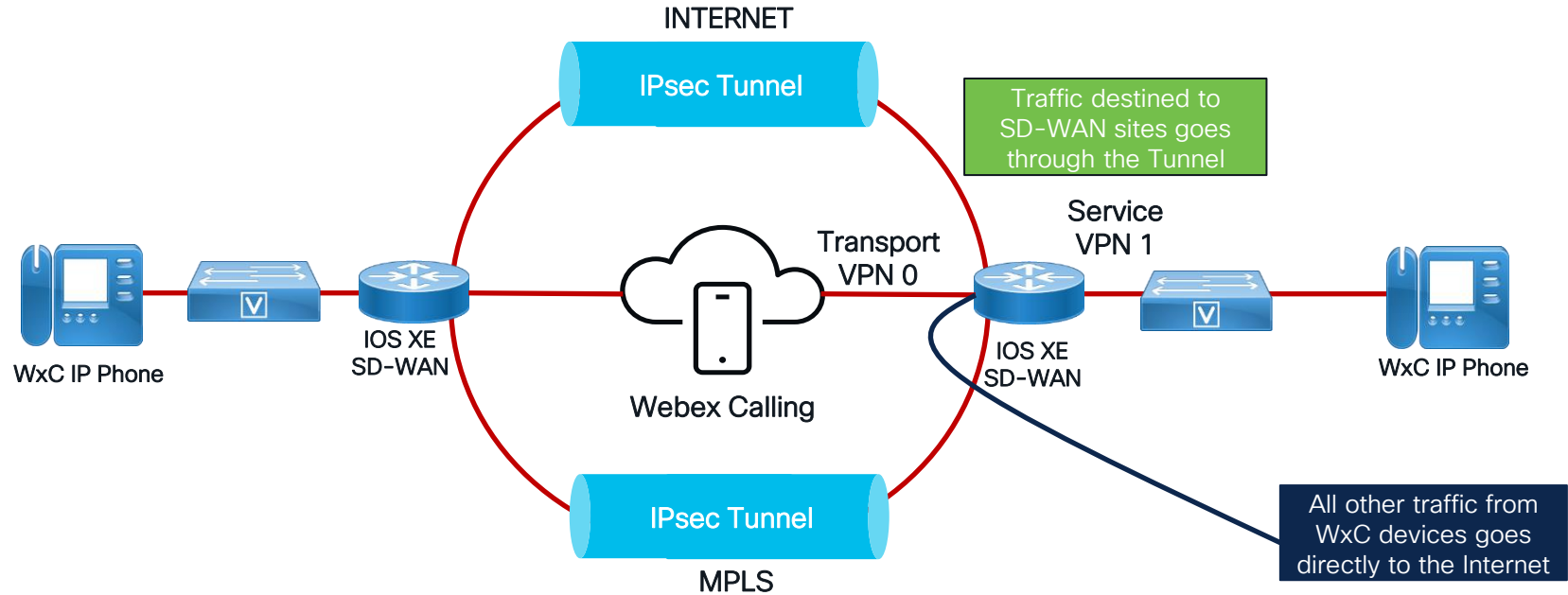


cEdge <-> cEdge

Centralized Policy Defines Data Plane Connectivity



Direct Internet Access (DIA)



Define Components of Centralized Policy

Configuration > Policies > Centralized Policy > Custom Options



Site List

Group sites of same types into a list

Centralized Policy > Define Lists

Custom Options

Select a list type on the left and start creating your groups of interest

New Site List

Name	Entries	Reference Count	Updated By	Last Updated	Action
Central-Sites	120001	5	admin	31 May 2024 6:01:49 PM E...	Edit Delete
Type1-Branches	121001	5	admin	31 May 2024 6:36:37 PM E...	Edit Delete
Type2-Branches	122001	5	admin	31 May 2024 7:16:59 PM E...	Edit Delete

VPN List

Group related VPNs into a list

Catalyst SD-WAN

Centralized Policy > Define Lists

Custom Options

Select a list type on the left and start creating your groups of interest

New VPN List

Name	Entries	Reference Count	Updated By	Last Updated	Action
Management_VPN_List	512	1	admin	16 Apr 2024 5:48:48 AM E...	Edit Delete
Service_VPN_List	1	4	admin	16 Apr 2024 5:48:30 AM E...	Edit Delete
Transport_VPN_List	0	1	admin	16 Apr 2024 5:48:14 AM EDT	Edit Delete

Sample Topology

Choose Mesh Topology – All sites talk with each other

Topology Name = SDWAN-UC

Centralized Policy > Topology > Add Mesh Region

Monitor
Configuration
Tools
Maintenance
Administration
Workflows
Reports
Analytics
Explore

Name* SDWAN-UC

Description* SDWAN UC Network Topology

VPN List* Service_VPN_List

New Mesh Region

Mesh Region Name	Site List	Action
SDWAN-UC-Sites	Type1-Branches, Central-Sites, Type2-Branches	 

* Full mesh is the default topology type

VPN Membership

Define the sites and VPNs which should receive routes from Controllers

Policy Name = SDWAN-UC-VPN-Membership

Centralized Policy > VPN Membership

Monitor
Configuration
Tools
Maintenance
Administration
Workflows
Reports
Analytics
Explore

Specify your network topology

Topology VPN Membership

Q Search

Add VPN Membership Policy (Choose and add VPNs to specific site lists)

Name	Type	Description	Mode
SDWAN-UC-VPN-Member...	VPN Membership	SDWAN-UC-VPN-Member...	-

Edit

Update VPN Membership Policy

VPN Membership Name* SDWAN-UC-VPN-Membership

Description* SDWAN-UC-VPN-Membership

Site List* VPN Lists*

Type1-Branches Transport_VPN_List x Service_VPN_List x Manageme

Central-Sites Transport_VPN_List x Service_VPN_List x Manageme

Type2-Branches Management_VPN_List x Service_VPN_List x Transp

Add List

Cancel Save

Define Traffic Data Policy To Enable DIA

Data Prefix list is used to group IP address prefixes

Name	Description	IP Address Type	Entries
SDWAN_Overlay_Traffic	Traffic destined to SD-WAN sites	Destination IP address	172.16.106.0/24,172.16.107.0/24, 172.16.108.0/24,172.16.109.0/24, 172.16.110.0/24,172.16.115.0/24, 172.16.116.0/24,172.16.117.0/24, 172.16.118.0/24,172.16.119.0/24, 172.16.126.0/24,172.16.127.0/24, 172.16.128.0/24,172.16.129.0/24
SDWAN_UC_WxC	All other traffic from WxC devices	Source IP address	172.16.106.0/24,172.16.107.0/24, 172.16.116.0/24,172.16.117.0/24, 172.16.126.0/24,172.16.127.0/24

Define Traffic Data Policy To Enable DIA

Redirect specific traffic from Service VPN to Transport VPN

View Data Policy

Policies

Name* WxC_DIA

Description* Route traffic from WxC IP phones to WxC Cloud

Custom

1 Match Conditions

Destination Data Prefix List: SDWAN_Overlay_Traffic

Destination: IP

2 Match Conditions

Source Data Prefix List: SDWAN_UC_WxC

Source: IP

Actions

Accept

NAT VPN: 0

Match: External traffic from WxC devices

Action: Send to VPN 0

Associate Components to Centralized Policy

Traffic Data Policy Association

Centralized Policy > Edit Policy

Policy Application | Topology | Traffic Rules

Add policies to sites and VPNs

Policy Name* Centralized_Policy_SDWAN_UC_DIA

Policy Description* Centralized Policy SDWAN UC DIA

Topology | Application-Aware Routing | **Traffic Data** | Cflowd | Role Mapping for Regions

WxC_DIA

New Site/Region List and VPN List

Site/Region List	Region ID	VPN List
Type1-Branches, Central-Sites, Type2-...	N/A	Service_VPN_1

Edit WxC_DIA

Direction

☒ From Service ☐ From Tunnel ☐ All

Select Site List

Type1-Branches x Central-Sites x Type2-Branches x

Select VPN List

Service_VPN_1 x

Policy is applied to traffic coming from Service VPN 1

Update Cancel

Associate Components to Centralized Policy

Topology and VPN Membership Policy Association

The screenshot displays the 'Centralized Policy > Edit Policy' configuration page. The 'Policy Application' tab is active, showing a form to add policies to sites and VPNs. The 'Policy Name' is 'Centralized_Policy_SDWAN_UC_DIA' and the 'Policy Description' is 'Centralized Policy SDWAN UC DIA'. Below the form, the 'Topology' sub-tab is selected, displaying a table titled 'SDWAN-UC-VPN-Membership' (highlighted with a red box). This table maps sites to VPN lists. The table has two columns: 'Site List' and 'VPN List'. The rows are: 'Type1-Branches' mapping to 'Transport_VPN_List, Service_VPN_List, Management_VPN_List'; 'Central-Sites' mapping to 'Transport_VPN_List, Service_VPN_List, Management_VPN_List'; and 'Type2-Branches' mapping to 'Management_VPN_List, Service_VPN_List, Transport_VPN_List'. At the bottom, a 'SDWAN-UC' section (also highlighted with a red box) shows a 'VPN List' and a 'MESH' button. A 'Feedback' button is visible on the right side of the table.

Centralized Policy > Edit Policy

Policy Application | Topology | Traffic Rules

Add policies to sites and VPNs

Policy Name* Centralized_Policy_SDWAN_UC_DIA

Policy Description* Centralized Policy SDWAN UC DIA

Topology | Application-Aware Routing | Traffic Data | Cflowd | Role Mapping for Regions

SDWAN-UC-VPN-Membership VPN MEMBERSHIP

Site List	VPN List
Type1-Branches	Transport_VPN_List, Service_VPN_List, Management_VPN_List
Central-Sites	Transport_VPN_List, Service_VPN_List, Management_VPN_List
Type2-Branches	Management_VPN_List, Service_VPN_List, Transport_VPN_List

SDWAN-UC MESH

VPN List

Deploy Global Centralized Policy

Activate and Apply Policy to SD-WAN Controllers

The screenshot displays the Cisco SD-WAN configuration interface. On the left is a navigation sidebar with icons for Monitor, Configuration, Tools, Maintenance, Administration, Workflows, Reports, Analytics, and Explore. The main content area is titled 'Centralized Policy > Edit Policy'. It features three tabs: 'Policy Application' (active), 'Topology', and 'Traffic Rules'. Below the tabs, there are input fields for 'Policy Name*' (containing 'Centralized_Policy_SDWAN_UC_DIA') and 'Policy Description*' (containing 'Centralized Policy'). A modal dialog box titled 'Activate Policy' is centered on the screen. It contains the text 'Policy will be applied to the reachable devices: 1.1.1.3, 1.1.1.5' and two buttons: 'Activate' and 'Cancel'. In the background, under the 'Topology' tab, there is a section for 'SDWAN-UC-VPN-Membership' with a 'Site List' table. The table has columns for 'Type' and 'List'. The rows show 'Type1-Branches' with 'Central-Sites', 'Type2-Branches' with 'Management_VPN_List, Service_VPN_List, Transport_VPN_List', and 'VPN MEMBERSHIP' with 'Management_VPN_List'.

Centralized Policy > Edit Policy

Policy Application Topology Traffic Rules

Add policies to sites and VPNs

Policy Name* Centralized_Policy_SDWAN_UC_DIA

Policy Description* Centralized Policy

Topology Application-Aware Routing

SDWAN-UC-VPN-Membership

Site List

Type	List
Type1-Branches	Central-Sites
Type2-Branches	Management_VPN_List, Service_VPN_List, Transport_VPN_List
VPN MEMBERSHIP	Management_VPN_List

Activate Policy

Policy will be applied to the reachable devices:
1.1.1.3, 1.1.1.5

Activate Cancel

Preview Save Policy Changes Cancel

Deploy Global Centralized Policy

Activate and Apply Policy to SD-WAN Controllers

Monitor

Configuration

Tools

Maintenance

Administration

Workflows

Reports

Analytics

Explore

Push vSmart Policy | Validation success

Initiated By: admin

Total Task: 2 | Success : 2

Device Group (2)

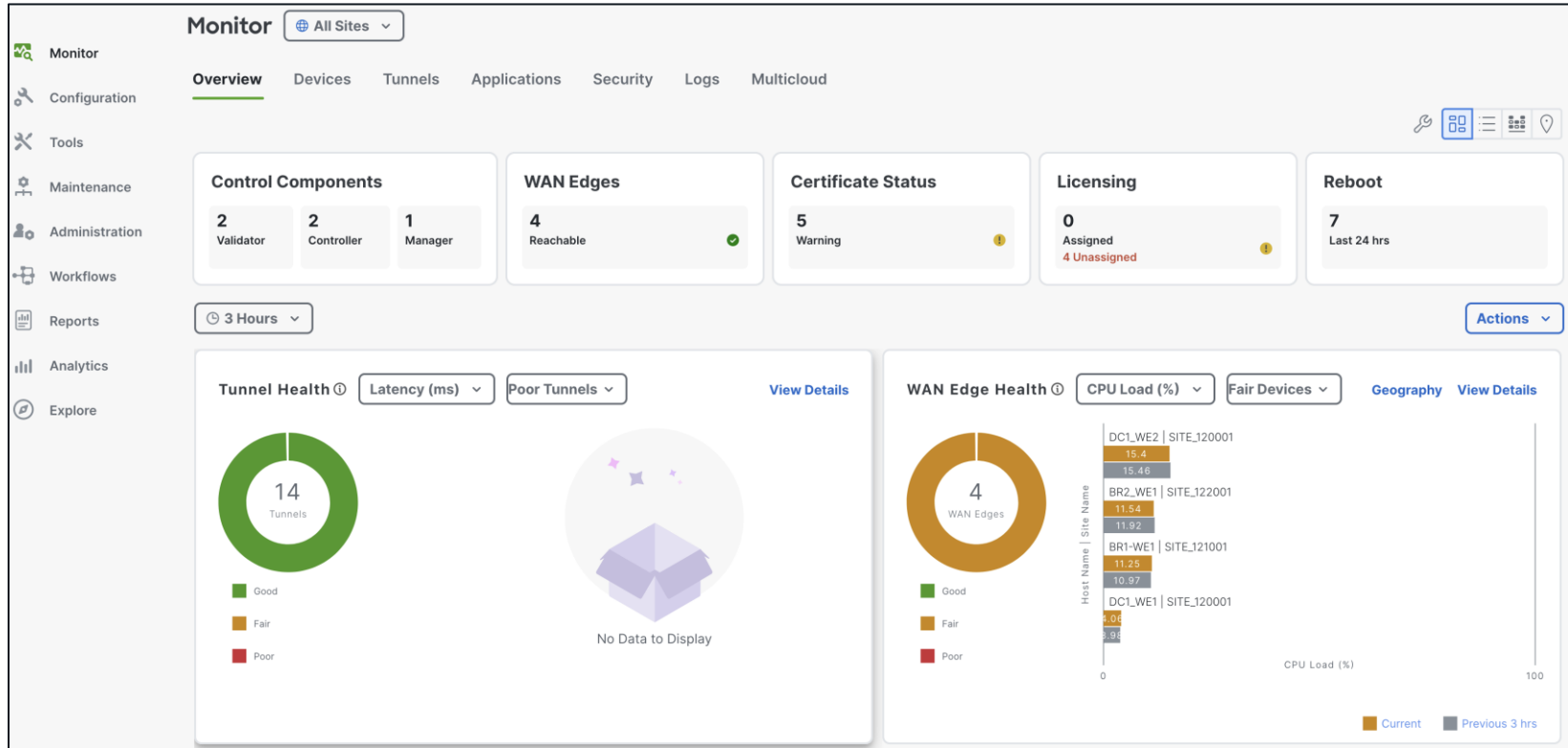
Q Search Table

As of: Apr 25, 2024 03:13 AM

Status	Message	Hostname	System IP	Site ID	vManage IP	Action
Success	Done - Push vSmart Policy	vsmart-east	1.1.1.5	1014	1.1.1.7	
Success	Done - Push vSmart Policy	vsmart-west	1.1.1.3	1013	1.1.1.7	

Items per page: 10 1 ~ 2 of 2 |< < > >

SD-WAN Tunnels Are UP!



SD-WAN Tunnels Are UP!

Monitor All Sites

Overview Devices **Tunnels** Applications Security Logs Multicloud

SD-WAN Tunnels SIG/SSE Tunnels

3 Hours

SD-WAN Tunnels (14) Export

Search Table

As of: Jun 01, 2024 04:17 PM

Tunnel Endpoints	Health	State	QoE	Avg. Latency (ms)	Avg. Loss (%)	Avg. Jitter (ms)	Local IP	Remote IP	Action
DC1_WE2:biz-internet-BR2_WE1:biz-internet	✓	↑	10.00	22.00	0.01	0.00	192.168.1.154	192.168.3.193	...
BR1_WE1:biz-internet-BR2_WE1:biz-internet	✓	↑	10.00	21.12	0.00	0.00	192.168.2.187	192.168.3.193	...
DC1_WE1:biz-internet-BR2_WE1:biz-internet	✓	↑	10.00	21.00	0.00	0.00	192.168.1.153	192.168.3.193	...
DC1_WE2:public-internet-BR2_WE1:biz-internet	✓	↑	10.00	21.00	0.00	0.00	192.168.1.154	192.168.3.193	...
BR2_WE1:biz-internet-DC1_WE1:biz-internet	✓	↑	10.00	21.00	0.01	0.00	192.168.3.193	192.168.1.153	...

Traffic Data Policy Pushed by Controller to Edge

```
BR1-WE1# show sdwan policy from-vsmart
from-vsmart data-policy _Service_VPN_1_BR_DIA
direction from-service
vpn-list Service_VPN_1
sequence 1
match
  destination-data-prefix-list SDWAN_Overlay_Traffic
action accept
sequence 11
match
  source-data-prefix-list SDWAN_UC_WxC
action accept
  nat use-vpn 0
  no nat fallback
default-action drop
```

Traffic Data Policy Pushed by Controller to Edge

```
from-vsmart lists data-prefix-list SDWAN_Overlay_Traffic
```

```
ip-prefix 172.16.106.0/24
```

```
ip-prefix 172.16.107.0/24
```

```
ip-prefix 172.16.108.0/24
```

```
ip-prefix 172.16.109.0/24
```

```
ip-prefix 172.16.110.0/24
```

```
ip-prefix 172.16.115.0/24
```

```
ip-prefix 172.16.116.0/24
```

```
....
```

```
ip-prefix 172.16.119.0/24
```

```
ip-prefix 172.16.126.0/24
```

```
...
```

```
ip-prefix 172.16.129.0/24
```

```
from-vsmart lists data-prefix-list SDWAN_UC_WxC
```

```
ip-prefix 172.16.106.0/24
```

```
ip-prefix 172.16.107.0/24
```

```
ip-prefix 172.16.116.0/24
```

```
ip-prefix 172.16.117.0/24
```

```
ip-prefix 172.16.126.0/24
```

```
ip-prefix 172.16.127.0/24
```

```
BR1-WE1#
```

IP NAT Translations: WxC IP Phone Registration

```
BR1-WE1# show ip nat translations
```

Pro	Inside global	Inside local	Outside local	Outside global
...				
udp	203.0.113.187:5167	172.16.117.14:52750	8.8.8.8:53	8.8.8.8:53
...				
tcp	203.0.113.187:5079	172.16.117.14:5074	23.89.40.110:8934	23.89.40.110:8934
...				
...				

BR1-WE1#

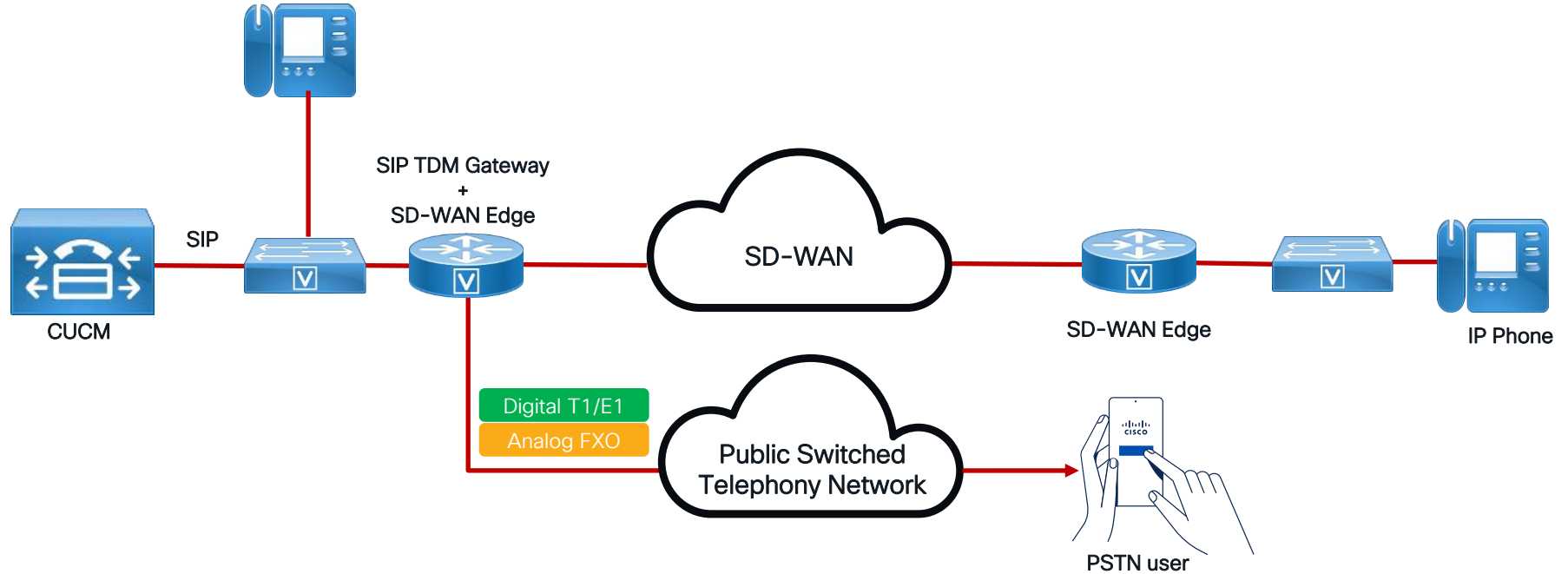
DNS

SIP

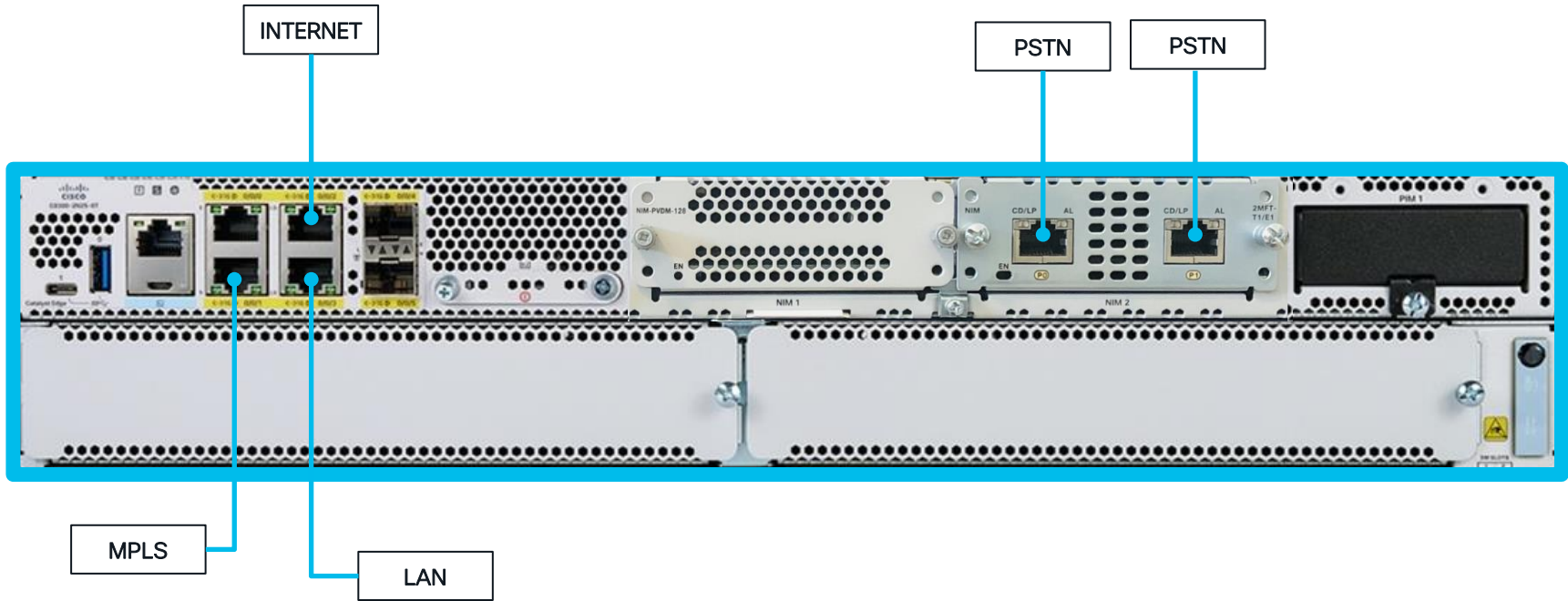
SIP TDM Gateway

Calling via PSTN

On-Premise UC: External Calls Via PSTN

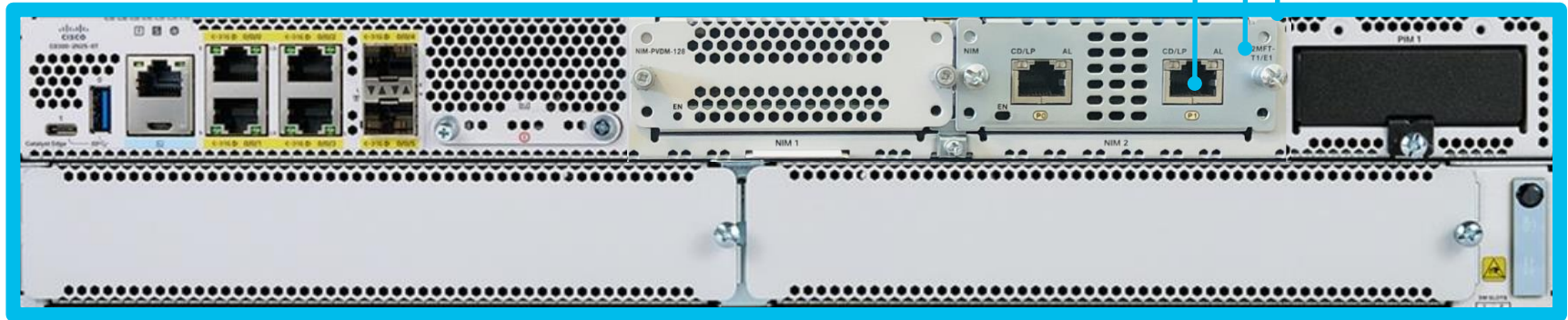


Catalyst 8300 as SIP TDM Gateway

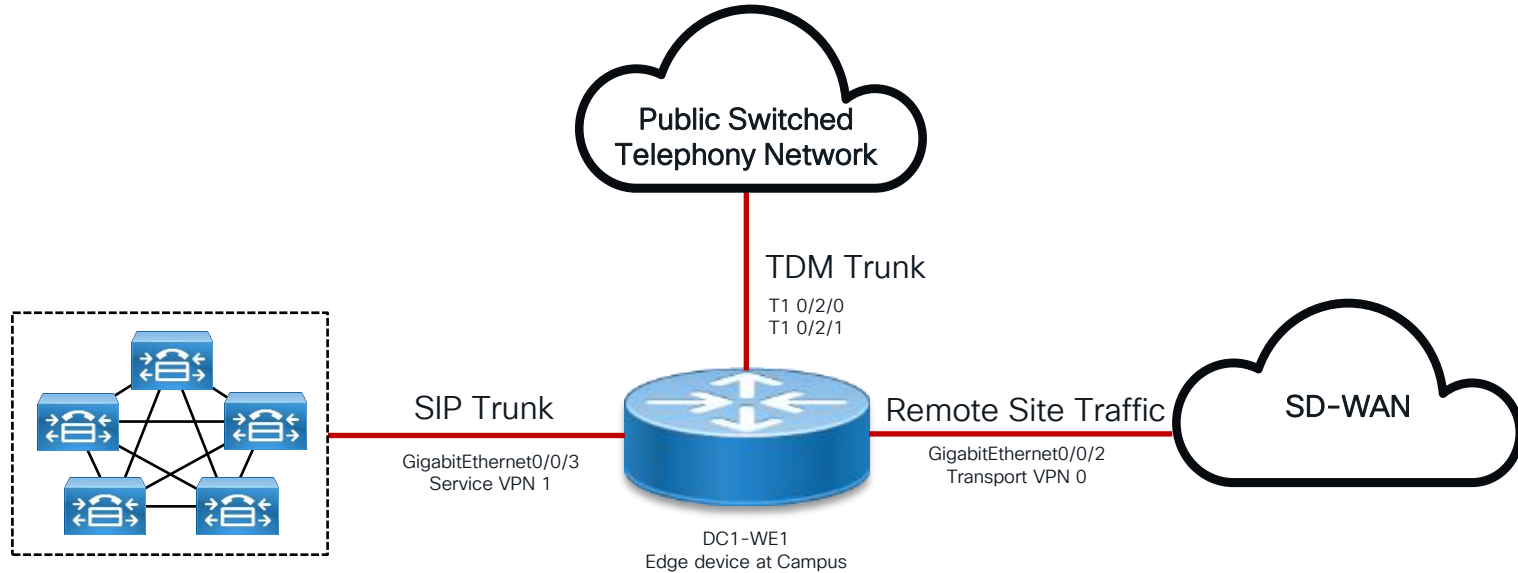


Catalyst 8300 as SIP TDM Gateway

Slot 0
voice-card {slot/sub-slot}
controller t1 {slot/sub-slot/port}
D-channel interface {slot/sub-slot/port}:23

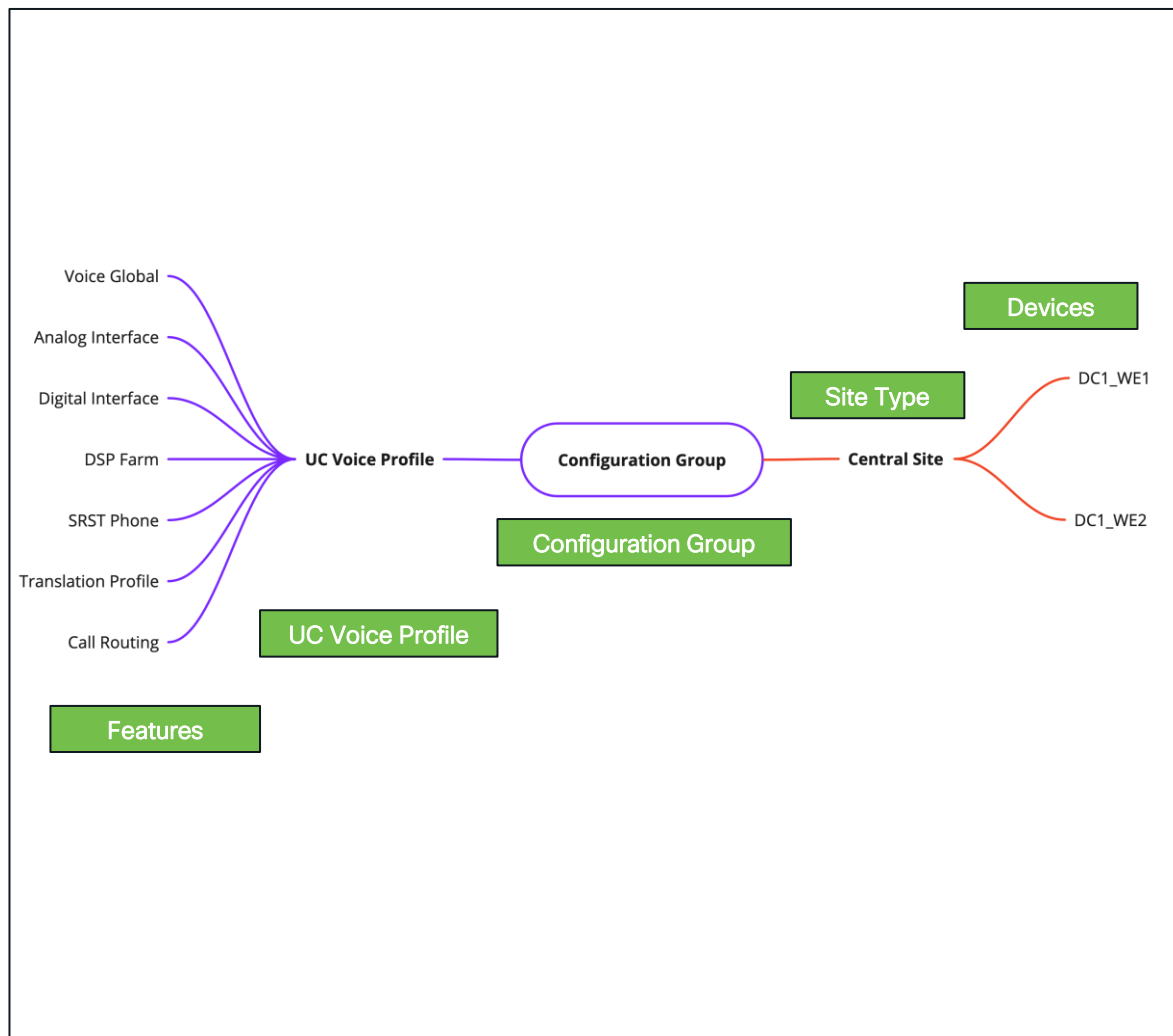


SIP and PSTN Integrations @ SD-WAN Edge



Voice Configuration Blocks

Define once *easily*,
Reuse everywhere!



SIP Gateway Configuration in SD-WAN Manager

Create a UC Voice Profile within the Configuration Group

The screenshot displays the Cisco Catalyst SD-WAN Manager interface. The top navigation bar includes the Cisco logo, the text "Catalyst SD-WAN", and user information "admin". The left sidebar contains navigation icons for Monitor, Configuration, Analytics, Workflows, Tools, Reports, and Maintenance. The main content area is titled "Configuration Groups" and shows a dropdown menu set to "SD-WAN". Below this, a breadcrumb trail indicates the current path: Configuration Groups (3) > System Profile (4) > Transport & Management Profile (4) > Policy Profile (1) > Service Profile (4) > CLI Add-on Profile (1) > UC Voice Profile (1). A search bar and filters for "Last Updated" and "Status" are present, along with buttons for "Create Configuration Group", "Export", and "Import".

Name	Type	Profile	Provisioning Status In Sync Devices / Associated Devices	Origin	Updated By	Last Updated On	Actions
DC1_WE_Config_Group_1	WAN Edge devices in Data Center				Updated by admin	Updated Jan 25, 2025	

Below the table, the configuration details for "DC1_WE_Config_Group_1" are shown. The "Type" is "Dual Device". The "System Profile" is set to "DC1_WE_Config_Group_1_Basic". The "Service Profile (optional)" is set to "DC1_WE_Config_Group_1_LAN". The "Transport & Management Profile" is set to "DC1_WE_Config_Group_1_WAN". A "Deployment" section on the right shows tags "DC1_WE1, DC1_WE2", associated "2 devices", and a provisioning status of "0 out of sync". A "Deploy" button is available.

A callout box highlights the "+ Add Profile" button, which opens a menu with the following options:

- Policy Profile
- CLI Add-on Profile
- UC Voice Profile
- Other Profile

SIP Gateway Configuration in SD-WAN Manager

Add UC Features

Cisco Catalyst SD-WAN

← Configuration Groups (DC1_WE1_Config_Group_1 - UC Voice Profile)

UC_SIP_TDM_GW [Edit](#)

Description: UC SIP TDM Gateway

Type	Device Name	DEVICE SOLUTION	MODIFIED BY	LAST UPDATED	SHARED
Dual Device	DC1_WE1, DC1_WE2	SD-WAN	Admin	Jan 26, 2025 11:44:34	1 Group

[All](#) [DC1_WE1](#) [DC1_WE2](#)

[+ Add New Feature](#)

Add new features to begin building your profile

Add Feature

- << DSP Farm
- << Analog Interface
- << Voice Global
- << Translation Profile
- << SRST Phone
- << Call Routing
- << Digital Interface

[Back](#) No Changes Made

SIP Gateway Configuration in SD-WAN Manager

Add Digital Interface for T1/E1 PRI Ports

Voice Interface Template

Module Location

Use DSP

Port and Clock Selector

Basic Settings

ISDN Timer

ISDN Map

Line Params

Outgoing IE

Associations

Digital Interface

Name: T1_PRI_0/2/0_0/2/1

Description (optional): T1 PRI Interfaces

Voice Interface Templates

Voice Interface Templates: 2 Port T1

Module Location: 0/2

Use DSP: ☒

Port and Clock Selector

Selected 2

0 1 2 3 4 5 6 7

Basic Configuration | ISDN Timer | ISDN Map | Shutdown | Line Params | Outgoing IE | Associations

Basic Settings

+ Add Basic Settings

Port Range	Time slots (optional)	Cable Length Type (optional)	Cable Length (optional)	Action
0-1	1-24	long	0	

ISDN Timer

Cancel Save

SIP Gateway Configuration in SD-WAN Manager

Specify Clock Sources for T1/E1 PRI ports

Voice Interface Template

Module Location

Use DSP

Port and Clock Selector

Basic Settings

ISDN Timer

ISDN Map

Line Params

Outgoing IE

Associations

The screenshot shows the Cisco SD-WAN Manager interface for configuring a UC SIP TDM Gateway. The configuration group is named 'UC_SIP_TDM_GW' and is described as 'UC SIP TDM Gateway'. It is a 'Dual Device' with device names 'DC1_WE1' and 'DC1_WE2'. The configuration is for 'T1 PRI Interfaces'. A 'Port and Clock Selector' dialog box is open, showing options for 'All', 'Port 0', and 'Port 1' under 'Line' and 'Network' tabs. The 'Line' tab is selected, and 'Port 0' and 'Port 1' are both checked. The 'Primary Clock' is set to 'Line' and the 'Secondary Clock' is set to 'Network'. The dialog box also shows a table of port ranges, time slots, cable length types, and cable lengths.

Port Range	Time slots (optional)	Cable Length Type (optional)	Cable Length (optional)	Action
0-1	1-24	long	0	

SIP Gateway Configuration in SD-WAN Manager

Voice Global Configuration

Trusted IPv4 / IPv6 List

SIP Source Interface

Network Clock

The screenshot displays the 'Voice Global' configuration page in the Cisco SD-WAN Manager. The left sidebar shows navigation options like Monitor, Configuration, Analytics, Workflows, Tools, Reports, Maintenance, Administration, and Explore. The main content area is divided into two tabs: 'Call Routing' (active) and 'Network Clock'. The 'Call Routing' tab shows a 'Trusted IPv4/IPv6 Prefix List' with the value '2.16.110.67/32,172.16.110.68/32,172.16.110.69/32' and a 'Source Interface' dropdown set to 'sourceInterfa_source_intf'. The 'Network Clock' tab shows a 'Participation' toggle and 'Clock Priority Sorting' options. The top of the page shows the 'Catalyst SD-WAN' header and user information.

SIP Gateway Configuration in SD-WAN Manager

Add Translation Profile for Calling Number Translation

Calling
Number
Translation

The screenshot shows the Cisco Catalyst SD-WAN Manager interface. The top navigation bar includes the Cisco logo, the title 'Catalyst SD-WAN', and user information. The left sidebar contains navigation icons for Monitor, Configuration, Analytics, Workflows, Tools, Reports, Maintenance, Administration, and Explore. The main content area is titled 'Configuration Groups (DC1_WE1_Config_Group_1a - UC Voice Profile)' and displays the configuration for 'UC_SIP_TDM_GW'. The configuration details include a description 'UC SIP TDM Gateway', a type of 'Dual Device', and a table with columns for Device Name, Device Solution, Modified By, Last Updated, and Shared. The table shows two devices, DC1_WE1 and DC1_WE2, both using the SD-WAN device solution. Below the table, there is a search bar and a 'Digital Interface' section with a dropdown menu showing 'T1_PRI_0/2/0_0/2/1'. A 'Back' button indicates 'No Changes Made'. On the right, an 'Add Feature' panel lists various features, with 'Translation Profile' highlighted.

Type	Device Name	DEVICE SOLUTION	MODIFIED BY	LAST UPDATED	SHARED
Dual Device	DC1_WE1, DC1_WE2	SD-WAN	Admin	Jan 26, 2025 11:56:09	1 Group

Digital Interface

T1_PRI_0/2/0_0/2/1

Back No Changes Made

Add Feature

- << DSP Farm
- << Analog Interface
- << Voice Global
- << Translation Profile
- << SRST Phone
- << Call Routing
- << Digital Interface

SIP Gateway Configuration in SD-WAN Manager

Add Translation Rule

Calling
Number
Translation

The screenshot displays the Cisco Catalyst SD-WAN Manager interface. The top navigation bar includes the Cisco logo, the title "Catalyst SD-WAN", and user information for "admin". The left sidebar contains navigation icons for Monitor, Configuration, Analytics, Workflows, Tools, Reports, Maintenance, Administration, and Explore. The main content area is titled "Configuration Groups (DC1_WE1_Config_Group_1a - UC Voice Profile)" and shows the configuration for "UC_SIP_TDM_GW". Below the title, there is a description "UC SIP TDM Gateway" and a table with configuration details: Type (Dual Device), Device Name (DC1_WE1, DC1_WE2), DEVICE SOLUTION (SD-WAN), MODIFIED BY (Admin), LAST UPDATED (Jan 26, 2025 11:56:09), and SHARED (1 Group). A search bar is present above the configuration sections. The "Translation Profile" section has a dropdown menu labeled "Select Translation Profile" and a "+" button. The "Digital Interface" section has a dropdown menu labeled "T1_PRI_0/2/0_0/2/1" and a "+" button. At the bottom left, there is a "Back" button and the text "No Changes Made". On the right side, there is a "Add Feature" dialog box with a "+" button and the text "Translation Rule".

SIP Gateway Configuration in SD-WAN Manager

Define Translation Rule Set

Calling
Number
Translation

The screenshot shows the 'Translation Rule' configuration page in the Cisco SD-WAN Manager. The page is titled 'Catalyst SD-WAN' and shows a sidebar with navigation options like Monitor, Configuration, Analytics, Workflows, Tools, Reports, Maintenance, and Administration. The main content area is titled 'Translation Rule' and includes fields for Name, Description, and Basic Settings. A table at the bottom lists the translation rules.

Translation Rule

Name: Calling_DN_Translation_Rule_1

Description (optional): Calling Translation Rule for Main DN

Basic Settings

Translation rule set number: 1

Rule number	Matching pattern	Action	Replacement pattern	Action
1	/919392..../	Replace	/9193926000/	

SIP Gateway Configuration in SD-WAN Manager

Associate Translation Rule to Translation Profile

Calling
Number
Translation

Translation Profile

Name: Calling_DN_Translation_Profile_1

Description (optional): Calling DN Translation Profile

Basic Settings

Name (optional): Calling_DN_Translation_Profile_1

Translation Profile configuration

+ Add Translation Profile configuration

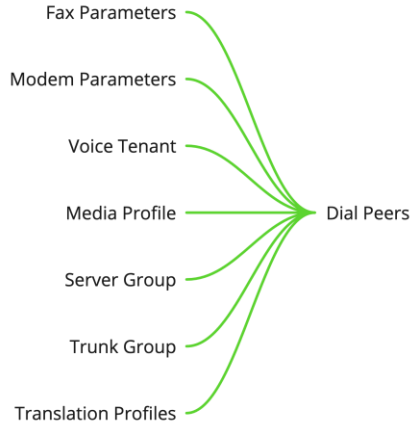
Select Call type	Select Translation rule	Action
calling	Calling_DN_Translation_Rule_1	View Rule

[Back](#) All Changes Saved

[Cancel](#) [Save](#)

SIP Gateway Configuration in SD-WAN Manager

Call Routing :: Dial Peers



The screenshot displays the Cisco SD-WAN Manager interface for configuring a SIP Gateway. The main configuration area is titled 'Call Routing' and is part of the 'UC_SIP_TDM_GW' configuration group. The 'Dial Peer' tab is selected, showing a table of dial peers. The table has columns for Tag, Dial Peer Type, Direction, Description, Number Pattern, Forward Digits Type, Transport Protocol, and Action. Four dial peers are listed: 401 (pots, incoming), 402 (pots, outgoing), 301 (sip, incoming), and 302 (sip, outgoing). The 'Add Dial Peer' button is highlighted with a blue box.

Call Routing Configuration Details:

- Name:** SIP_TDM_GW_Call_Routing
- Description (optional):** Dial-peers for SIP TDM Gateway
- Dial Peer Tag Prefix:** <system default>
- Description:** <system default>
- Voice Module Location Parcel Name:** <system default>

Tag	Dial Peer Type	Direction	Description	Number Pattern	Forward Digits Type	Transport Prot	Action
401	pots	incoming	Inbound calls from PSTN	{{SIP_TDM_GW_Inbound_PSTN_DN_Pattern}}	N/A	N/A	...
402	pots	outgoing	Outbound calls to PSTN	{{SIP_TDM_GW_Outbound_PSTN_DN_Pattern}}	all	N/A	...
301	sip	incoming	Inbound calls from CUCM	{{SIP_TDM_GW_Inbound_SIP_DN_Pattern}}	N/A	tcp	...
302	sip	outgoing	Outbound calls to CUCM	{{SIP_TDM_GW_Outbound_SIP_DN_Pattern}}	N/A	tcp	...

SIP Gateway Configuration in SD-WAN Manager

Dial-Peer :: Translation profile, Media profile, Voice Tenant Association

Fax Parameters

Modem Parameters

Voice Tenant

Media Profile

Server Group

Trunk Group

Translation Profiles

Dial Peers

UC_SIP_TDM_GW

Description: UC Voice Profile for SIP

Type: Dual Device

Device Name: DC1_WE1, DC1_WE2

Translation Profile

Calling_DN_Translation_Profile_1

Digital Interface

T1_PRI_0/2/0_0/2/1

Call Routing

SIP_TDM_GW_Call_Routing

Media Profile / SIP_TDM_GW_Media_Profile_1

Translation Profile / Calling_DN_Translation_Profile_1

Voice Global

SIP_TDM_GW_Global_UC_Config

Back All Changes Saved

Cancel Save

Call Routing

Name: SIP_TDM_GW_Call_Routing

Description (optional): Dial-peers for SIP TDM Gateway

Dial Peer Fax Modem Association

Dial Peer Range	Primary Protocol	Fallback Protocol	Low Speed Redundancy	High Speed Redundancy	Action
No data available					

Modem

+ Add Modem Pass-through

Dial Peer Range	Protocol	Action
No data available		

Association

+ Add Association

Dial Peer Range (optional)	Voice Tenant (optional)	Media Profile Name	Server Group (optional)	Action
301-302	<system default>	SIP_TDM_GW_Media_Profile_1	<system default>	
402	<system default>	<system default>	<system default>	

SIP Gateway Configuration in SD-WAN Manager

Dial-Peer :: Translation profile, Media profile, Voice Tenant Association

Fax Parameters

Modem Parameters

Voice Tenant

Media Profile

Server Group

Trunk Group

Translation Profiles

Dial Peers

UC_SIP_TDM_GW

Description: UC Voice Profile for SIP

Type: Dual Device

Device Name: DC1_WE1, DC1_WE2

Translation Profile

Calling_DN_Translation_Profile_1

Translation Rule / Calling

Digital Interface

T1_PRI_0/2/0_0/2/1

Call Routing

SIP_TDM_GW_Call_Routing

Media Profile / SIP_TDM

Translation Profile / Calling

Voice Global

SIP_TDM_GW_Global_UC_Config

Back

No Changes Made

Call Routing

Name: SIP_TDM_GW_Call_Routing

Description (optional): Dial-peers for SIP TDM Gateway

Dial Peer: Fax, Modem, Association

Modem

+ Add Modem Pass-through

Dial Peer Range

Protocol

Action

Association

+ Add Association

Trunk Group Priority

default>

<system default>

default>

<system default>

Translation Profile

<system default>

Calling_DN_Translation_Profile

Translation Profile Direction

<system default>

outgoing

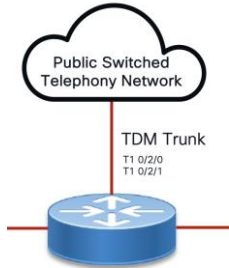
Action

Cancel

Save

SIP Gateway Configuration in SD-WAN Manager

Define Device Specific Voice Parameters and Deploy!



Cisco Catalyst SD-WAN

Deploy Configuration Group

Monitor

Configuration

Analytics

Workflows

Tools

Reports

Maintenance

Administration

Explore

Add and Review Device Configuration

We've autogenerated minimal configurations to make it easier for you to bring up your devices. Modify them as needed and directly edit the table to add System IP and Site IDs.

Search

Global

SITE_120001

UC_SIP_TDM_GW (uc-voice) 8 Optional

SIP_TDM_GW_Global_UC_Config (voice-global)

sourceInterfa_source_intf...

GigabitEthernet0/0/3

SIP_TDM_GW_Call_Routing (call-routing)

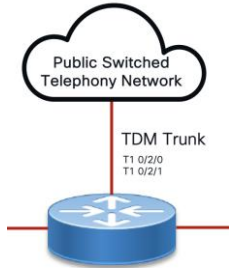
SIP_TDM_GW_Inbound_PSTN_D...	SIP_TDM_GW_Ca_1_voice_por...	SIP_TDM_GW_Outbound_PSTN_...
91943808..	0/2/0:23	9.....
SIP_TDM_GW_Ca_2_voice_por...	SIP_TDM_GW_Inbound_SIP_DN...	SIP_TDM_GW_Outbound_SIP_D...
0/2/0:23	1222.....	91943808..
SIP_TDM_GW_Ca_4_voice_des...		
ipv4:172.16.110.67		

Cancel

Review Back Next

SIP Gateway Configuration in SD-WAN Manager

Deployment Successful!



Catalyst SD-WAN

← Back

Deploy configuration group to associated devices | Validation success

Initiated By: admin From: 173.38.117.80

Total Task: 1 | Success : 1

Device Group (1)

Search Table

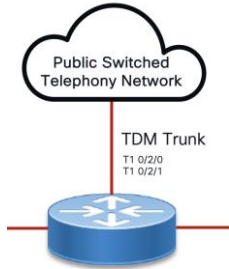
Status	Chassis Number	Hostname	Message	Start Time	System IP	Action
Success	C8300-2N2S-6T-FLM	DC1_WE2	Config Group successfully deployed to ...	Jan 26, 2025, 3:34:12 PM	192.168.1.154	

As of: Jan 26, 2025 03:34 PM

Items per page: 10 1 - 1 of 1

Feedback

T1 PRI Trunk is UP and Functional!



```
DC1_WE2# show isdn status
```

```
Global ISDN Switchtype = primary-ni
```

```
ISDN Serial0/2/0:23 interface
```

```
    dsl 0, interface ISDN Switchtype = primary-ni
```

```
Layer 1 Status:
```

```
    ACTIVE
```

```
Layer 2 Status:
```

```
    TEI = 0, Ces = 1, SAPI = 0, State = MULTIPLE_FRAME_ESTABLISHED
```

```
Layer 3 Status:
```

```
    0 Active Layer 3 Call(s)
```

```
Active dsl 0 CCBs = 0
```

```
The Free Channel Mask: 0x807FFFFFFF
```

```
Number of L2 Discards = 0, L2 Session ID = 4
```

```
.....
```

```
DC1_WE2#
```

CUCM IP Phone User → PSTN User

BR1-WE1# show call active voice brief

```
0      : 32209 2037291360ms.1 (11:59:01.188 EDT Fri May 31 2024) +1250 [pid:301] Answer [9193922001] active
dur 00:00:14 tx:0/0 rx:0/0 dscp:0 media:0 audio tos:0xB8 video tos:0x0
IP 172.16.109.11:28138 SRTP: off rtt:0ms pl:0/0ms lost:0/0/0 delay:0/0/0ms g711ulaw TextRelay: off Transcoded: No ICE: Off
media inactive detected:n media contrl rcvd:n/a timestamp:n/a
long duration call detected:n long duration call duration:n/a timestamp:n/a
LostPacketRate:0.00 OutOfOrderRate:0.00
LocalUUID:3a856cf900105000a0000c15bd110c90
RemoteUUID:2c309b6475d75ab7ab030081ab95e075
VRF: 1
0      : 32210 2037291390ms.1 (11:59:01.212 EDT Fri May 31 2024) +1220 [pid:402] Originate [12223334444] active
dur 00:00:14 tx:190/0 rx:712/0 dscp:0 media:0 audio tos:0x0 video tos:0x0
Tele 0/2/0:23 (32210) [0/2/0.23] tx:14340/3850/0ms g711ulaw noise:-54 acom:3 i/0:-51/-57 dBm
```

Is there a problem?

CUCM IP Phone User → PSTN User

BR1-WE1# **show call active voice brief**

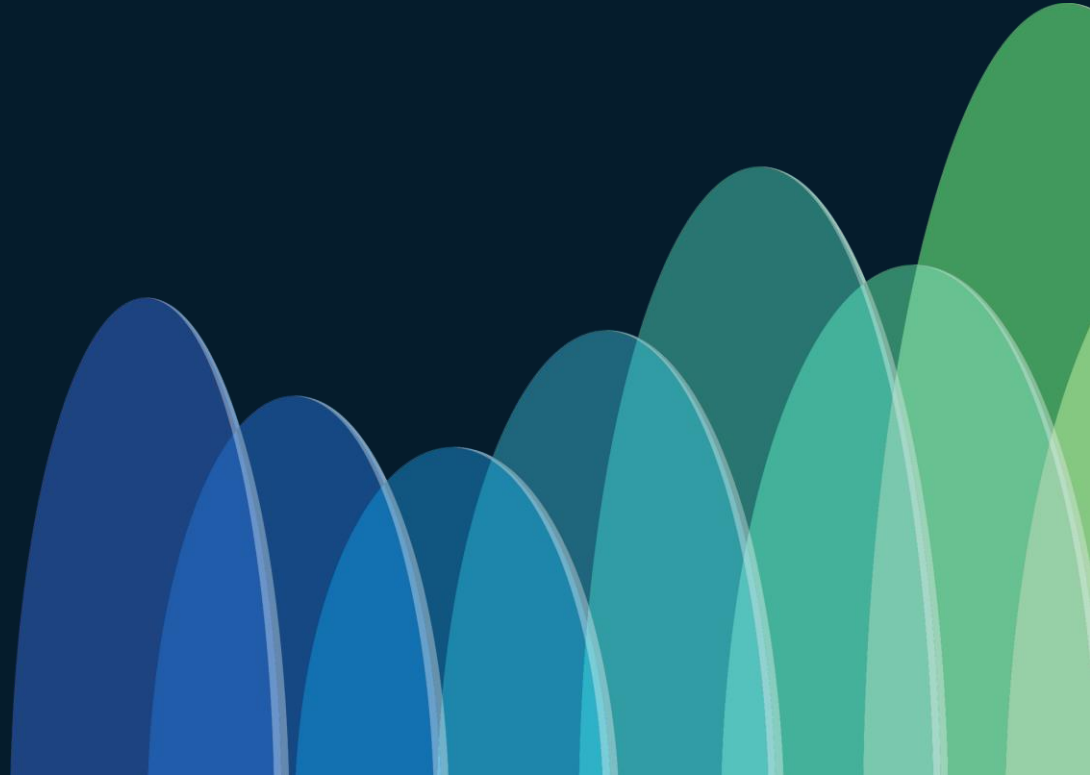
```
0      : 32227 2038269980ms.1 (12:15:19.811 EDT Fri May 31 2024) +1530 [pid:301] Answer [9193922001] active
dur 00:00:25 [tx:745/149000 rx:1263/252600] dscp:0 media:0 audio tos:0xB8 video tos:0x0
IP 172.16.109.11:24786 SRTP: off rtt:0ms pl:0/0ms lost:0/0/0 delay:0/0/0ms g711ulaw TextRelay: off Transcoded: No ICE: Off
media inactive detected:n media contrl rcvd:n/a timestamp:n/a
long duration call detected:n long duration call duration:n/a timestamp:n/a
LostPacketRate:0.00 OutOfOrderRate:0.00
LocalUUID:506a7c2300105000a000c75bd110c90
RemoteUUID:a401e82aa7725b4883dd9d3de4ab9aaf
VRF: 1

0      : 32228 2038270010ms.1 (12:15:19.837 EDT Fri May 31 2024) +1500 [pid:402] Originate [12223334444] active
dur 00:00:25 tx:760/0 rx:1277/0 dscp:0 media:0 audio tos:0x0 video tos:0x0
Tele 0/2/0:23 (32228) [0/2/0:23] tx:25650/15250/0ms g711ulaw noise:-52 acom:11 i/0:-49/-55 dBm
```

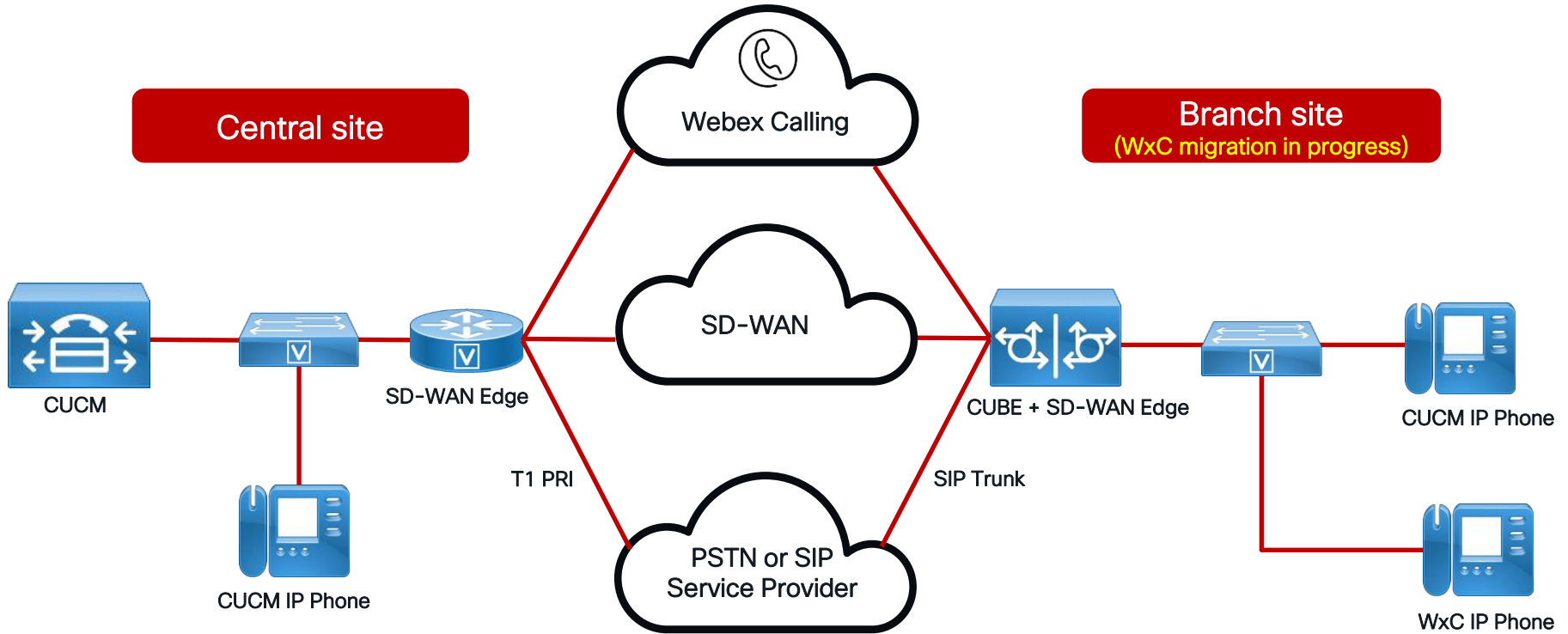
After adding a **CLI Add-On Template** with:
voice service voip
media-statistics

Webex Calling Local Gateway

Calling via SIP Trunks

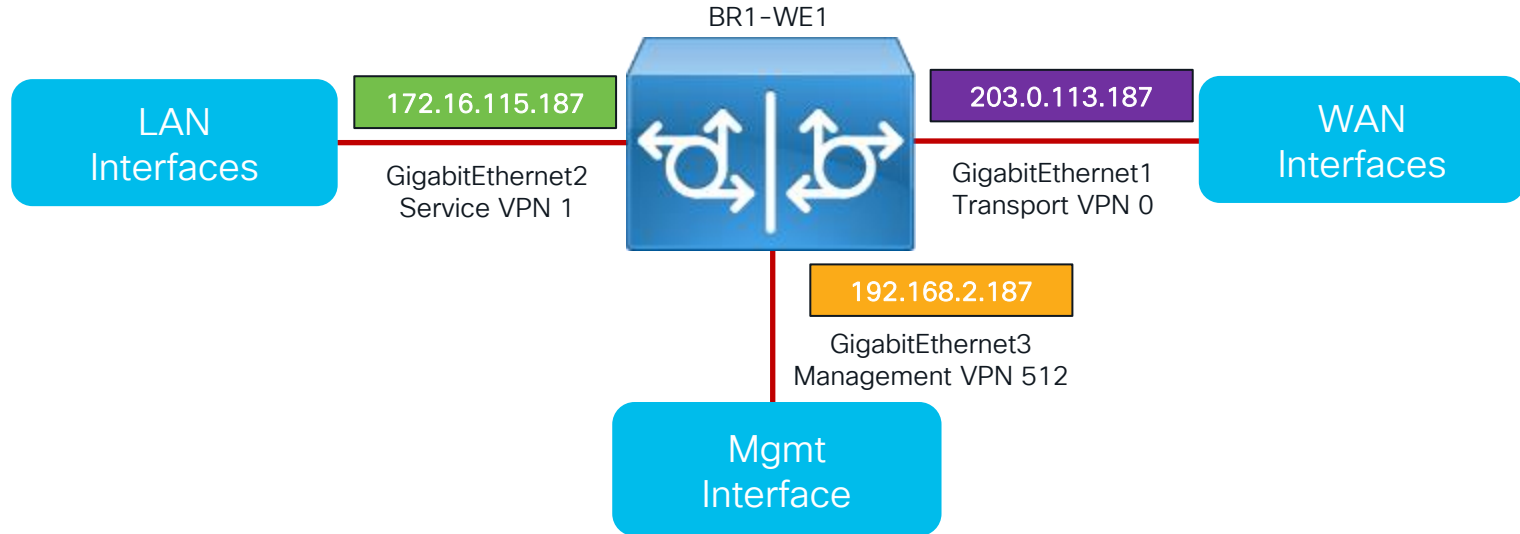


Webex Calling: Local Gateway at Branch Site



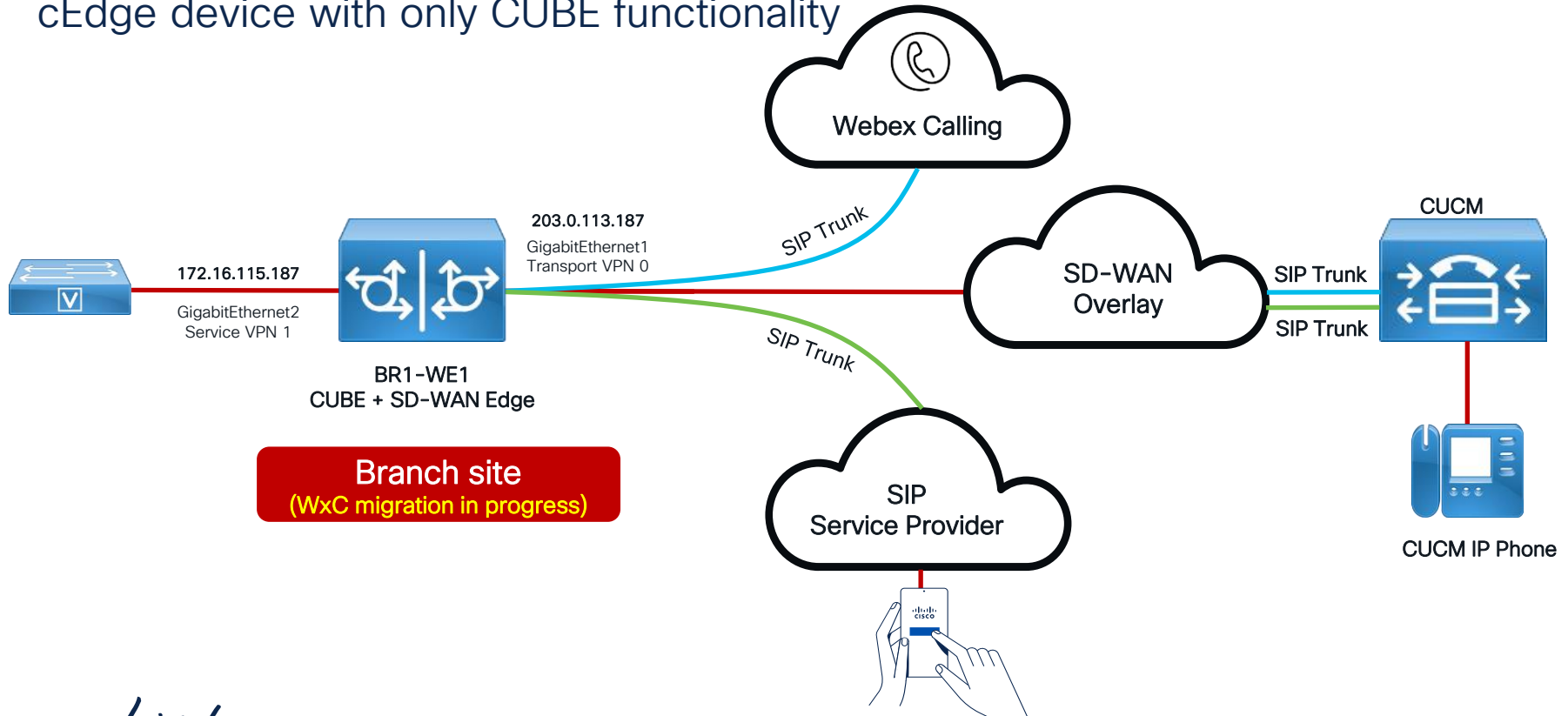
Catalyst 8000v as Session Border Controller

cEdge device with only CUBE functionality



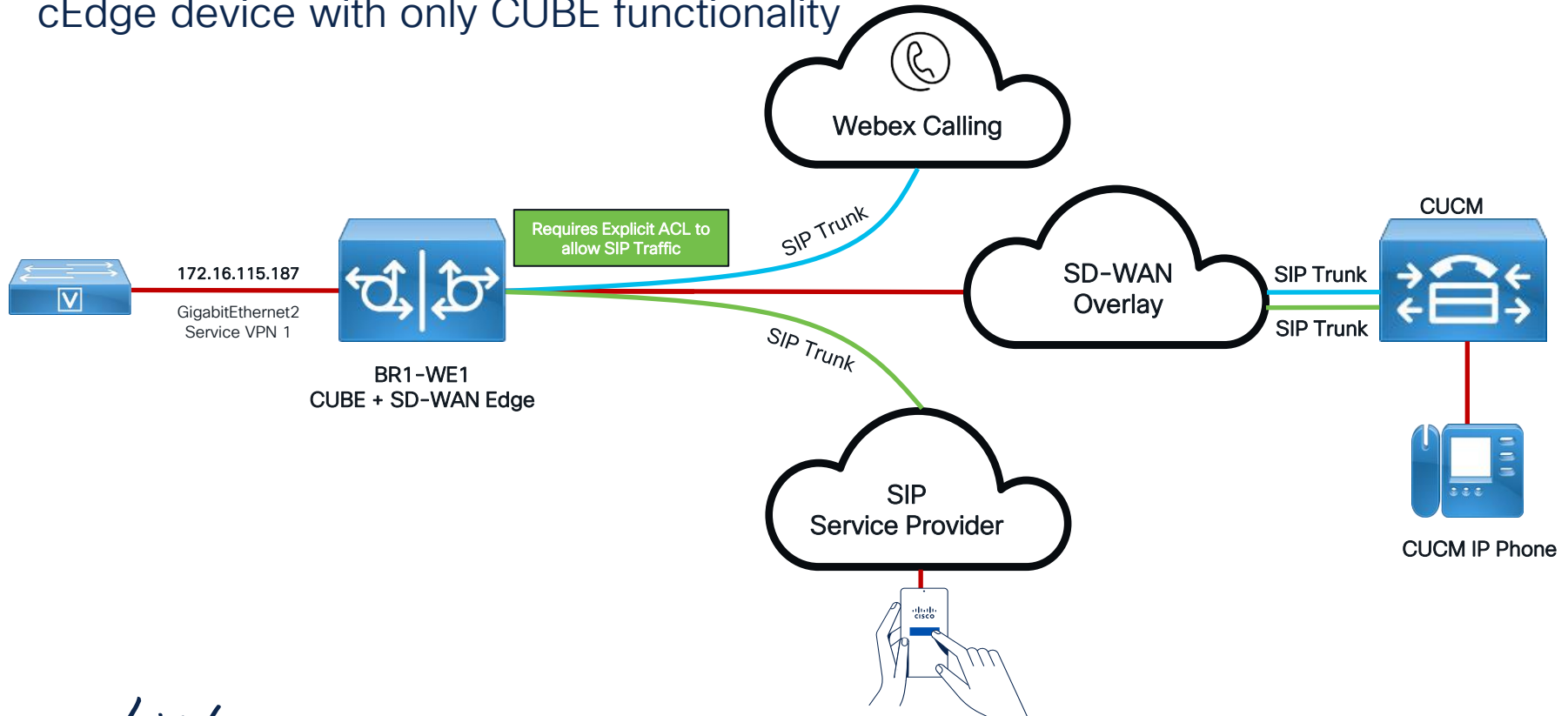
Local Gateway: Network View

cEdge device with only CUBE functionality



Local Gateway: Network View

cEdge device with only CUBE functionality



Create ACL To Allow Inbound SIP Traffic

Allow SIP Traffic from Webex Calling and SIP Provider

The screenshot displays the Cisco Catalyst SD-WAN configuration interface. The main configuration area is titled "BR1_WE_Config_Group_1_WAN" with a description "BR1_WE_Config_Group_1 WAN Transport Profile". It shows the configuration for a "Transport VPN" (VPN0) and a "Management VPN" (VPN512). The "Transport VPN" section is expanded, showing the "Ethernet Interface" set to "Internet". The "Management VPN" section is also expanded, showing the "Management Ethernet Interface" set to "Management".

On the right side, the "Add New Feature" panel is open, showing a list of features to add. The "ACL IPv4" option is highlighted with a red box, indicating it is the selected feature to be added to the configuration.

At the bottom left, there is a "Back" button and a status message "No Changes Made".

Create ACL To Allow Inbound SIP Traffic

Allow SIP Traffic from Webex Calling and SIP Provider

The screenshot shows the Cisco Catalyst SD-WAN configuration interface. The main configuration area is titled 'ACL IPv4'. The 'Name' field is 'LGW_Allow_SIP_Traffic' and the 'Description' is 'Allow SIP and SIPS Traffic'. Below the description, there are links for 'Add ACL Sequence' and 'Import ACL'. The 'Ordered Sequences' section lists three sequences: '1 LGW_SIP_ACL_1', '2 LGW_SIP_ACL_2', and '3 LGW_SIP_ACL_3'. A callout box on the left points to these sequences with the text '3 Conditions and Actions'. The 'Sequence Rules (1)' section shows a table with columns for 'ORDER', 'CONDITION', and 'ACTION'. The table contains one rule with 'ORDER' 1, 'CONDITION' 'Source Data Prefix List : WxC_IPv4_List', and 'ACTION' 'Accept'. The rule is 'Enabled'.

ORDER	CONDITION	ACTION
1	Source Data Prefix List : WxC_IPv4_List	Accept

Create ACL To Allow Inbound SIP Traffic

Add ACL Sequence for Webex Calling

The screenshot displays the Cisco Catalyst SD-WAN configuration page for a configuration group named **BR1_WE_Config_Group**. The left sidebar shows navigation options: Monitor, Configuration, Analytics, Workflows, Tools, and Reports. The main content area is titled **Configuration Groups (BR1_WE_Config_Group)** and shows the **DEVICE SOLUTION SD-WAN** tab. The **Transport VPN** section is expanded, showing **VPN0** and **Ethernet Interface**. The **ACL Sequence Name** is **LGW_SIP_ACL_1**. The **Condition** section shows **Order** 1, **Source Data Prefix** (selected), and **Source Port** (selected). The **Action Type** is **Accept**. The **Source Data Prefix List** section shows **Source Data Prefix List Name** as **WxC_IPv4_List** and **Source IP Prefix** as **5062**. The **Source Port** section shows **Value*** as **5062**. The **Enable Variables** toggle is off. The **Back** button is labeled **No Changes Made**. The **Cancel** and **Save** buttons are at the bottom right.

Name
LGW-Allow_SIP_Traffic

Description(optional)
Allow SIP and SIPS Traffic

ACL Sequence Name
LGW_SIP_ACL_1

Order
1

Condition
Source Data Prefix
Source Port

Action Type
Accept

Accept Actions

Source Data Prefix List
Source Data Prefix List Name
WxC_IPv4_List

Source IP Prefix
5062

Enable Variables

Source Port
Value*
5062

Back No Changes Made

Cancel **Save**

Match: SIP Traffic from
WxC Cloud with
source port 5062

Action: Accept

Create ACL To Allow Inbound SIP Traffic

Add ACL Sequence for Webex Calling

The screenshot displays the Cisco Catalyst SD-WAN configuration page for a configuration group named **BR1_WE_Config_G**. The left sidebar shows navigation options: Monitor, Configuration, Analytics, Workflows, Tools, and Reports. The main content area is titled **BR1_WE_Config_G** and includes a description: **BR1_WE_Config_Group_2**. Below this, there are tabs for **DEVICE SOLUTION SD-WAN** and **MODIFIED BY Admin**. The **Transport VPN** section is active, showing a list of VPNs with **VPN0** selected. The **Ethernet Interface** section is also visible, showing **Internet** and **Management Ethernet In** interfaces. The **ACL Sequence Name** is **LGW_SIP_ACL_2**. The **Condition** section shows **Order** 1, **Source Data Prefix** (selected), and **Source Port** (selected). The **Action Type** is **Accept**. The **Source Data Prefix List** section shows **Source Data Prefix List Name** as **WxC_IPv4_List** and **Source IP Prefix** as **8934**. The **Source Port** section shows **Value*** as **8934**. The **Enable Variables** toggle is off. The **Accept Actions** dropdown is set to **Accept**. The **Back** button is labeled **No Changes Made**. The **Cancel** and **Save** buttons are at the bottom right.

Name
LGW-Allow_SIP_Traffic

Description(optional)
Allow SIP and SIPS Traffic

ACL Sequence Name
LGW_SIP_ACL_2

Order
1

Condition
Source Data Prefix X
Source Port X

Action Type
Accept
Accept Actions

Source Data Prefix List
Source Data Prefix List Name
WxC_IPv4_List
Source IP Prefix

Source Port
Value*
8934

Enable Variables
Enable Variables

Accept Actions
Accept

Back No Changes Made

Cancel **Save**

Match: SIP Traffic from
WxC Cloud with
source port 8934

Action: Accept

Create ACL To Allow Inbound SIP Traffic

Add ACL Sequence for SIP Provider

The screenshot displays the Cisco Catalyst SD-WAN configuration page for a configuration group named **BR1_WE_Config_Group**. The left sidebar shows navigation options: Monitor, Configuration, Analytics, Workflows, Tools, and Reports. The main content area is titled **Configuration Groups (BR1_WE_Config_Group)** and shows the **DEVICE SOLUTION SD-WAN** tab. The **ACL Sequence Name** is **LGW_SIP_ACL_3**. The **Condition** section shows **Order** 1, **Source Data Prefix**, and **Source Port**. The **Action Type** is **Accept**. The **Source Data Prefix List** section shows **Source Data Prefix List Name** as **SIP_PSTN_IPv4_List** and **Source IP Prefix**. The **Source Port** section shows **Value** as **5060**. The **Accept Actions** section shows **Accept**. The **Enable Variables** toggle is off. The **Source Port** section shows **Value** as **5060**. The **Accept Actions** section shows **Accept**. The **Back** button is labeled **No Changes Made**. The **Cancel** and **Save** buttons are at the bottom right.

Name
LGW-Allow_SIP_Traffic

Description(optional)
Allow SIP and SIPs Traffic

ACL Sequence Name
LGW_SIP_ACL_3

Order
1

Condition
Source Data Prefix
Source Port

Action Type
Accept

Accept Actions
Accept Actions

Source Data Prefix List
Source Data Prefix List Name
SIP_PSTN_IPv4_List

Source IP Prefix

Enable Variables

Source Port
Value
5060

Accept Actions
Accept

Back No Changes Made

Cancel **Save**

Match: SIP Traffic from
SIP Provider with
source port 5080

Action: Accept

Apply ACL to Transport Interface

The screenshot displays the Cisco Catalyst SD-WAN configuration interface. The left sidebar shows the navigation menu with options like Monitor, Configuration, Analytics, Workflows, Tools, Reports, Maintenance, Administration, and Explore. The main content area is titled 'Ethernet Interface' and shows the configuration for the 'Internet' interface. The 'ACL/QoS' tab is selected, and the 'ACL' section is highlighted with a red box, showing 'ACL IPv4 Ingress' and 'LGW_Allow_SIP_Traffic'. The 'QoS' section is also visible, showing 'Adaptive QoS' and 'Shaping Rate (Kbps)'. The 'Save' button is at the bottom right.

Deploy Configuration Group Changes

Deploy Configuration Group

Select Devices to Deploy

Select the devices below to deploy

Search

Global (1 of 7 nodes)

SITE_121001

Edge Devices (1)

Search Table

1 selected

Chassis Numbers	Site Name	Hostname	System IP	Tags	Last Configured
C8K-16F0E5D6-C5D5-0A3D-DE4E-F591584CC8B5	SITE_121001	BR1-WE1	192.168.2.246	None	Jan 29, 2025, 12:22:24 AM

Items per page: 25 1 - 1 of 1

Cancel Next

Localized Policy Deployed to Device!

Catalyst SD-WAN

← Back

Deploy configuration group to associated devices | ● Validation success

Total Task: 1 | Success : 1

Device Group (1)

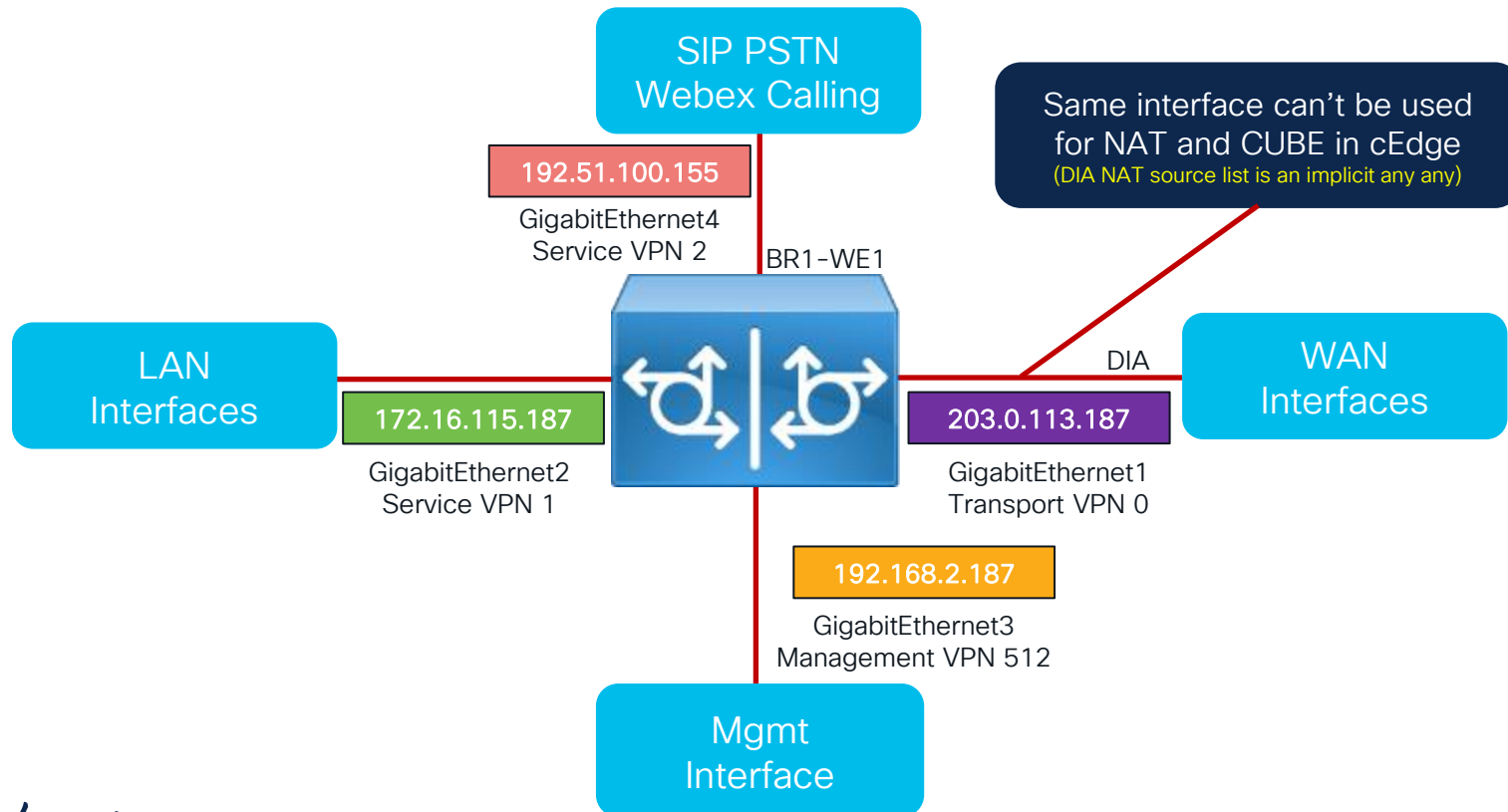
Q Search Table

Status	Chassis Number	Hostname	Message	Start Time
● Success	C8K-16F0E5D6-C5D5-0A3D-DE4E-F591584CC8B5	BR1-WE1	Config Group successfully deployed to ...	Jan 29, 2025, 12:27:4

```
sdwan
interface GigabitEthernet1
  tunnel-interface
  encapsulation ipsec weight 1
  no border
  color biz-internet
  no last-resort-circuit
  no low-bandwidth-link
  no vbond-as-stun-server
  vmanage-connection-preference 5
  port-hop
  carrier
  nat-refresh-interval          default
                                5
  hello-interval                1000
  hello-tolerance               12
  no allow-service all
  no allow-service bgp
  allow-service dhcp
  allow-service dns
  allow-service icmp
  no allow-service sshd
  no allow-service netconf
  allow-service ntp
  no allow-service ospf
  no allow-service stun
  allow-service https
  no allow-service snmp
  no allow-service bfd
exit
access-list LGW-Allow-SIP-Traffic in
```

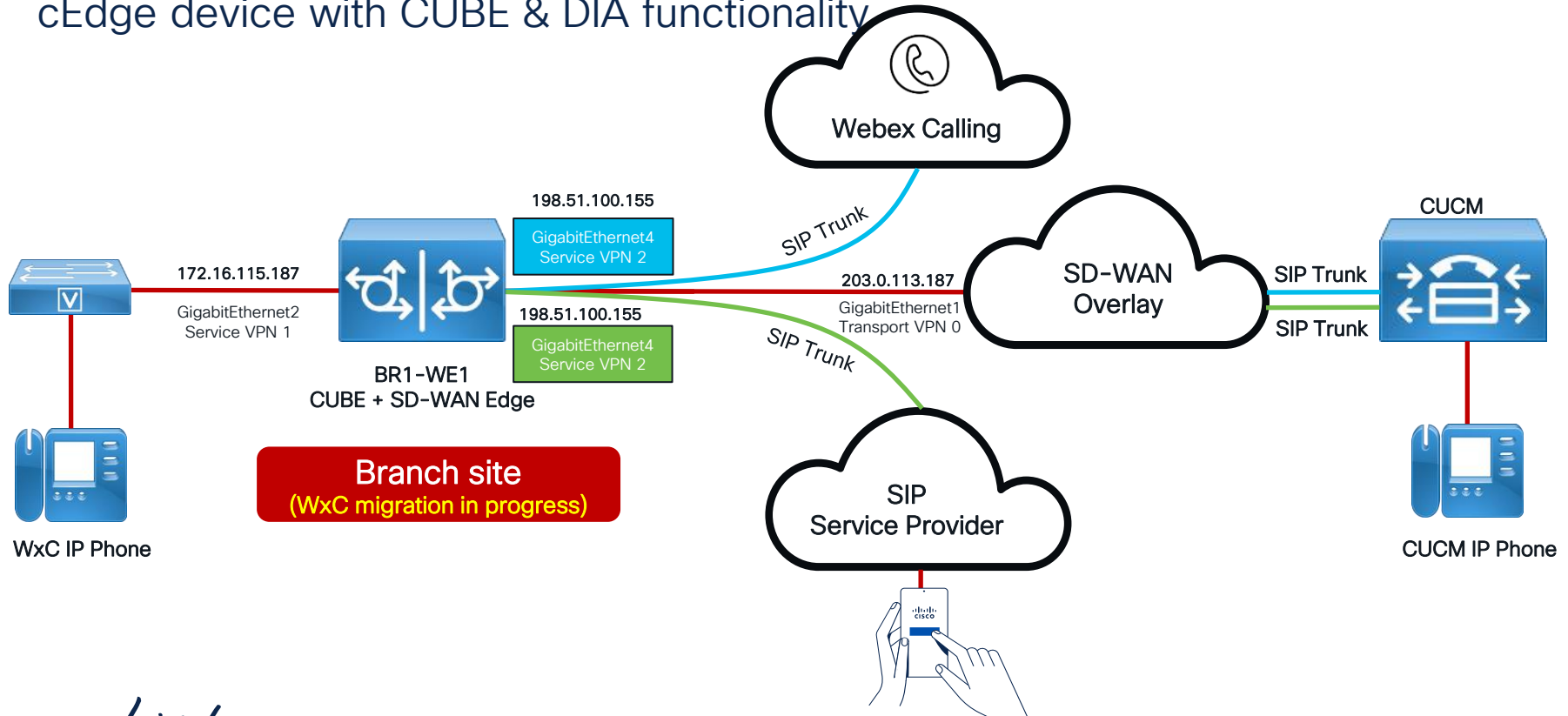
Catalyst 8000v as Session Border Controller

cEdge device with CUBE & DIA functionality



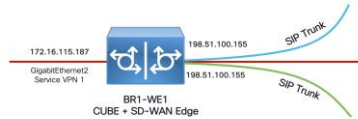
Local Gateway: Network View

cEdge device with CUBE & DIA functionality



Local Gateway Configuration in SD-WAN Manager

Create a CLI Add-On Profile



Configuration Groups SD-WAN

← **Configuration Groups** 3 System Profile 4 Transport & Management Profile 4 Policy Profile 1 Service Profile 4 CLI Add-on Profile 1 UC Voice Profile 2 0 →

Search Last Updated Status Create Configuration Group Export Import

Name	Type	Profile	Provisioning Status	Origin	Updated By	Last Updated On	Actions
BR1_WE_Config_Group_1	cEdge devices in Branch sites		In Sync Devices / Associated Devices		Updated by admin	Updated Jan 29, 2025	

Type: Single Router

System Profile

BR1_WE_Config_Group_1_Basic

Transport & Management Profile

BR1_WE_Config_Group_1_WAN

Policy Profile (optional)

Default_Policy_Object_Profile

Service Profile (optional)

BR1_WE_Config_Group_1_LAN

Deployment

Associated 1 device

Provisioning 0 out of sync

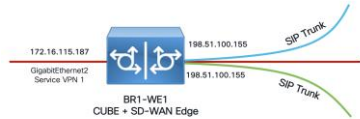
Deploy

+ Add Profile

- CLI Add-on Profile
- Other Profile
- UC Voice Profile

Local Gateway Configuration in SD-WAN Manager

CLI Add-On Template references Config variables



Catalyst SD-WAN

Configuration Groups (BR1_WE_Config_Group_1 - CLI Add-on Profile)

BR1_WE_Config_Group_1_UC_CUBE

Description: BR1_WE_Config_Group_1_UC_CUBE

DEVICE SOLUTION	MODIFIED BY	LAST UPDATED	SHARED
SD-WAN	Admin	Jan 31, 2025 04:50:43	1 Group

Search CLI + Create Variable Encrypt Type6 Import Config File Delete

Config Preview

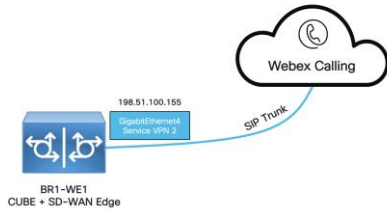
```
1 voice service voip
2 ip address trusted list
3   ipv4 {{sip_pstn_trusted_ipv4_addr_1}} /{{(sip_pstn_trusted_ipv4_addr_1.subnet_mask_bits)}}
4
5   ipv4 {{(wxc_trusted_ipv4_addr_1)}} /{{(wxc_trusted_ipv4_addr_1.subnet_mask_bits)}}
6   ipv4 {{(wxc_trusted_ipv4_addr_2)}} /{{(wxc_trusted_ipv4_addr_2.subnet_mask_bits)}}
7   ipv4 {{(wxc_trusted_ipv4_addr_3)}} /{{(wxc_trusted_ipv4_addr_3.subnet_mask_bits)}}
8   ipv4 {{(wxc_trusted_ipv4_addr_4)}} /{{(wxc_trusted_ipv4_addr_4.subnet_mask_bits)}}
9   ipv4 {{(wxc_trusted_ipv4_addr_5)}} /{{(wxc_trusted_ipv4_addr_5.subnet_mask_bits)}}
10  ipv4 {{(wxc_trusted_ipv4_addr_6)}} /{{(wxc_trusted_ipv4_addr_6.subnet_mask_bits)}}
11  ipv4 {{(wxc_trusted_ipv4_addr_7)}} /{{(wxc_trusted_ipv4_addr_7.subnet_mask_bits)}}
12  ipv4 {{(wxc_trusted_ipv4_addr_8)}} /{{(wxc_trusted_ipv4_addr_8.subnet_mask_bits)}}
13  ipv4 {{(wxc_trusted_ipv4_addr_9)}} /{{(wxc_trusted_ipv4_addr_9.subnet_mask_bits)}}
14  ipv4 {{(wxc_trusted_ipv4_addr_10)}} /{{(wxc_trusted_ipv4_addr_10.subnet_mask_bits)}}
15  ipv4 {{(wxc_trusted_ipv4_addr_11)}} /{{(wxc_trusted_ipv4_addr_11.subnet_mask_bits)}}
16  ipv4 {{(wxc_trusted_ipv4_addr_12)}} /{{(wxc_trusted_ipv4_addr_12.subnet_mask_bits)}}
17
18  ipv4 {{(wxc_trusted_ipv4_addr_13)}} /{{(wxc_trusted_ipv4_addr_13.subnet_mask_bits)}}
19  ipv4 {{(wxc_trusted_ipv4_addr_14)}} /{{(wxc_trusted_ipv4_addr_14.subnet_mask_bits)}}
20  ipv4 {{(wxc_trusted_ipv4_addr_15)}} /{{(wxc_trusted_ipv4_addr_15.subnet_mask_bits)}}
21  ipv4 {{(wxc_trusted_ipv4_addr_16)}} /{{(wxc_trusted_ipv4_addr_16.subnet_mask_bits)}}
22  ipv4 {{(cucm_trusted_ipv4_addr_1)}} /{{(cucm_trusted_ipv4_addr_1.subnet_mask_bits)}}
23  ipv4 {{(cucm_trusted_ipv4_addr_2)}} /{{(cucm_trusted_ipv4_addr_2.subnet_mask_bits)}}
24  ipv4 {{(cucm_trusted_ipv4_addr_3)}} /{{(cucm_trusted_ipv4_addr_3.subnet_mask_bits)}}
25
26 allow-connections sip to sip
27 media bulk-stats
28 no supplementary-service sip refer
29
```

Configuration variables such as wxc_trusted_ipv4_addr_1

Save Done

Webex Calling SIP Trunk Registration

Three Webex Calling Locations



webex Control Hub

Search

Overview
Alerts center

MONITORING
Analytics
Troubleshooting
Reports

MANAGEMENT
Users
Groups
Locations
Workspaces
Devices
Apps
Account
Security
Organization Settings

SERVICES
Updates & Migrations

Locations

What's a location?

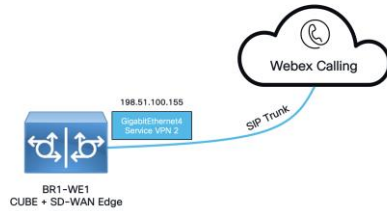
List Map Beta

Search by name or addr... Filter 7 locations Manage location

Location name	Address	Country / Region
BR1	Apex, NC, USA	United States
BR2	San Jose, CA, USA	United States
DC1	7025 Kit Creek Rd, Morrisville, NC 27560, USA	United States

Webex Calling SIP Trunk Registration

BR1 Location has Premises-based PSTN connection using BR1_WE1_LGW1



webex Control Hub

Search

Overview Alerts center

MONITORING

Analytics Troubleshooting Reports

MANAGEMENT

Users Groups Locations Workspaces Devices Apps Account Security Organization Settings

SERVICES

Updates & Migrations

< Locations

BR1
Location ID: 0283519f-dd50-4593-9dbf-8737677b7e6
3 users • 0 workspaces

Actions

Overview Floors **Calling**

Calling connection

PSTN connection Premises-based PSTN Manage

Route to BR1_WE1_LGW1

Main number +19193925556

Emergency calling

Callback number Location main number: +19193925556

Emergency call notification Off

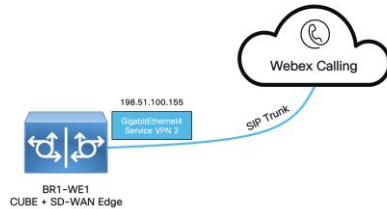
Enhanced emergency calling Off

Dialing

Internal dialing

Webex Calling SIP Trunk Registration

BR1_WE1_LGW1 SIP Trunk



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Calling

Numbers Virtual Lines **Call Routing** Managed Gateways Features PSTN Settings

Trunk Route Group Dial Plans Verify Call Routing Zone Trusted Network Edge

Trunk

SIP trunks provide connectivity to a customer-owned PSTN service and to an on-premises IP PBX deployment. The Gateway configuration page.

Search

Name	Location	Trunk Type	In Use
BR1_WE1_LGW1	BR1	Registration based	Yes
DC1_WE2_LGW2	DC1	Registration based	Yes

BR1_WE1_LGW1

Trunk

Details

Trunk Info Manage

In Use

Calls to On-Premises Extensions 0 Locations

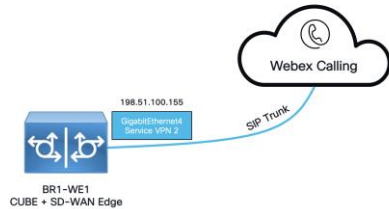
Dial Plans 0 Dial Plans

PSTN Connection 1 Locations

Route Group 1 Route Groups

Webex Calling SIP Trunk Registration

WxC_To_CUCM_RG Route group contains BR1_WE1_LGW1 SIP Trunk



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Calling

Numbers Virtual Lines **Call Routing** Managed Gateways Features PSTN Settings

Trunk Route Group Dial Plans Verify Call Routing Zone Trusted Network Edge

Trunk

SIP trunks provide connectivity to a customer-owned PSTN service and to an on-premises IP PBX deployment. The Gateway configuration page.

Search

Name	Location	Trunk Type	In Use
BR1_WE1_LGW1	BR1	Registration based	Yes
DC1_WE2_LGW2	DC1	Registration based	Yes

BR1_WE1_LGW1

Status
● Online

Trunk Type
Registration based

Device
Cisco CUBE Local Gateway

Registrar Domain
47846594.cisco-bcld.com

Trunk Group OTG/DTG
br1_we1_lgw11570029035_lgw

Line/Port
BR1_WE1_LGW11419173906_LGW 47846594.cisco-bcld.com

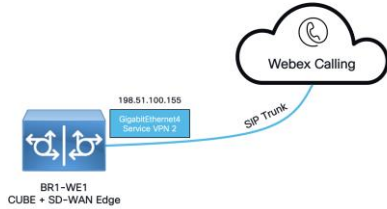
Outbound Proxy Address
jfk10.sipconnect-us.bcld.webex.com

Authentication Information
Retrieve the username and password for **BR1_WE1_LGW1**. Each time authentication information is retrieved, a new password is generated. During the password generation, PSTN is disrupted until the new password is saved.

SIP Line number

Webex Calling SIP Trunk Registration

WxC_CUCM_PSTN_Dialplan uses WxC_To_CUCM_RG Route group



webex Control Hub

Calling

Numbers Virtual Lines **Call Routing** Managed Gateways Features PSTN Set

Trunk Route Group **Dial Plans** Verify Call Routing Zone Trusted Network Edge

Dial Plans
Dial plans allow you to route calls to on-premises extensions via your trunk or route group.

Search dial plan

Name	Routing Choice
WxC_CUCM_PSTN_Dialplan	WxC_To_CUCM_RG

WxC_CUCM_PSTN_Dialplan

Dial Plan

Routing Choice
When internal calls match with a pattern, calls will be routed to the specified trunk or route group.

WxC_To_CUCM_RG

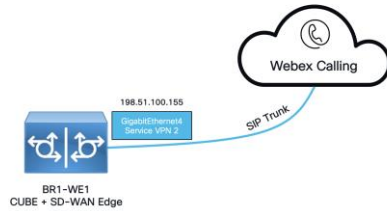
Dial Patterns
Add dial patterns Import CSV

4 dial patterns Delete All

1919XXXXXX	Delete
2XXX	Delete
3XXX	Delete
4XXX	Delete

Webex Calling SIP Trunk Registration

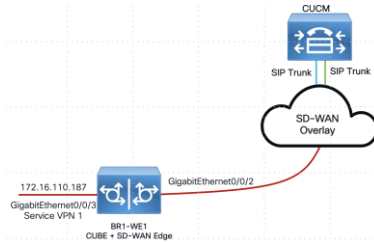
SIP Trunk parameters used in SD-WAN Manager CLI Add-on Template



Variable Name	Description	Value
wxc_trusted_ipv4_addr_1	Trusted List - 1st Network address of Webex Calling	23.89.0.0
wxc_trusted_ipv4_addr_1_subnet_mask_bits	Trusted List - 1st Network mask of Webex Calling	16
...		
wxc_trusted_ipv4_addr_16	Trusted List - 16th Network address of Webex Calling	199.19.199.0
wxc_trusted_ipv4_addr_16_subnet_mask_bits	Trusted List - 16th Network mask of Webex Calling	24
wxc_sip_control_ifname	Name of the Local gateway interface used for SIP control traffic	GigabitEthernet4
wxc_sip_media_ifname	Name of the Local gateway interface used for RTP media traffic	GigabitEthernet4
wxc_sip_registrar_domain	Domain of the SIP Registrar within Webex Calling Cloud	47846594.cisco-bcld.com
wxc_sip_line_number	Line number used in SIP registration	BR1_WE1_LGW11419173906_LGU
wxc_sip_username	Username used in SIP registration and authentication	BR1_WE1_LGW11570029035_LGU
wxc_sip_pwd	Password used in SIP registration and authentication	XXXXXXXX
sip_trustpoint_label	Name of the Trustpoint label in Local Gateway to be used for SIP registration	TP-self-signed-3019638781
wxc_inbound_DN_dest_pattern	Destination pattern that matches the directory number of WxC IP Phones	81.....
wxc_sip_uri_pattern	SIP URI pattern to match calls from WxC	lgw11570029035
wxc_sip_outbound_proxy	Outbound proxy server for calls destined to WxC	jfk10.sipconnect-us.bcld.webex.com
wxc_sip_trunk_group_OTG_DTG	Name of Trunk Group OTG/DTG. This is used in SIP Profiles.	br1_we1_lgw11570029035_lgu
wxc_stun_agent_id	ID of the STUN agent running in Local Gateway	1
wxc_stun_agent_pwd	Shared secret used by the STUN agent	XXXXXXXX

CUCM SIP Trunk Integration

SIP Trunk Security Profile with Listen Port on 5060



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administrator About Logout

System Call Routing Media Resources Advanced Features Device Application User Management Bulk Administration Help

SIP Trunk Security Profile Configuration Related Links: Back To Find/List Go

Save Delete Copy Reset Apply Config Add New

SIP Trunk Security Profile Information

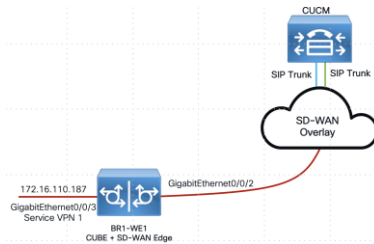
Name * CUBE SIP Trunk To SIP PSTN
Description CUBE SIP Trunk To SIP PSTN
Device Security Mode Non Secure
Incoming Transport Type * TCP+UDP
Outgoing Transport Type TCP
☐ Enable Digest Authentication
Nonce Validity Time (mins) * 600
Secure Certificate Subject or Subject Alternate Name
Incoming Port * 5060
☐ Enable Application level authorization
☐ Accept presence subscription
☐ Accept out-of-dialog refer**
☐ Accept unsolicited notification
☐ Accept replaces header
☐ Transmit security status
☐ Allow charging header
SIP V.150 Outbound SDP Offer Filtering * Use Default Filter

Save Delete Copy Reset Apply Config Add New

SIP Listen port is set to 5060

CUCM SIP Trunk Integration

SIP Trunk #1 (5060): For calls coming in from SIP PSTN via CUBE



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Trunk Configuration

Related Links: Back To Find/List Go

Save Delete Reset Add New

Status
Status: Ready

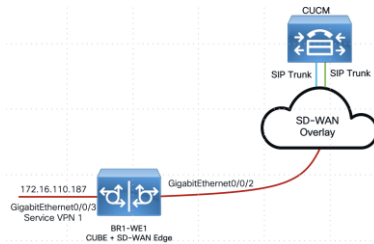
SIP Trunk Status
Service Status: Full Service
Duration: Time In Full Service: 0 day 0 hour 4 minutes

Device Information

Product:	SIP Trunk
Device Protocol:	SIP
Trunk Service Type	None(Default)
Device Name *	BR1 WE1 CUBE SIP Trunk To PSTN
Description	BR1 WE1 CUBE SIP Trunk To PSTN
Device Pool *	Default
Common Device Configuration	< None >
Call Classification *	Use System Default
Media Resource Group List	< None >
Location *	Hub None
AAR Group	< None >
Tunneled Protocol *	None
QSIG Variant *	No Changes
ASN.1 ROSE OID Encoding *	No Changes
Packet Capture Mode *	None
Packet Capture Duration	0

CUCM SIP Trunk Integration

SIP Trunk #1 (5060): For calls coming in from SIP PSTN via CUBE



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Trunk Configuration Related Links: Back To Find/List Go

Save Delete Reset Add New

SIP Information

☐ Destination Address is an SRV

	Destination Address	Destination Address IPv6	Destination Port	Status	Status Reason	Duration
1 *	172.16.115.187		5060	up		4 minutes

MTP Preferred Originating Codec* 711ulaw

BLF Presence Group* Standard Presence group

SIP Trunk Security Profile* CUBE_SIP_Trunk_To_SIP_PSTN

Rerouting Calling Search Space < None >

Out-Of-Dialog Refer Calling Search Space < None >

SUBSCRIBE Calling Search Space < None >

SIP Profile* CUBE_SIP_Trunk_With_OPTIONS_PING View Details

DTMF Signaling Method* No Preference

Normalization Script

Normalization Script < None >

☐ Enable Trace

Parameter Name	Parameter Value
1	

Recording Information

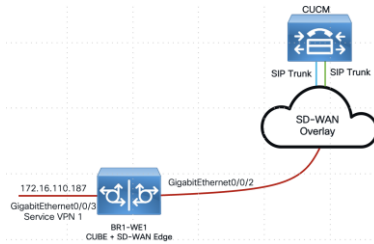
☒ None

☐ This trunk connects to a recording-enabled gateway

☐ This trunk connects to other clusters with recording-enabled gateways

CUCM SIP Trunk Integration

SIP Trunk Security Profile with Listen Port on 5065



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SIP Trunk Security Profile Configuration Related Links: Back To Find/List ▾ Go

Save Delete Copy Reset Apply Config Add New

SIP Trunk Security Profile Information

Name * CUBE SIP Trunk To WxC

Description CUBE SIP Trunk To WxC

Device Security Mode Non Secure ▾

Incoming Transport Type * TCP+UDP ▾

Outgoing Transport Type TCP ▾

☐ Enable Digest Authentication

Nonce Validity Time (mins) * 600

Secure Certificate Subject or Subject Alternate Name

Incoming Port * 5065

☐ Enable Application level authorization

☐ Accept presence subscription

☐ Accept out-of-dialog refer**

☐ Accept unsolicited notification

☐ Accept replaces header

☐ Transmit security status

☐ Allow charging header

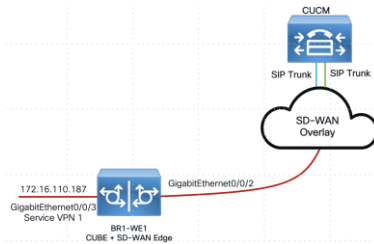
SIP V.150 Outbound SDP Offer Filtering * Use Default Filter ▾

Save Delete Copy Reset Apply Config Add New

SIP Listen port is set to 5065

CUCM SIP Trunk Integration

SIP Trunk #2 (5065): For calls coming in from Webex Calling Cloud via CUBE



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System Call Routing Media Resources Advanced Features Device Application User Management Bulk Administration Help

Trunk Configuration

Related Links: Back To Find/List Go

Save Delete Reset Add New

Status
Status: Ready

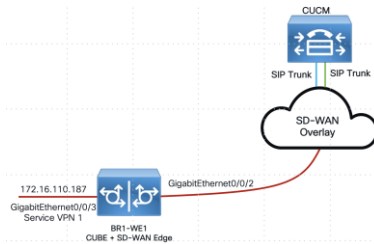
SIP Trunk Status
Service Status: Full Service
Duration: Time In Full Service: 0 day 0 hour 10 minutes

Device Information

Product:	SIP Trunk
Device Protocol:	SIP
Trunk Service Type	None(Default)
Device Name *	BR1 WE1 CUBE SIP Trunk To WxC
Description	BR1 WE1 CUBE SIP Trunk To WxC
Device Pool *	Default
Common Device Configuration	< None >
Call Classification *	Use System Default
Media Resource Group List	< None >
Location *	Hub_None
AAR Group	< None >
Tunneled Protocol *	None
QSIG Variant *	No Changes
ASN.1 ROSE OID Encoding *	No Changes
Packet Capture Mode *	None
Packet Capture Duration	0
☐ Media Termination Point Required	

CUCM SIP Trunk Integration

SIP Trunk #2 (5065): For calls coming in from Webex Calling Cloud via CUBE



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System Call Routing Media Resources Advanced Features Device Application User Management Bulk Administration Help

Trunk Configuration

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Save Delete Reset Add New

SIP Information

Destination

☐ Destination Address is an SRV

	Destination Address	Destination Address IPv6	Destination Port	Status	Status Reason	Duration
1 *	172.16.115.187		5060	up		Time Up: 10 minutes

MTP Preferred Originating Codec* 711ulaw

BLF Presence Group* Standard Presence group

SIP Trunk Security Profile* CUBE SIP Trunk To WxC

Rerouting Calling Search Space < None >

Out-Of-Dialog Refer Calling Search Space < None >

SUBSCRIBE Calling Search Space < None >

SIP Profile* CUBE SIP Trunk With OPTIONS_PING [View Details](#)

DTMF Signaling Method* No Preference

Normalization Script

Normalization Script < None >

☐ Enable Trace

	Parameter Name	Parameter Value
1		

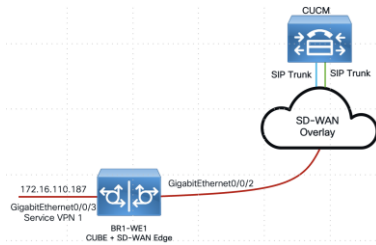
Recording Information

☒ None

☐ This trunk connects to a recording-enabled gateway

☐ This trunk connects to other clusters with recording-enabled gateways

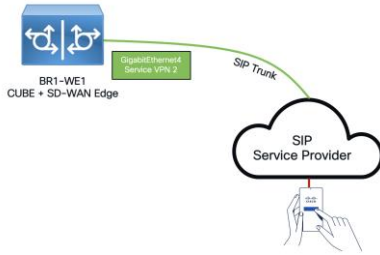
CUCM SIP Trunk Integration



SIP Trunk parameters used in SD-WAN Manager CLI Add-on Template

Parameter Name	Description	Value
cucm_trusted_ipv4_addr_1	Trusted List - IPv4 address of CUCM Node #1	172.16.110.67
cucm_trusted_ipv4_addr_1_subnet_mask_bits	Trusted List - IPv4 address mask of CUCM Node #1	32
cucm_trusted_ipv4_addr_2	Trusted List - IPv4 address of CUCM Node #2	172.16.110.68
cucm_trusted_ipv4_addr_2_subnet_mask_bits	Trusted List - IPv4 address mask of CUCM Node #2	32
cucm_trusted_ipv4_addr_3	Trusted List - IPv4 address of CUCM Node #3	172.16.110.69
cucm_trusted_ipv4_addr_3_subnet_mask_bits	Trusted List - IPv4 address mask of CUCM Node #3	32
cucm_sip_ipv4_addr_pattern	IPv4 address pattern to be used in SIP URI pattern configuration	172.16.110.6[7-9]
cucm_sip_control_ifname	Name of the Local gateway interface used for SIP control traffic	GigabitEthernet2
cucm_sip_media_ifname	Name of the Local gateway interface used for RTP media traffic	GigabitEthernet2

PSTN SIP Trunk Integration



SIP Trunk parameters used in SD-WAN Manager CLI Add-on Template

Parameter Name	Description	Value
pstn_sip_uri_host	Host pattern to be used in SIP URI pattern configuration	s1.sp1.com
pstn_sip_registrar_domain	Domain of the SIP Registrar within Service provider	sp1.com
pstn_sip_line_number	Line number used in SIP registration	19193926000
pstn_sip_uname	Username used in SIP registration and authentication	19193926000
pstn_sip_pwd	Password used in SIP registration and authentication	XXXXXXXX
pstn_sip_realm	Realm used in SIP registration	sp1.com
pstn_sip_control_ifname	Name of the Local gateway interface used for SIP control traffic	GigabitEthernet1
pstn_sip_media_ifname	Name of the Local gateway interface used for RTP media traffic	GigabitEthernet1
sip_pstn_inbound_DN_dest_pattern	Destination pattern that matches the inbound DID number	19193926000
sip_pstn_trusted_ipv4_addr_1	Trusted List - 1st Network address of Service provider	X.X.X.X
sip_pstn_trusted_ipv4_addr_1_subnet_mask_bits	Trusted List - 1st Network mask of Service provider	XXXXXXXX
sip_pstn_trusted_ipv4_addr_2	Trusted List - 2nd Network address of Service provider	X.X.X.X
sip_pstn_trusted_ipv4_addr_2_subnet_mask_bits	Trusted List - 2nd Network mask of Service provider	XX

CLI Add-On Template for Local Gateway

Global level configuration

```
voice service voip
ip address trusted list
ipv4 {{sip_pstn_trusted_ipv4_addr_1}} /{{sip_pstn_trusted_ipv4_addr_1_subnet_mask_bits}}
ipv4 {{wxc_trusted_ipv4_addr_1}} /{{wxc_trusted_ipv4_addr_1_subnet_mask_bits}}
...
ipv4 {{cucm_trusted_ipv4_addr_1}} /{{cucm_trusted_ipv4_addr_1_subnet_mask_bits}}
....
allow-connections sip to sip
media statistics
media bulk-stats
no supplementary-service sip refer
no supplementary-service sip handle-replaces
sip
g729 annexb-all
early-offer forced
asymmetric payload full
stun
stun flowdata agent-id {{wxc_stun_agent_id}} boot-count 4
stun flowdata shared-secret 0 {{wxc_stun_agent_pwd}}

voice class tls-profile 200
trustpoint {{sip_trustpoint_label}}
cn-san validate server
sip-ua
crypto signaling default tls-profile 200
transport tcp tls v1.2
```

Trusted IPv4 address list

Enable CUBE functionality

Enable STUN agent

TLS profile

Default TLS profile for
SIP user agent

CLI Add-On Template for Local Gateway

SIP PSTN Trunk related configuration

```
voice class sip-profiles 100
rule 1 request ANY sdp-header Audio-Attribute modify "a=inactive" "a=sendrecv"
rule 2 response ANY sdp-header Audio-Attribute modify "a=inactive" "a=sendrecv"
rule 3 response 200 sdp-header Video-Attribute modify "a=sendrecv" ""
```

SIP Profile for SIP PSTN Trunk

```
voice class tenant 100
registrar dns:{{pstn_sip_registrar_domain}} expires 3600
credentials username {{pstn_sip_uname}} password 0 {{pstn_sip_pwd}} realm {{pstn_sip_realm}}
authentication username {{pstn_sip_uname}} password 0 {{pstn_sip_pwd}} realm {{pstn_sip_realm}}
session transport tcp
url sip
connection-reuse
bind control source-interface {{pstn_sip_control_ifname}}
bind media source-interface {{pstn_sip_media_ifname}}
error-passthru
asserted-id pai
no pass-thru content custom-sdp
sip-profiles 100
sip-server dns:{{pstn_sip_registrar_domain}}
```

SIP PSTN Trunk registration
and authentication parameters

SIP transport method is TCP

Source interface for SIP control
and media traffic

[Cisco IOS XE Catalyst SD-WAN Qualified Command Reference](#)

CLI Add-On Template for Local Gateway

Webex Calling SIP Trunk related configuration

```
voice class srtp-crypto 200  
crypto 1 AES_CM_128_HMAC_SHA1_80
```

Crypto suite for Secure RTP

```
voice class sip-profiles 200  
rule 9 request ANY sip-header SIP-Req-URI modify "sips:(.*)" "sip:\x5c1"  
rule 10 request ANY sip-header To modify "<sips:(.*)" "<sip:\x5c1"  
rule 11 request ANY sip-header From modify "<sips:(.*)" "<sip:\x5c1"  
rule 12 request ANY sip-header Contact modify "<sips:(.*)>" "<sip:\x5c1;transport=tls>"  
rule 13 response ANY sip-header To modify "<sips:(.*)" "<sip:\x5c1"  
rule 14 response ANY sip-header From modify "<sips:(.*)" "<sip:\x5c1"  
rule 15 response ANY sip-header Contact modify "<sips:(.*)" "<sip:\x5c1"  
rule 20 request ANY sip-header From modify ">" ";otg={{wxc_sip_trunk_group_OTG_DTG}}>"  
rule 30 request ANY sip-header P-Asserted-Identity modify "sips:(.*)" "sip:\x5c1"
```

SIP Profile for interoperability
with Webex Calling

\x5c represents '\'

CLI Add-On Template for Local Gateway

Webex Calling SIP Trunk related configuration

```
voice class tenant 200
  registrar dns:{{wxc_sip_registrar_domain}} scheme sips expires 240 refresh-ratio 50 tcp tls
  credentials number {{wxc_sip_line_number}} username {{wxc_sip_uname}} password 0 {{wxc_sip_pwd}} realm BroadWorks
  authentication username {{wxc_sip_uname}} password 0 {{wxc_sip_pwd}}
  sip-server dns:{{wxc_sip_registrar_domain}}
  connection-reuse
  srtp-crypto 200
  session transport tcp tls
  url sips
  error-passthru
  asserted-id pai
  no pass-thru content custom-sdp
  sip-profiles 200
  outbound-proxy dns:{{wxc_sip_outbound_proxy}}
  privacy-policy passthru
  bind control source-interface {{wxc_sip_control_ifname}}
  bind media source-interface {{wxc_sip_media_ifname}}
```

WxC Trunk registration and authentication parameters

SIP Server

SIP transport method is TLS

SIP outbound proxy server

Source interface for SIP control and media traffic

CLI Add-On Template for Local Gateway

CUCM SIP Trunk related configuration

```
voice class tenant 300
 session transport tcp
 url sip
 error-passthru
 no pass-thru content custom-sdp
 bind control source-interface {{cucm_sip_control_ifname}}
 bind media source-interface {{cucm_sip_media_ifname}}
```

```
voice class server-group 301
 ipv4 {{cucm_trusted_ipv4_addr_1}}
 ipv4 {{cucm_trusted_ipv4_addr_2}}
 ipv4 {{cucm_trusted_ipv4_addr_3}}
```

```
voice class server-group 303
 ipv4 {{cucm_trusted_ipv4_addr_1}} port 5065
 ipv4 {{cucm_trusted_ipv4_addr_2}} port 5065
 ipv4 {{cucm_trusted_ipv4_addr_3}} port 5065
```

No registration!

SIP transport method is TCP

Source interface for SIP control
and media traffic

Server group containing three
CUCM nodes for SIP Trunk #1

Server group containing three
CUCM nodes for SIP Trunk #2

CLI Add-On Template for Local Gateway

Call routing related configuration

```
voice class uri 100 sip  
pattern {{pstn_sip_uri_pattern}}
```

SIP URI to match calls
from SIP PSTN

```
voice class uri 200 sip  
pattern {{wxc_sip_uri_pattern}}
```

SIP URI to match calls
from Webex Calling

```
voice class uri 300 sip  
pattern {{cucm_sip_ipv4_addr_pattern}}:5060
```

SIP URI to match calls
from CUCM SIP Trunk #1
(for calls destined to PSTN)

```
voice class uri 301 sip  
pattern :5065
```

SIP URI to match calls
from CUCM SIP Trunk #2
(for calls destined to WxC)

CLI Add-On Template for Local Gateway

Call routing and connection related configurations

```
voice translation-rule 1
rule 1 /{{sip_pstn_inbound_did_1}}/ /{{sip_pstn_inbound_translated_dn_1}}/
```

Translation profile to change inbound PSTN DID to a WxC phone directory number

```
voice translation-profile Replace-DID
translate called 1
```

```
voice class codec 99
codec preference 2 g711ulaw
codec preference 3 g711alaw
```

List of codec advertised by LGW during call setup

```
voice class stun-usage 200
stun usage firewall-traversal flowdata
stun usage ice lite
```

Enable Interactive Connection Establishment (ICE) protocol for NAT traversal. This is used in calls setup between LGW and WxC

CLI Add-On Template for Local Gateway

Call routing related configuration [SIP PSTN → LGW → CUCM]

```
! Start of configuration for routing calls from/to SIP PSTN
dial-peer voice 301 voip
description "Outgoing dial-peer to CUCM server group for calls from SIP PSTN"
destination-pattern {{sip_pstn_inbound_DN_dest_pattern}}
session protocol sipv2
session server-group 301
voice-class codec 99
dtmf-relay rtp-nte
voice-class sip tenant 300
no vad
```

Outbound dial-peer to CUCM for calls from SIP PSTN

```
! SIP PSTN --> LGW --> CUCM
voice class dpg 301
dial-peer 301 preference 1
```

Dial-peer group to route calls coming from SIP PSTN (DP 100) directly to CUCM (DP 301)

```
! Inbound dial-peer for calls from SIP PSTN. These calls are sent to CUCM.
dial-peer voice 100 voip
description Incoming dial-peer for calls from SIP PSTN
session protocol sipv2
incoming uri via 100
destination dpg 301
voice-class codec 99
dtmf-relay rtp-nte
voice-class sip tenant 100
no vad
translation-profile incoming Replace-DID
```

Inbound dial-peer for calls from SIP PSTN. These calls are sent to CUCM via DPG 301.

CLI Add-On Template for Local Gateway

Call routing related configuration [SIP PSTN ← LGW ← CUCM]

```
! Outgoing dial-peer for calls to SIP PSTN
dial-peer voice 101 voip
description "Outgoing dial-peer for calls to SIP PSTN"
destination-pattern +1[2-9]..[2-9].....$
session protocol sipv2
session target sip-server
session transport udp
voice-class codec 99
dtmf-relay rtp-nte
voice-class sip tenant 100
no vad
```

Outbound dial-peer for calls to SIP PSTN

```
voice class dpg 101
dial-peer 101 preference 1
```

Dial-peer group to route calls coming from CUCM Trunk #1 (DP 300) directly to SIP PSTN (DP 101).

```
! Inbound dial-peer for calls from SIP PSTN. These calls are sent to CUCM.
dial-peer voice 300 voip
description Incoming dial-peer for calls from CUCM destined to SIP PSTN
session protocol sipv2
incoming uri via 300
destination dpg 101
voice-class codec 99
dtmf-relay rtp-nte
voice-class sip tenant 300
no vad
```

Inbound dial-peer for calls from CUCM Trunk #1 (for calls destined to SIP PSTN).

CLI Add-On Template for Local Gateway

Call routing related configuration [WxC → LGW → CUCM]

```
dial-peer voice 303 voip
description "Outgoing dial-peer to CUCM server group for calls from WxC"
destination-pattern {{wxc_inbound_DN_dest_pattern}}
session protocol sipv2
session server-group 303
voice-class codec 99
dtmf-relay rtp-nte
voice-class sip tenant 300
no vad
```

Outbound dial-peer to CUCM for calls from Webex Calling Cloud

```
voice class dpg 303
dial-peer 303 preference 1
```

Dial-peer group to route calls coming from WxC (DP 200201) directly to CUCM (DP 303)

```
dial-peer voice 200201 voip
description Inbound/Outbound Webex Calling
max-conn 250
session protocol sipv2
incoming uri request 200
destination dpg 303
destination-pattern {{wxc_inbound_DN_dest_pattern}}
session target sip-server
voice-class codec 99
dtmf-relay rtp-nte
voice-class sip tenant 200
voice-class stun-usage 200
srtp
no vad
```

Inbound dial-peer for calls from Webex Calling Cloud. These calls are sent to CUCM via DPG 303

Enable STUN flowdata and ICE

CLI Add-On Template for Local Gateway

Call routing related configuration [WxC ← LGW ← CUCM]

```
dial-peer voice 200201 voip
description Inbound/Outbound Webex Calling
max-conn 250
session protocol sipv2
incoming uri request 200
destination dpd 303
destination-pattern {{wxc_inbound_DN_dest_pattern}}
session target sip-server
voice-class codec 99
dtmf-relay rtp-nte
voice-class sip tenant 200
voice-class stun-usage 200
srtp

voice class dpd 201
dial-peer 200201 preference 1

dial-peer voice 302 voip
description Incoming dial-peer for calls from CUCM destined to WxC
session protocol sipv2
incoming uri via 301
destination dpd 201
voice-class codec 99
dtmf-relay rtp-nte
voice-class sip tenant 300
no vad
```

Outbound dial-peer for calls to Webex Calling

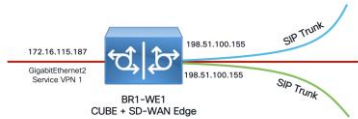
Enable STUN flowdata and ICE

Dial-peer group to route calls coming from CUCM Trunk #2 (DP 302) directly to Webex Calling Cloud (DP 200201)

Inbound dial-peer for calls from CUCM Trunk #2 (for calls destined to Webex Calling Cloud)

Local Gateway Configuration in SD-WAN Manager

Configuration pushed to the Local Gateway!



Cisco Catalyst SD-WAN

Monitor
Configuration
Analytics
Workflows
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← Back

Deploy configuration group to associated devices | Validation success

Initiated By: admin From: 173.38.117.82

Total Task: 1 | Success : 1

Device Group (1)

Q Search Table

As of: Feb 01, 2025 01:31 PM

Status	Chassis Number	Hostname	Message	Start Time	System IP	Action
Success	CBK-16F0E5D6-C5D5-0A3D-DE4E-F591584CC8B5	BR1-WE1	Config Group successfully deployed to ...	Feb 1, 2025, 1:31:17 PM	192.168.2.246	

Items per page: 10 1 - 1 of 1

SIP Registrations

```
BR1-WE1# show sip register status
```

```
Tenant: 100
```

```
----- Registrar-Index 1 -----  
Line                peer      expires(sec)  reg survival  P-Associ-URI  
=====
```

19193926000	-1	31	yes	normal	
-------------	----	----	-----	--------	--

```
=====
```

```
Tenant: 200
```

```
----- Registrar-Index 1 -----  
Line                peer      expires(sec)  reg survival  P-Associ-URI  
=====
```

BR1_WE1_LGW11419173906_LGU	-1	30	yes	normal	
----------------------------	----	----	-----	--------	--

```
=====
```

```
BR1-WE1#
```

IP NAT Translations: WxC IP Phone Registration

```
BR1-WE1# show ip nat translations
```

Pro	Inside global	Inside local	Outside local	Outside global
...				
udp	203.0.113.187:5167	172.16.117.14:52750	8.8.8.8:53	8.8.8.8:53
...				
tcp	203.0.113.187:5079	172.16.117.14:5074	23.89.40.110:8934	23.89.40.110:8934
...				
udp	203.0.113.187:5547	172.16.117.14:19624	198.51.100.155:8370	198.51.100.155:8370
...				
...				

```
BR1-WE1#
```

DNS

SIP

RTP

SIP PSTN Caller → WxC IP Phone User

BR1-WE1# show call active voice brief

```
D6F : 22525 265563610ms.1 (22:35:22.643 EDT Tue May 28 2024) +1610 [pid:100] Answer [19193921234] connected
dur 00:00:23 tx:1147/229400 rx:1156/231200 dscp:0 media:0 audio tos:0xB8 video tos:0x0
IP 199.87.144.74:12598 SRTP: off rtt:0ms pl:0/0ms lost:0/0/0 delay:0/0/0ms g711ulaw TextRelay: off Transcoded: No ICE: Off
media inactive detected:n media contrl rcvd:n/a timestamp:n/a
long duration call detected:n long duration call duration:n/a timestamp:n/a
LostPacketRate:0.00 OutOfOrderRate:0.00
LocalUUID:deace46a51515dfabd6476d45645197f
RemoteUUID:326a670f00105000a000cc98915497da
VRF: 2
```

PSTN -> CUBE

```
D6F : 22526 265563610ms.2 (22:35:22.643 EDT Tue May 28 2024) +1600 [pid:301] Originate [6003] connected
dur 00:00:23 tx:1156/231200 rx:1147/229400 dscp:0 media:0 audio tos:0xB8 video tos:0x0
IP 172.16.115.187:8368 SRTP: off rtt:0ms pl:0/0ms lost:0/0/0 delay:0/0/0ms g711ulaw TextRelay: off Transcoded: No ICE: Off
media inactive detected:n media contrl rcvd:n/a timestamp:n/a
long duration call detected:n long duration call duration:n/a timestamp:n/a
LostPacketRate:0.00 OutOfOrderRate:0.00
LocalUUID:326a670f00105000a000cc98915497da
RemoteUUID:deace46a51515dfabd6476d45645197f
VRF: 1
```

CUBE -> CUCM

SIP PSTN Caller → WxC IP Phone User

```
0      : 22527 265563630ms.1 (22:35:22.663 EDT Tue May 28 2024) +1560 pid:302 Answer 19193921234 connected
dur 00:00:23 tx:1147/229400 rx:1156/231200 dscp:0 media:0 audio tos:0xB8 video tos:0x0
IP 172.16.115.187:8366 SRTP: off rtt:0ms pl:0/0ms lost:0/0/0 delay:0/0/0ms g711ulaw TextRelay: off Transcoded: No ICE: Off
media inactive detected:n media contrl rcvd:n/a timestamp:n/a
long duration call detected:n long duration call duration:n/a timestamp:n/a
LostPacketRate:0.00 OutOfOrderRate:0.00
LocalUUID:deace46a51515dfabd6476d45645197f
RemoteUUID:326a670f00105000a000cc98915497da
VRF: 1
```

CUCM → CUBE

```
0      : 22528 265563630ms.2 (22:35:22.663 EDT Tue May 28 2024) +1560 pid:200201 Originate 8116003 connected
dur 00:00:23 tx:1156/242760 rx:1147/240870 dscp:0 media:0 audio tos:0xB8 video tos:0x0
IP 203.0.113.187:5547 SRTP: on rtt:0ms pl:0/0ms lost:0/0/0 delay:0/0/0ms g711ulaw TextRelay: off Transcoded: No ICE: On
media inactive detected:n media contrl rcvd:n/a timestamp:n/a
long duration call detected:n long duration call duration:n/a timestamp:n/a
LostPacketRate:0.00 OutOfOrderRate:0.00
LocalUUID:326a670f00105000a000cc98915497da
RemoteUUID:deace46a51515dfabd6476d45645197f
VRF: 2
```

CUBE → WxC

SIP PSTN Caller → WxC IP Phone User @ Branch

show voip rtp connections

```
BR1-WE1# show voip rtp connections
```

```
VoIP RTP Port Usage Information:
```

```
Max Ports Available: 19999, Ports Reserved: 101, Ports in Use: 4
```

```
Port range not configured
```

Media-Address Range	Min Port	Max Port	Ports Available	Ports Reserved	Ports In-use
Global Media Pool	8000	48198	19999	101	4

```
VoIP RTP active connections :
```

No.	CallId	dstCallId	LocalRTP	RmtRTP	LocalIP	RemoteIP	MPSS	VRF
1	22525	22526	8364	12598	192.51.100.155	199.87.144.74	NO	2
2	22526	22525	8366	8368	172.16.115.187	172.16.115.187	NO	1
3	22527	22528	8368	8366	172.16.115.187	172.16.115.187	NO	1
4	22528	22527	8370	5547	192.51.100.155	203.0.113.187	NO	2

```
Found 4 active RTP connections
```

```
BR1-WE1#
```

SIP PSTN Trunk IP Address

cEdge IP Address
(Service VPN 2)

cEdge Transport IP address
IP Phone's NAT-ed IP address

SIP PSTN Caller → WxC IP Phone User @ Home

show voip rtp connections

```
BR1-WE1#show voip rtp connections
```

```
VoIP RTP Port Usage Information:
```

```
Max Ports Available: 19999, Ports Reserved: 101, Ports in Use: 4
```

```
Port range not configured
```

Media-Address Range	Min Port	Max Port	Ports Available	Ports Reserved	Ports In-use
Global Media Pool	8000	48198	19999	101	4

```
VoIP RTP active connections :
```

No.	CallId	dstCallId	LocalRTP	RmtRTP	LocalIP	RemoteIP	MPSS	VRF
1	2037	2038	8068	14598	198.51.100.155	199.87.144.74	NO	2
2	2038	2037	8070	8072	172.16.115.187	172.16.115.187	NO	1
3	2039	2040	8072	8070	172.16.115.187	172.16.115.187	NO	1
4	2040	2039	8074	19572	198.51.100.155	203.0.113.7	NO	2

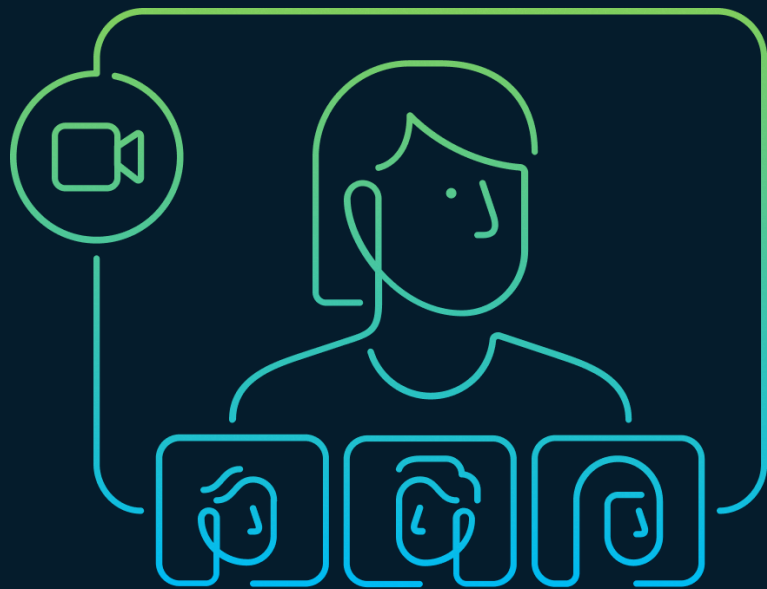
```
Found 4 active RTP connections
```

```
BR1-WE1#
```

SIP PSTN Trunk IP Address

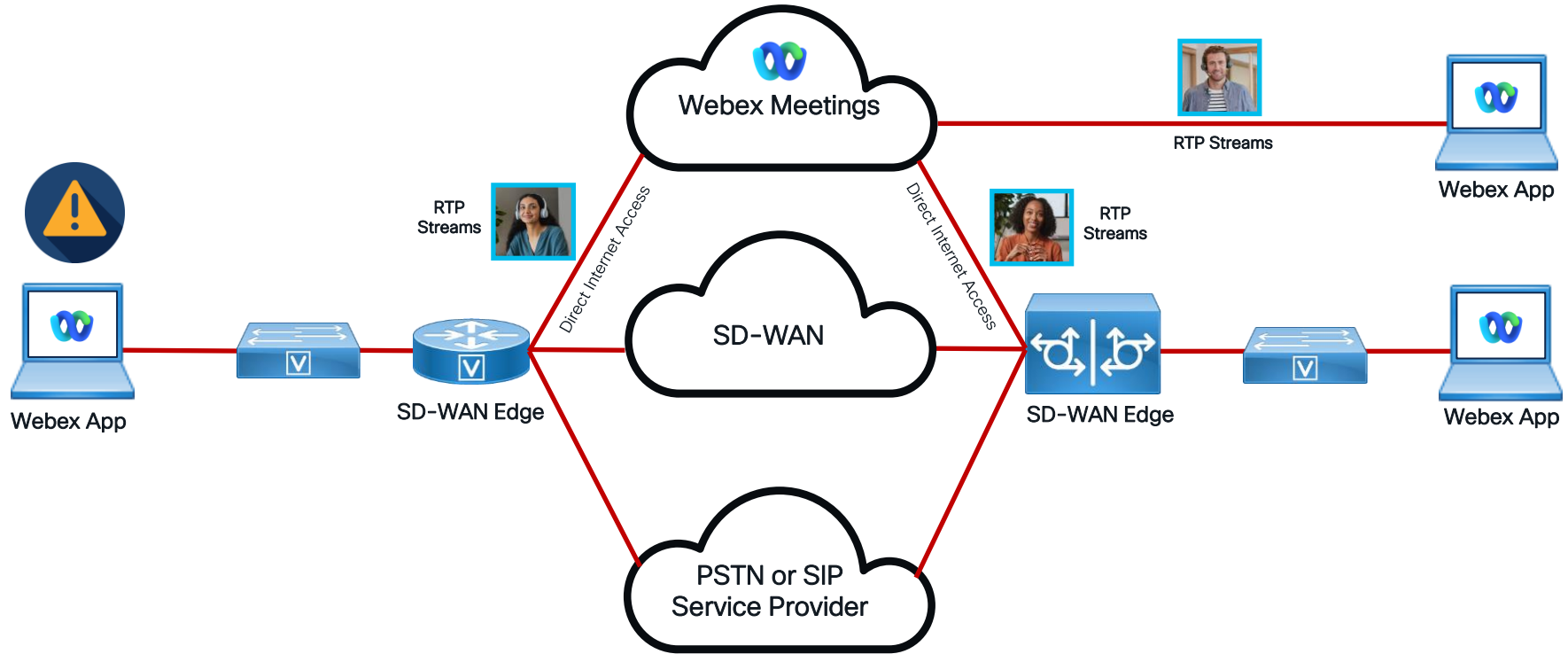
cEdge IP Address
(Service VPN 2)

Home router IP address
IP Phone's NAT-ed IP address



Path Insights & Optimization

Webex Meetings In SD-WAN Topology



Troubleshooting Using Webex Control Hub

webex Control Hub

Search

Troubleshooting

Meetings & Calls | Live Meetings | Status | Admin Activities | Connected UC | Logs | Watermark Analysis

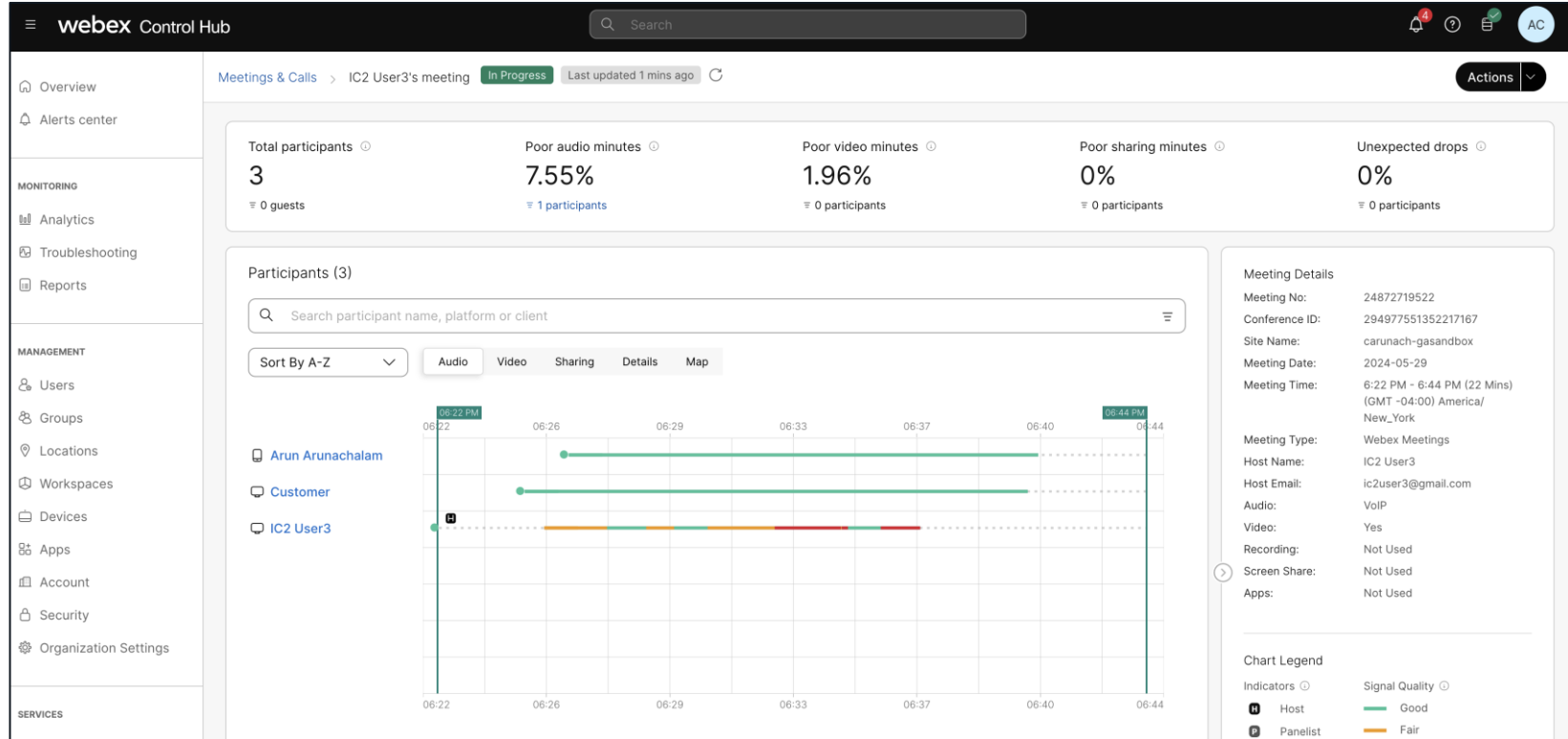
Q 2487 271 9522 x 1 Meeting

Last 7 Days | 05/23/2024 → 05/29/2024 | (GMT -04:00) America/New_York

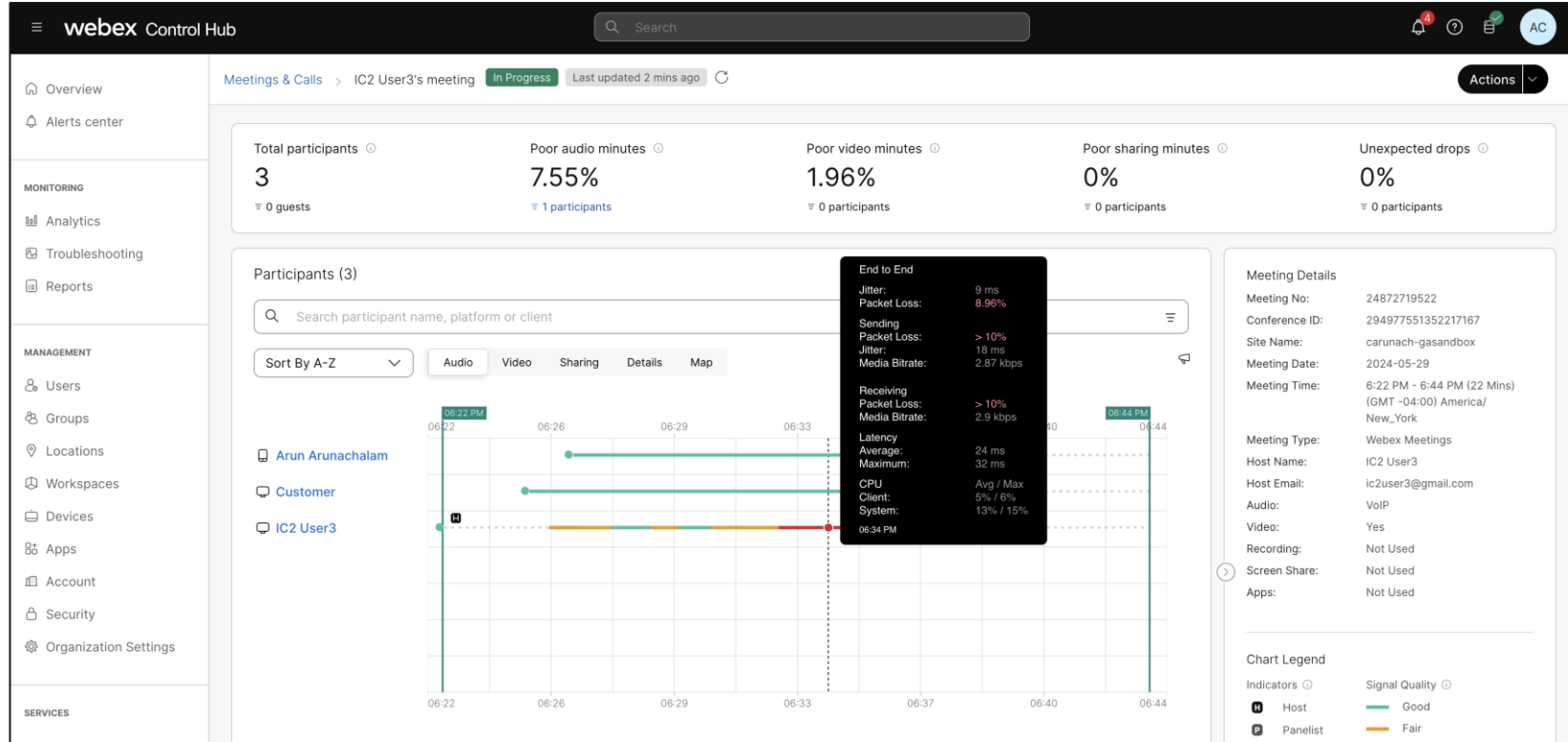
Quality	Service	Start Time	Meeting / Caller ...	Name	Host / Caller	Particip...	Duration	Site / Loc...	Conference / Call ID
None		2024-05-29 06:22:54 PM	24872719522	IC2 User3's meeting	ic2user3@gm...	3	In progr...	carunac...	294977551352217167

Find Meeting by ID

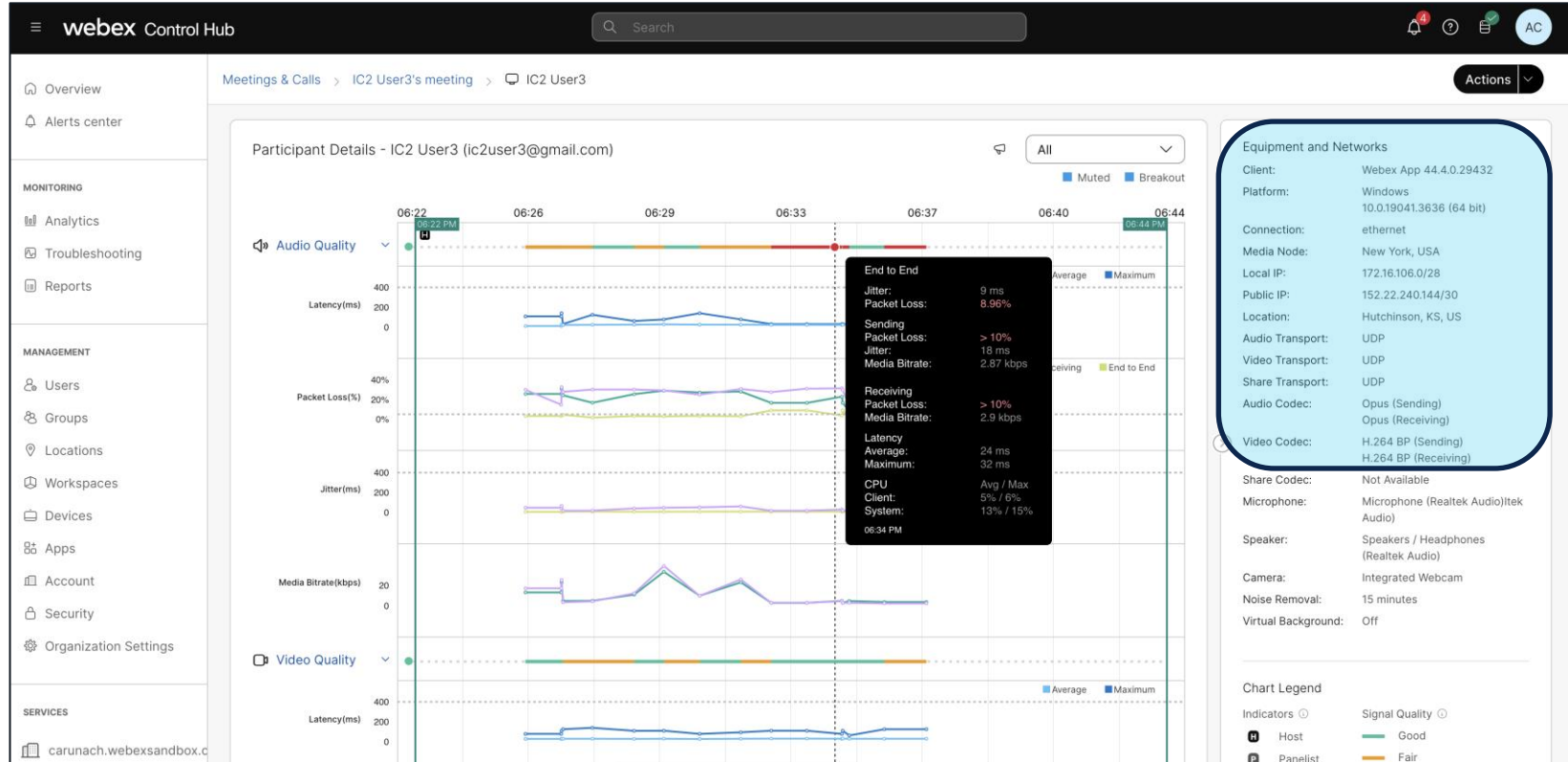
Webex Meeting Participants View



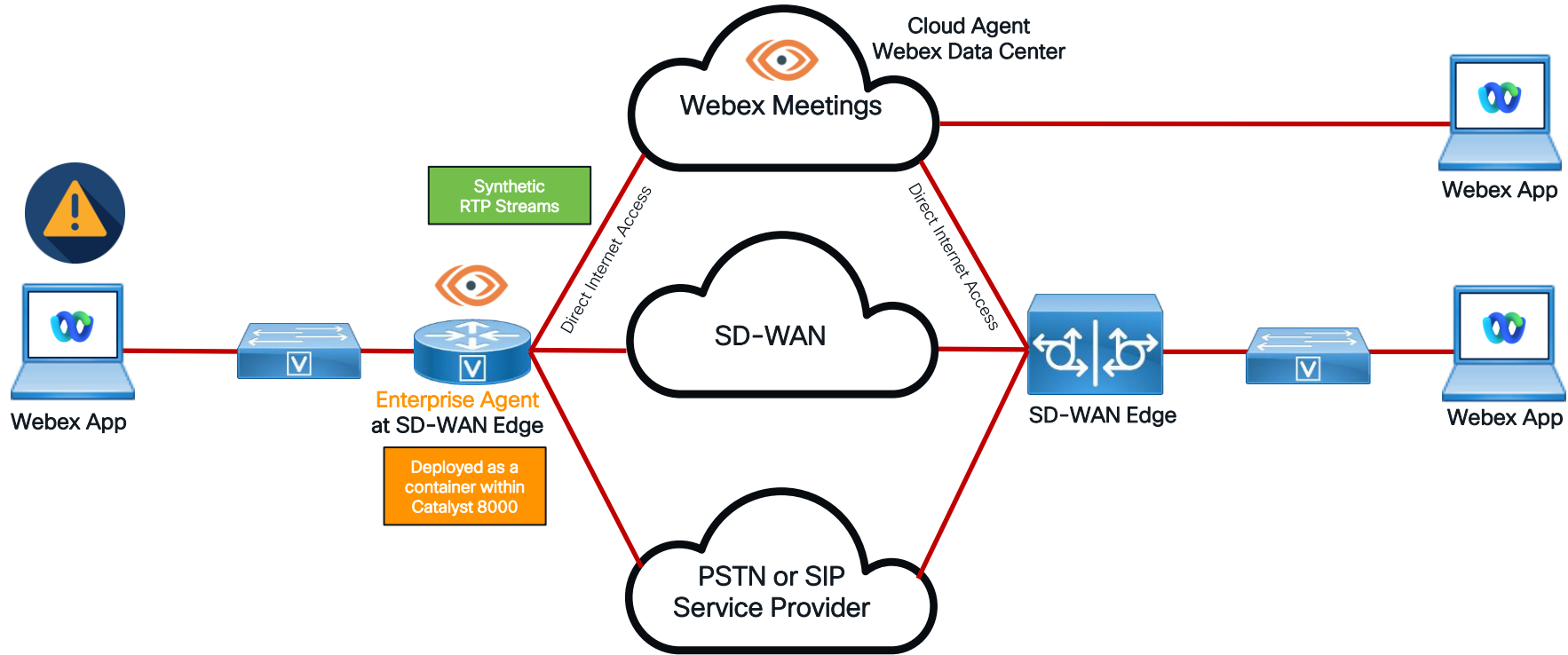
Webex Meeting Participant Media Statistics View



Webex Meeting Audio Quality Statistics



ThousandEyes Enterprise Agent @ Catalyst 8000



ThousandEyes Enterprise Agent Test Settings

The screenshot shows the ThousandEyes Enterprise Agent Test Settings interface. The left sidebar contains navigation links: Cloud & Enterprise Agents, Event Detection, Views, Test Settings (highlighted), Agent Settings, BGP Monitors, Endpoint Agents, Devices, Internet Insights, Dashboards, Alerts, Integrations, Sharing, and Account Settings. The main content area is titled 'Cloud & Enterprise Agents' and has tabs for Tests, Test Labels, and Credentials Repository. The Tests tab is active, showing a table of tests. A filter 'DC1' is applied. The table lists a test named 'DC1_WE2_Webex_Audio' with type 'Network - Agent to Agent', target 'New York, NY (Webex):9000', and status 'Enabled'. Below the table, the 'Basic Configuration' tab is selected, showing fields for Test Name, Test Description, Target Agent (New York, NY (Webex)), Interval (2 minutes), Agents (1 of 960 selected), Direction (Both Directions), Protocol (UDP), and Alerts (Enable). The 'Advanced Settings' tab is also visible. The 'General Information' section shows labels, sharing options, and creation/modification details. The 'Views Enabled for This Test' section lists Network, Overview, Path Visualization, and Routing. At the bottom, there are buttons for 'Cancel', 'Run Once', and 'Save Changes'.

Test Name	Type	Target	Alerts	Enabled
DC1_WE2_Webex_Audio	Network - Agent to Agent	New York, NY (Webex):9000	✓	✓

Basic Configuration

Test Name: DC1_WE2_Webex_Audio

Test Description: Optional

Target Agent: New York, NY (Webex)

Interval: 2 minutes

Agents: 1 of 960 selected

Direction: Both Directions

Protocol: UDP

Alerts: ☒ Enable

1 of 4 alert rules selected

General Information

Labels: Add Labels

Sharing: 0 of 28 account groups

Created: By Arun Arunachalam, 3 minutes ago

Modified: By Arun Arunachalam, Just now

Views Enabled for This Test

Network: Overview, Path Visualization, BGP Route Visualization

Target Agent: New York (Webex)
Source Agent: TE at cEdge device

Direction: Both Directions
Protocol: UDP

ThousandEyes Enterprise Agent Test Settings

Cloud & Enterprise Agents ▾

- Event Detection
- Views
- Test Settings**
- Agent Settings
- BGP Monitors

Endpoint Agents ▸

Devices ▸

Internet Insights

Dashboards

Alerts ▸

Integrations

Sharing ▸

Account Settings ▸

Tests | Test Labels | Credentials Repository

Add New Test ▾

Add a filter ▾

DC1

3 deleted tests

Test Name	Type	Target	Alerts	Enabled	
DC1_WE2_Webex_Audio	Network - Agent to Agent	New York, NY (Webex):9000	✓	✓	...

Basic Configuration | **Advanced Settings**

NETWORK

Server Port: 9000

Payload Size: **Auto** | Manual

☒ Collect BGP data
All public BGP monitors will be included

Transmission Rate: ☐ Enforce fixed packet rate

No. of Path Traces: ☒ Default (3)

DSCP: EF (DSCP 46)

General Information

Labels: Add Labels ▾

Sharing: 0 of 28 account groups ▾
Share with other customers

Created: By Arun Arunachalam
4 minutes ago

Modified: By Arun Arunachalam
1 minute ago

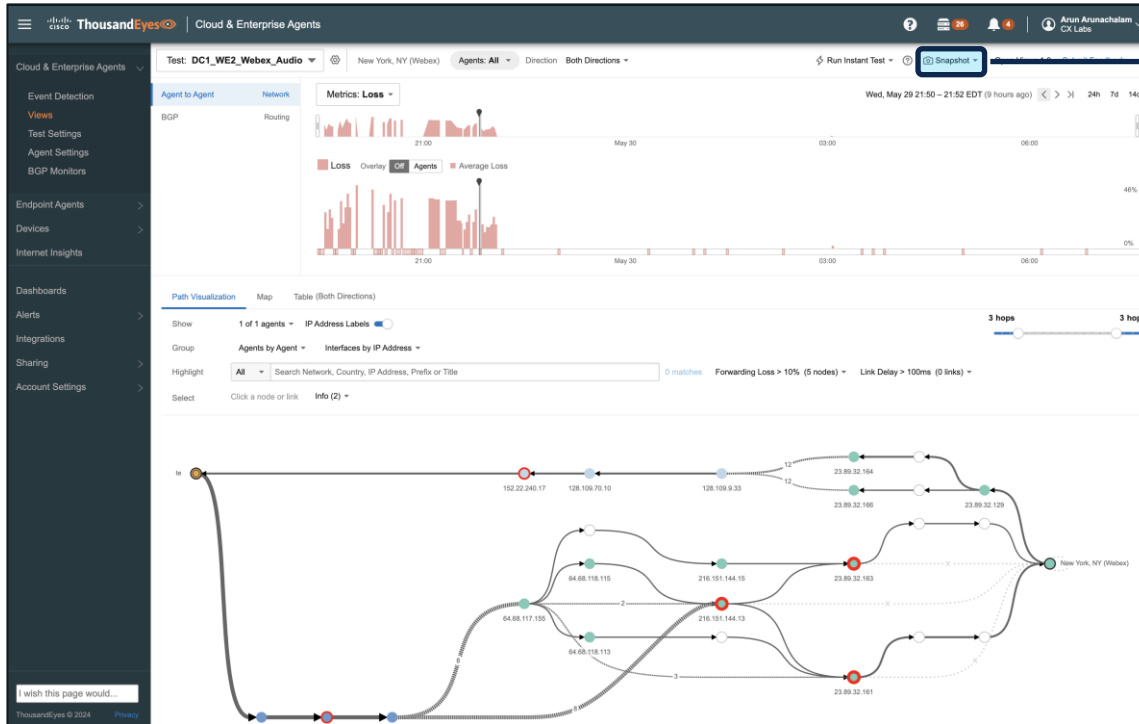
Views Enabled for This Test

Network: Overview
Path Visualization
BGP Route Visualization

Cancel | Run Once | Save Changes

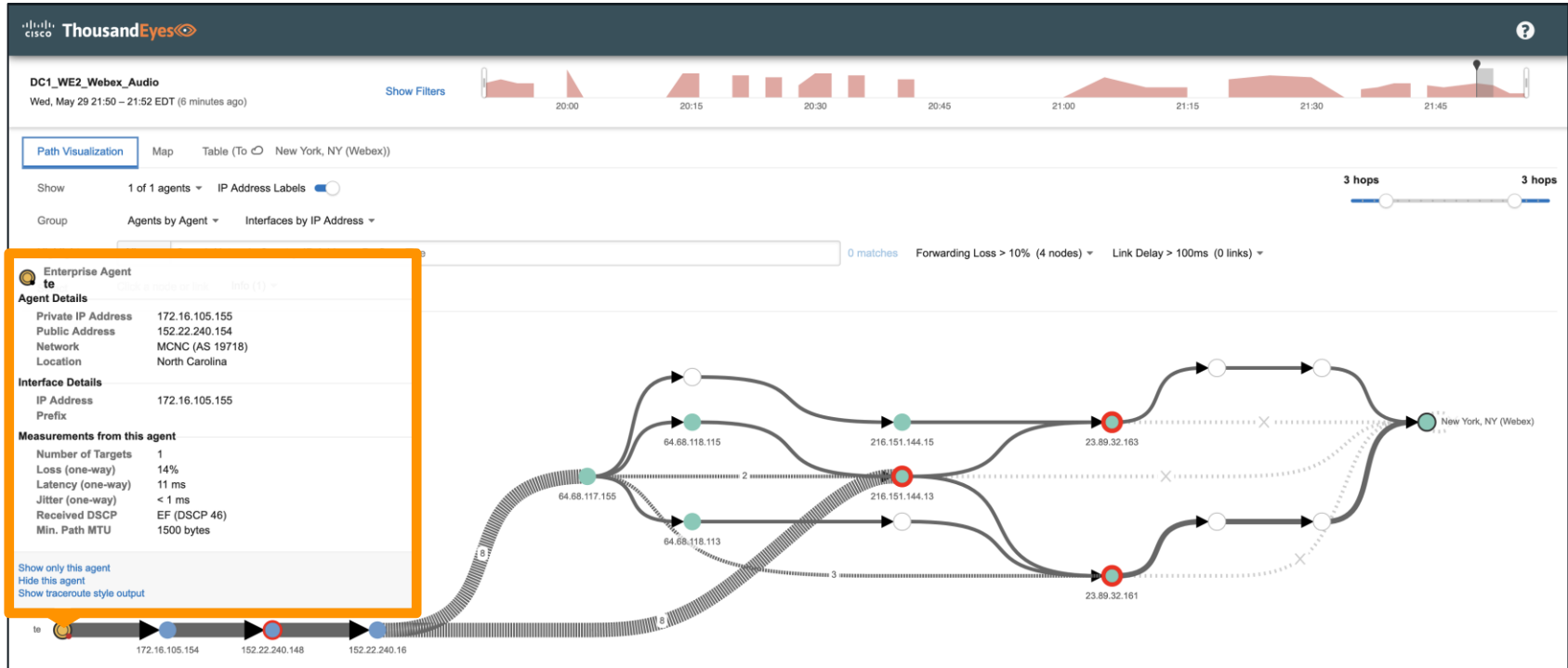
Server Port: 9000, 5004
DSCP for Audio: 46 (EF)
DSCP for Video: 34 (AF41)
Run Once for an Instant Test

End-to-End Bi-Directional Media Path Visualization

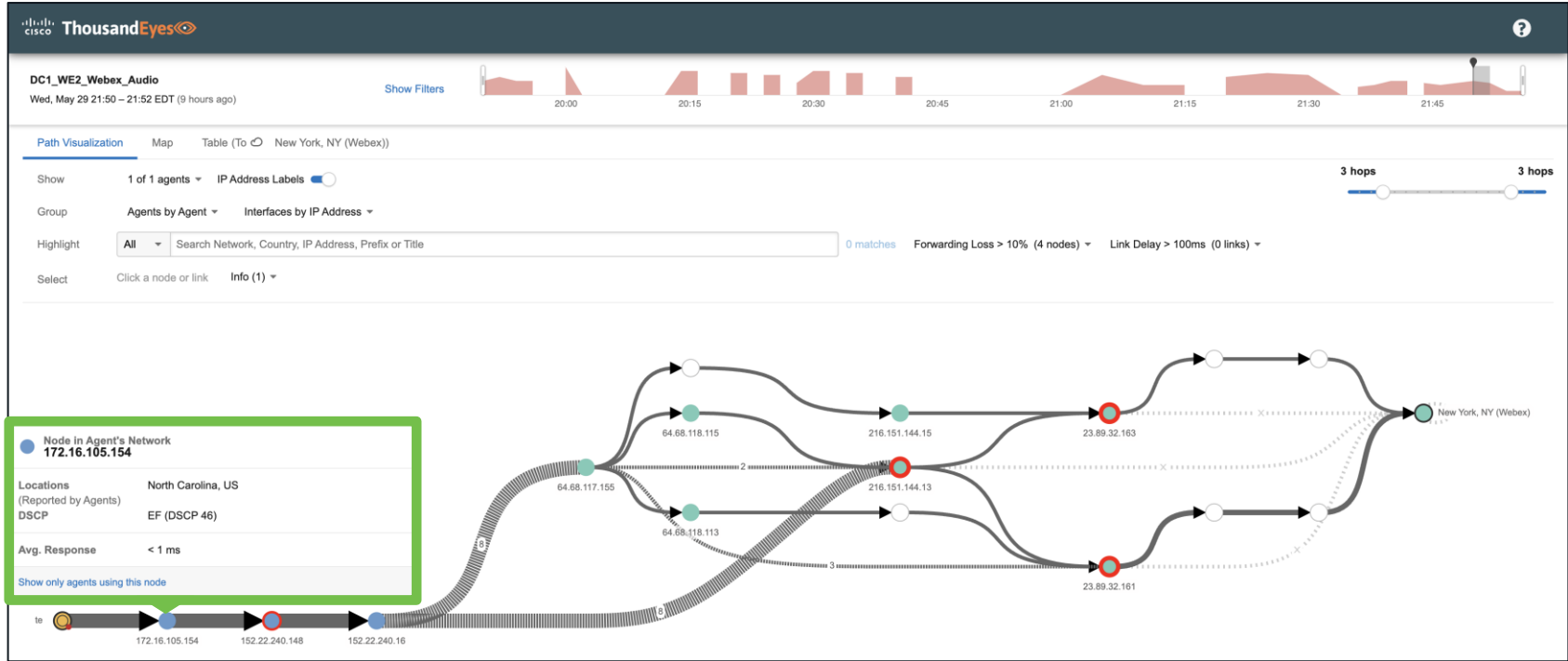


Create a **Snapshot** and share it with key collaborators!

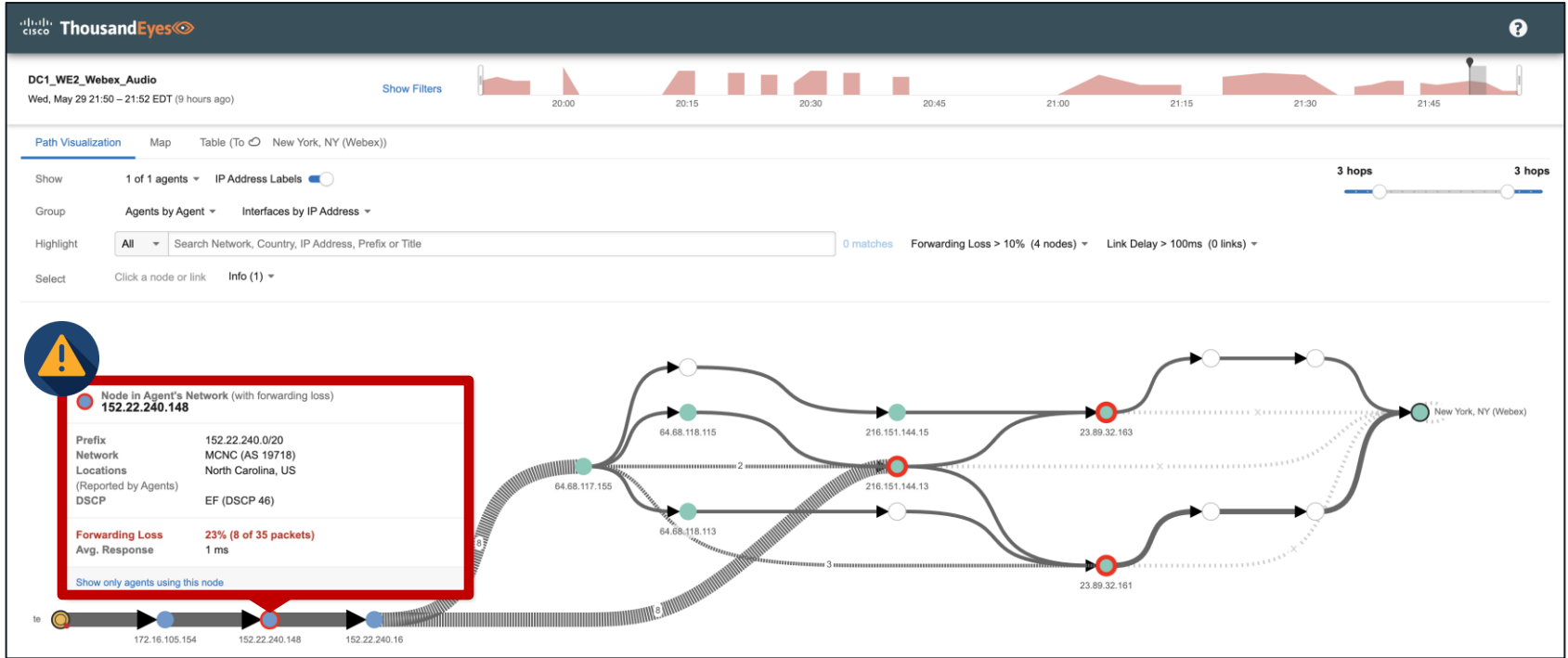
Measurements From Enterprise to Cloud Agent



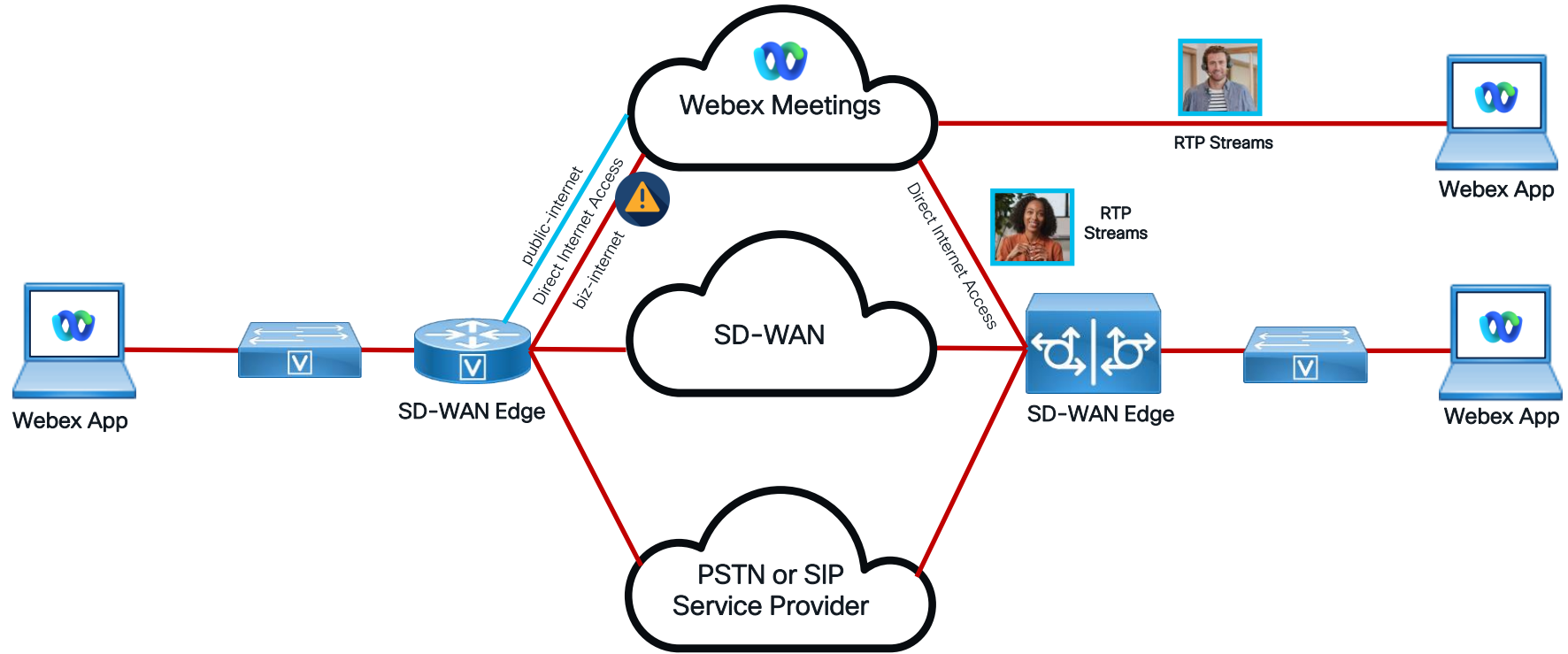
No Packet Loss At The LAN Interface @ cEdge



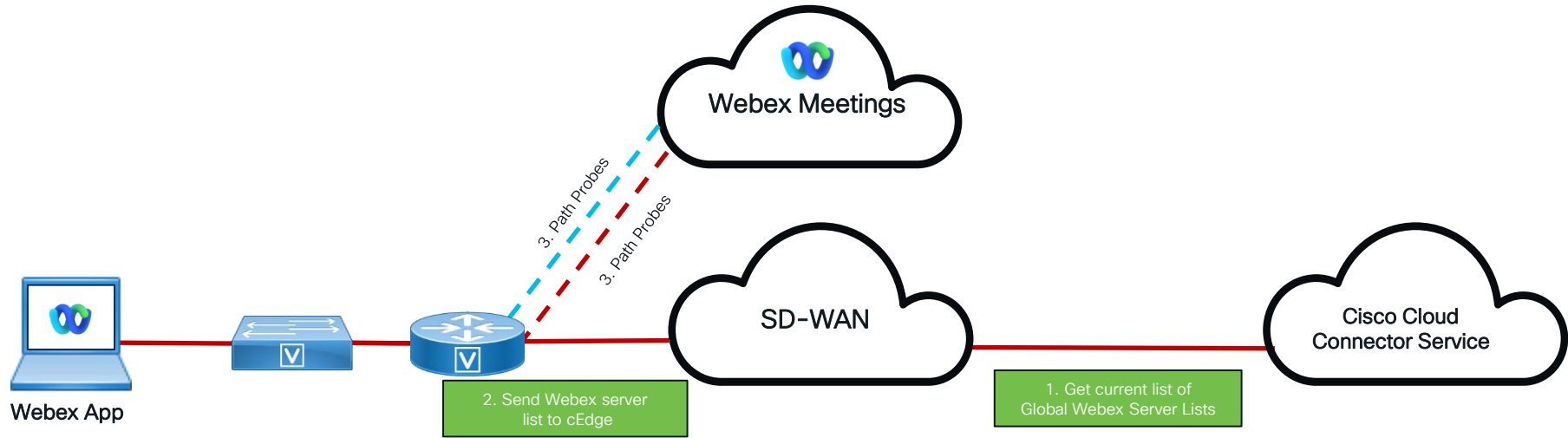
Upstream Packet Loss Towards Internet



What If We have Dual Internet Links?



Cloud OnRamp for Webex



Cloud OnRamp for Webex Determines Best Path

Cloud OnRamp for SaaS > Webex

✓ Bad (0-5) ✗ ✓ Average (5-8) ⚠ ✓ Good (8-10) ✓

Monitor

Configuration

Tools

Maintenance

Administration

Workflows

Reports

Analytics

Explore

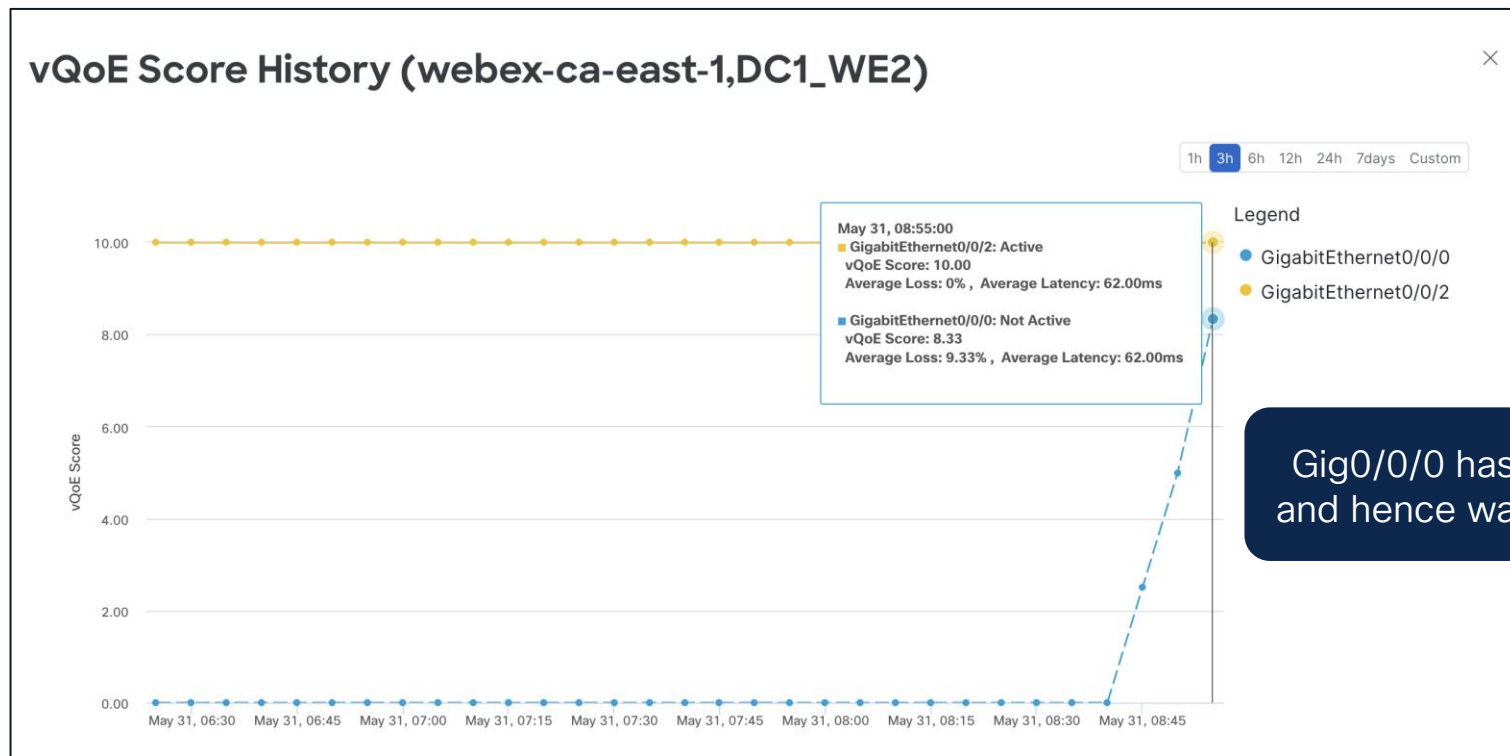
Q DC1_WE2 x Search

VPN List All

Total Rows: 11 of 33

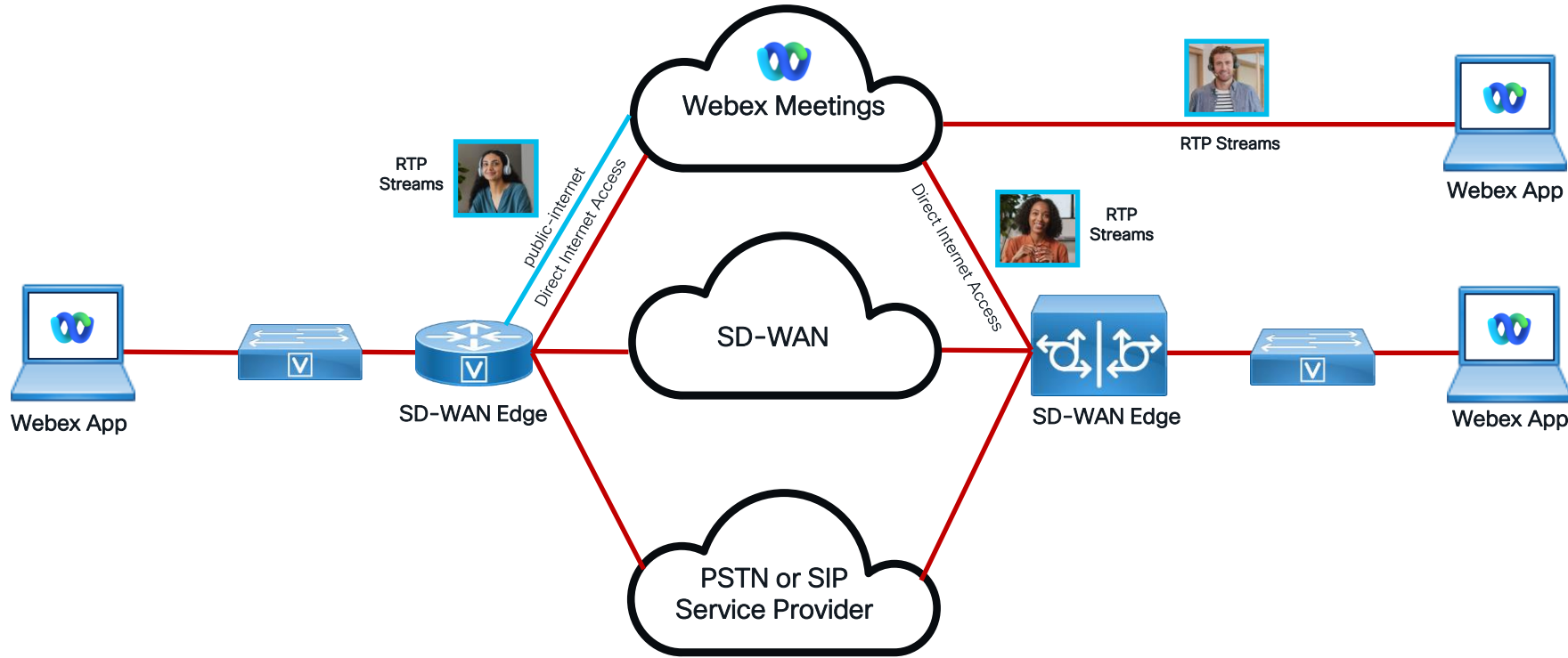
Sites List	Hostname	Webex region	vQoE Status	vQoE Score	DIA Status	Selected interface	Activated Gateway	Local Color	Remote Color	Applicatio
120001	DC1_WE1	-			-	-	-	-	-	
120001	DC1_WE2	us-west-1		5.0	local	GigabitEthernet0/0/0	N/A	N/A	N/A	View Us
120001	DC1_WE2	us-east-3		10.0	local	GigabitEthernet0/0/0	N/A	N/A	N/A	View Us
120001	DC1_WE2	us-central-1		10.0	local	GigabitEthernet0/0/0	N/A	N/A	N/A	View Us
120001	DC1_WE2	ca-east-1		10.0	local	GigabitEthernet0/0/2	N/A	N/A	N/A	View Us
120001	DC1_WE2	eu-central-1		7.0	local	GigabitEthernet0/0/0	N/A	N/A	N/A	View Us
120001	DC1_WE2	eu-central-2		7.0	local	GigabitEthernet0/0/0	N/A	N/A	N/A	View Us
120001	DC1_WE2	eu-west-1		0.0	local	GigabitEthernet0/0/2	N/A	N/A	N/A	View Us
120001	DC1_WE2	ap-south-2		5.0	local	GigabitEthernet0/0/2	N/A	N/A	N/A	View Us
120001	DC1_WE2	ap-southeast-1		6.0	local	GigabitEthernet0/0/0	N/A	N/A	N/A	View Us
120001	DC1_WE2	ap-southeast-2		6.0	local	GigabitEthernet0/0/0	N/A	N/A	N/A	View Us
120001	DC1_WE2	ap-northeast-1		6.0	local	GigabitEthernet0/0/0	N/A	N/A	N/A	View Us

Cloud OnRamp for Webex Determines Best Path

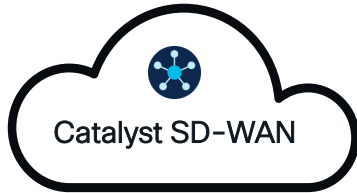


Gig0/0/0 has packet loss and hence wasn't selected

RTP Streams Are Routed Across the Best Path



Key Takeaways



Connect Users, Deliver Best Experience and Simplify Network Operations
using *integrated* Cisco platforms

Webex App

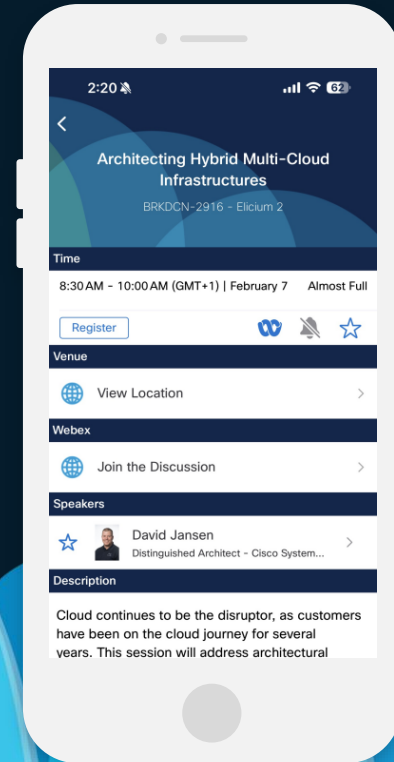
Questions?

Use the Webex app to chat with the speaker after the session

How

- 1 Find this session in the Cisco Events mobile app
- 2 Click “Join the Discussion”
- 3 Install the Webex app or go directly to the Webex space
- 4 Enter messages/questions in the Webex space

Webex spaces will be moderated by the speaker until February 28, 2025.



Fill Out Your Session Surveys



Participants who fill out a minimum of 4 session surveys and the overall event survey will get a unique Cisco Live t-shirt.

(from 11:30 on Thursday, while supplies last)



All surveys can be taken in the Cisco Events mobile app or by logging in to the Session Catalog and clicking the 'Participant Dashboard'



Content Catalog

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- Book your one-on-one Meet the Engineer meeting
- Attend the interactive education with DevNet, Capture the Flag, and Walk-in Labs
- Visit the On-Demand Library for more sessions at ciscolive.com/on-demand. Sessions from this event will be available from March 3.

Contact me at: carunach@cisco.com

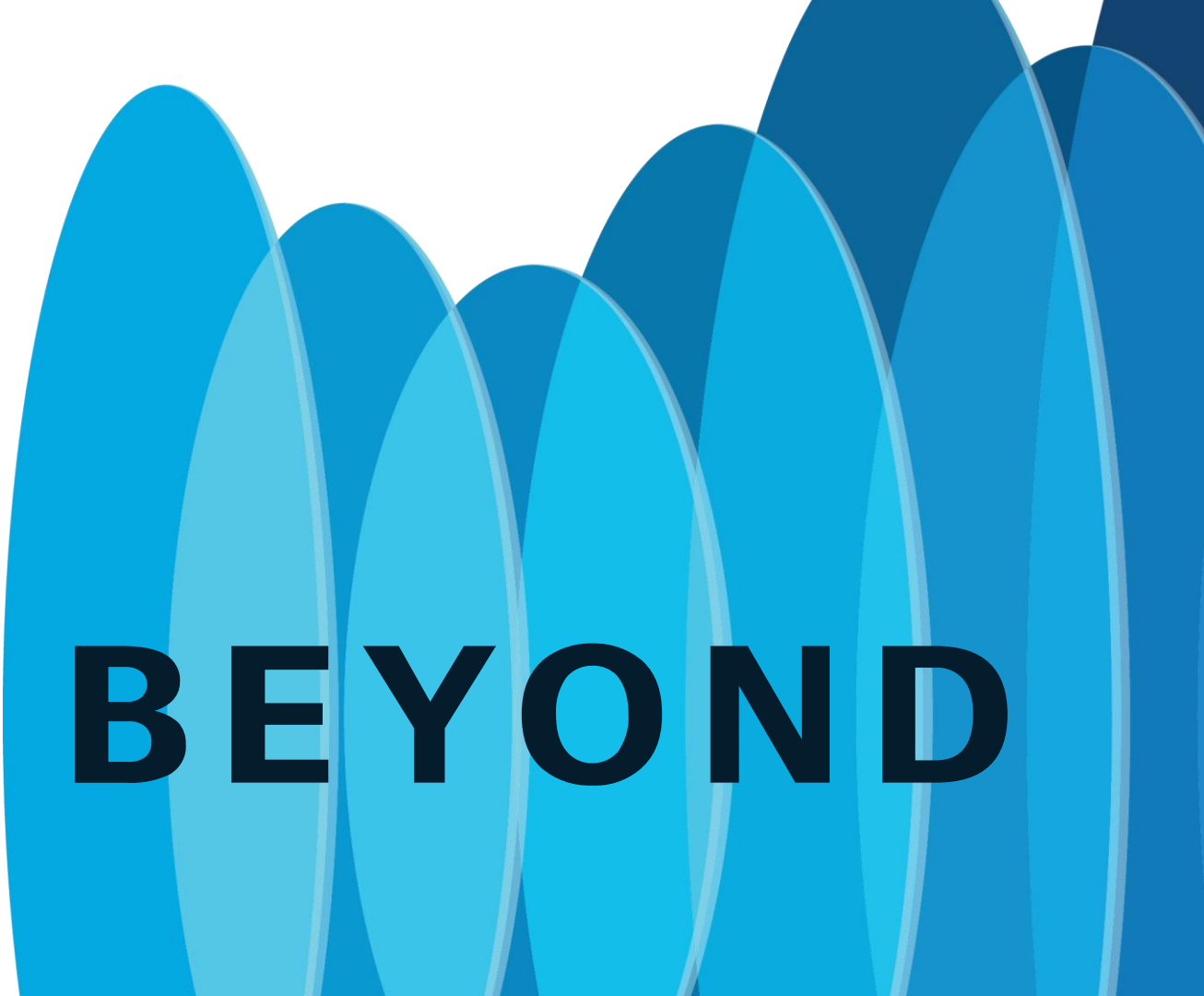


Thank you



CISCO *Live!*

GO BEYOND

The background of the slide features a series of overlapping, teardrop-shaped elements in various shades of blue, ranging from light sky blue to deep navy blue. These shapes are arranged in a way that creates a sense of depth and movement, resembling a stylized horizon or a series of waves. The overall composition is clean and modern, with the text 'GO BEYOND' prominently displayed in the center.